How Many Demonstrations Is a Task Description Worth? Specificity and Reliability in In-Context Learning

by

Trung Quan Nguyen

A thesis submitted in partial fulfillment of the requirements for the degree of

Master of Science

in

Statistics

Department of Mathematical and Statistical Sciences
University of Alberta

© Trung Quan Nguyen, 2026

Abstract

Prompts for in-context learning (ICL) can describe a task in addition to, or instead of, demonstrating it through examples. Yet how descriptions work remains largely underexplored. Through the Bayesian view of ICL, we model descriptions as informative priors, characterized by their specificity and reliability. We adapt the effective sample size (ESS) from Bayesian statistics to quantify a description's predictive value as an equivalent number of demonstrations. Our theoretical results on in-context linear regression tasks show that a reliable description's ESS grows with its specificity and with the task's dimension. However, the ESS falls sharply if the description may be wrong: in a 16-dimensional task with no demonstrations, a one-in-ten chance of being wrong cuts a highly specific description's ESS from 117 to 13.5. We then test whether Transformer networks and pretrained language models, including frontier models, behave like Bayesian predictors in these settings. Meta-trained Transformers come close to the Bayes-optimal ESS when test descriptions are as reliable as training ones. Pretrained language models, however, make too little use of highly specific descriptions when answering without reasoning, and most of them closely match a Bayesian reference only when reasoning is enabled. Taken together, these results extend the Bayesian analysis of ICL from demonstrations to descriptions.

Preface

This thesis is an original work by Trung Quan Nguyen, carried out under the supervision of Professor Linglong Kong and Professor Bei Jiang. The author formulated the research questions, designed and carried out the analysis and experiments, and wrote the text; the supervisors provided guidance and feedback throughout.

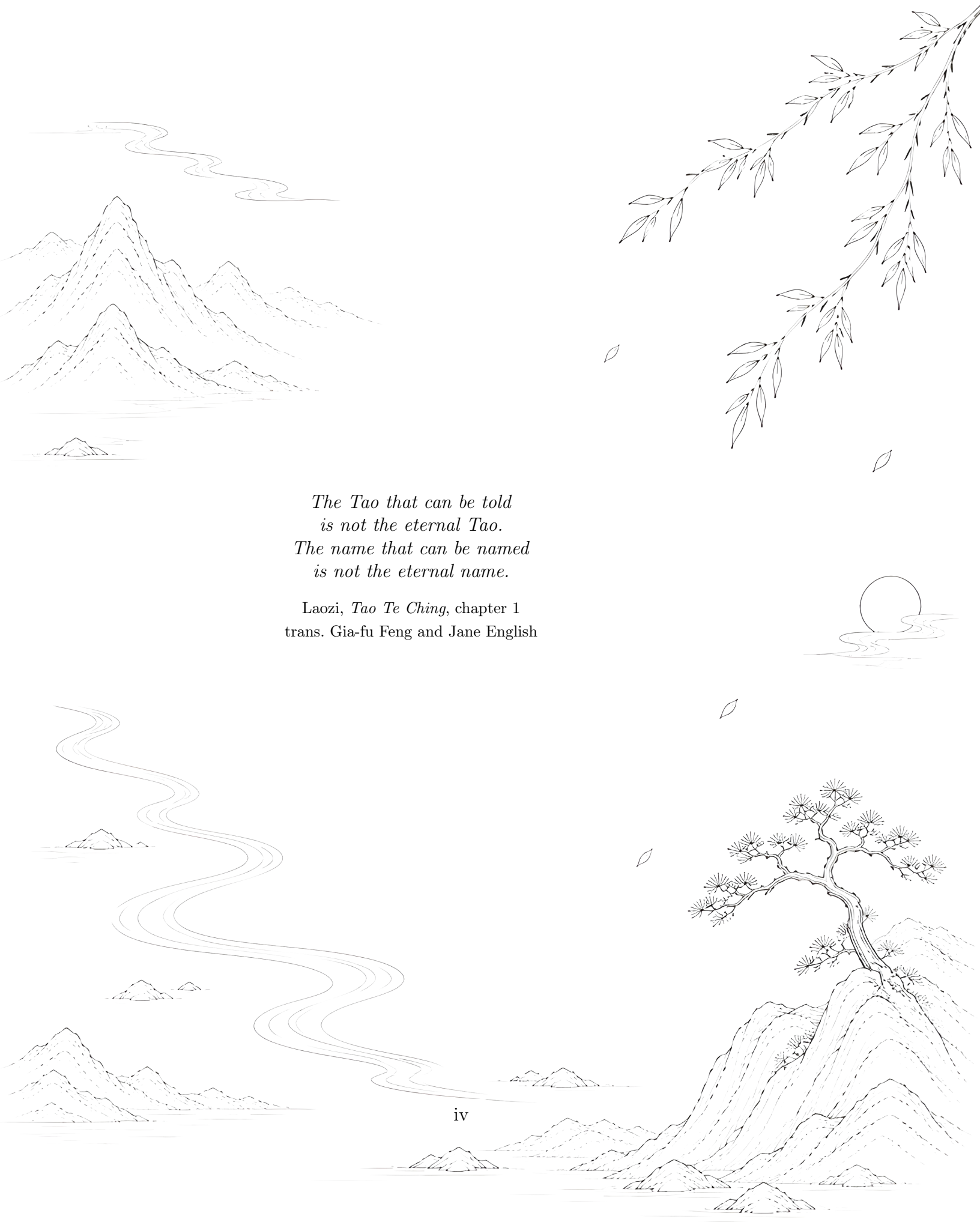
The thesis is based on the manuscript "How Many Demonstrations Is a Task Description Worth? Specificity and Reliability in In-Context Learning" by Bruce Nguyen, the name under which the author writes, prepared in the format of the International Conference on Machine Learning. Chapters 1 and 3 to 8 and Appendix A follow that manuscript's text, with its appendices placed in the chapters they support and some added detail; Sections 1.2 and 8.2 are new. Chapter 2 is new, except that it incorporates the manuscript's review of related work, placed in the sections whose topics it concerns. The manuscript has not yet been published.

The research involved no human participants, animals, or personal data, so no research ethics approval was required.

Beyond studying large language models in Chapter 7, the author used tools built on them, Anthropic's Claude Code and OpenAI's Codex, to help write and run experiment code, check derivations, and draft and edit text. The author designed the study, reviewed all code, results, and text, and takes responsibility for them.

The author's salary and the Claude Code and Codex accounts used in this research were paid for by Professor Andrew McGee of the Department of Economics, from funding for his own research.

The author founded Decaude, and this research is part of its research program; the manuscript lists the author's affiliations as Decaude and the University of Alberta.



The Tao that can be told
is not the eternal Tao.
The name that can be named
is not the eternal name.

Laozi, *Tao Te Ching*, chapter 1
trans. Gia-fu Feng and Jane English

Acknowledgements

I am deeply grateful to my family for their unconditional love, which carried me through these years.

I thank my supervisors, Professor Linglong Kong and Professor Bei Jiang, for their guidance and support throughout my program.

I am grateful to Professor Andrew McGee of the Department of Economics for his trust, patience, and generosity.

Finally, I thank God for granting me the strength and wisdom to overcome a life-altering illness during my studies and to complete this work.

Table of Contents

Abstract ii

Preface iii

Quotation iv

Acknowledgements v

List of Tables viii

List of Figures ix

1 Introduction 1

1.1 Main Findings 2

1.2 Outline 3

2 Background 5

2.1 In-Context Learning 5

2.2 In-Context Learning as Bayesian Inference 6

2.3 In-Context Linear Regression 7

2.4 Squared Error 8

2.5 The Effective Sample Size of a Prior 10

2.6 Robust Mixture Priors 12

2.7 Descriptions in Theories of ICL 13

2.8 Descriptions and Priors in Pretrained Language Models 13

3 Problem Setup 16

4 The Value of a Reliable Description **20**
4.1 Complete Descriptions . . . 20
4.2 Partial Descriptions . . . 24

5 The Cost of Unreliability **28**
5.1 Unreliable Descriptions . . . 28

6 Meta-Trained Transformers **34**
6.1 Training and Evaluation . . . 35
6.2 Matched Training and Test Reliability . . . 36
6.3 Reliability Shifts . . . 39

7 Pretrained Language Models **42**
7.1 Task and Evaluation . . . 42
7.2 Reasoning and Stated Precision . . . 47
7.3 Stated Reliability . . . 48
7.4 Descriptions in Words: Sorting Messages . . . 52
7.5 Additional Results and Evaluation Details . . . 53
7.5.1 Prompts, Tasks, and Answer Extraction . . . 53
7.5.2 ESS Estimation and the Bayesian Reference . . . 56
7.5.3 Results of the 2,000-Task Run . . . 58
7.5.4 Irrelevant Descriptions and Wording . . . 61
7.5.5 Reasoning Settings and Inference Details . . . 62
7.5.6 Prediction Rules in Individual Replies . . . 66
7.5.7 Wording with Reasoning Enabled . . . 72
7.5.8 Details of the Message-Sorting Experiment . . . 73

8 Discussion **77**
8.1 Scope and Limitations . . . 78
8.2 Future Work . . . 79

Bibliography **82**

A Language-Model Results at Every Number of Demonstrations **88**

List of Tables

4.1 Demonstrations can increase the value of a specific partial description 25

5.1 The Bayes-optimal value of a description, in demonstrations 30
5.2 The bound after many demonstrations is nearly attained 32
5.3 The cost of unreliability at other dimensions and baseline variances . 33

6.1 Transformer ESS and fitted description weights under matched and shifted reliability 38

7.1 Evaluated model configurations 44
7.2 Unreliable descriptions with the stated chance of irrelevance 50
7.3 Description ESS and weights in the 2,000-task run without reasoning 55
7.4 Sensitivity of median ESS to baseline-curve estimation 60
7.5 Description weights under four wordings without reasoning 63
7.6 Description ESS across reasoning settings 67
7.7 Absolute error with a description 68
7.8 Description weights under stated unreliability 70
7.9 Stating the baseline prior changes description weights 72
7.10 ESS of notes in the message-sorting experiment 76

A.1 ESS by description condition and demonstration count: distribution readout 89
A.2 ESS by description condition and demonstration count: written readout 91
A.3 Description weights by demonstration count: distribution readout . . 92

List of Figures

1.1 Measuring a task description in demonstrations 4

4.1 Without demonstrations, reliability limits the benefit of specificity; demonstrations recover much of it 27

6.1 Meta-trained Transformers approach Bayes-optimal ESS and weight descriptions by training reliability 36
6.2 Transformer ESS as a function of demonstration count 39

7.1 Language-model ESS under stated precision and reliability 46
7.2 Learning curves used to estimate language-model ESS 59

Chapter 1

Introduction

Large language models (LLMs) are capable of in-context learning (ICL): given a prompt that specifies a task through a natural-language description, demonstrations, or both, they can predict the output for a new query input (Brown et al., 2020). In practice, prompts consist mostly of system instructions, user instructions, agent skill files, and earlier conversation turns, which typically describe the task rather than demonstrate it (Mao et al., 2025; Patsakis et al., 2026). The balance shifts further toward description as LLMs carry out increasingly long-horizon tasks (Kwa et al., 2025), for which demonstrations are costly to provide.

Theories of ICL, however, have mostly focused on prompts consisting of demonstrations alone (Xie et al., 2022; Garg et al., 2022; Akyürek et al., 2023). The few that include descriptions study only particular types, such as a task descriptor giving the mean of the inputs (Huang & Ge, 2025), a hypothesis class listed in the prompt (Lin et al., 2025), or datasets from related tasks that inform the prior (Zhu et al., 2026); none offers a systematic treatment of descriptions. We therefore ask: how can we characterize natural-language descriptions in general, so as to understand and analyze them formally? And how well do learners, whether meta-trained Transformers or pretrained language models, use descriptions in practice?

We study these questions through the Bayesian view of ICL, in which a learner infers a latent task, the function that maps each input to its output, from the demonstrations (Xie et al., 2022). A description carries information about the task without being a demonstration of it, so the description can be modeled as a prior over tasks. Descriptions differ in how much they say about a task: "write this report as an accountant" leaves most of the task open, whereas a step-by-step workflow leaves little.

They also differ in whether they are about the right task: an outdated instruction, or an instruction meant for a different task, describes the wrong one. Neither property can be measured directly in a natural-language instruction, so, as a first step, we study a setting in which both are known exactly. We formalize these two properties in d-dimensional Gaussian linear regression, a standard controlled setting for ICL (Garg et al., 2022; Panwar et al., 2024), in which a task is a latent regression vector. A description gives a prior mean, the regression vector it points to. We call a description that is about the right task *relevant*. Given a relevant description, the task is Gaussian around its prior mean with standard deviation τ: the more a description says, the smaller τ and the higher the description's *specificity*. A description is relevant with probability p, its *reliability*; otherwise it is *irrelevant*, and its mean is independent of the task (Figure 1.1).

To compare descriptions with demonstrations, we follow Bayesian work that expresses a prior's information or impact as an equivalent sample size (Wiesenfarth & Calderazzo, 2020; Reimherr et al., 2021; Selby et al., 2025), and work that expresses a prompt's value in training data points (Le Scao & Rush, 2021). A description's *predictive effective sample size* (ESS) is the number of additional demonstrations that a learner would need, in place of the description, to reach the same expected error. For the Bayes-optimal predictor, ESS gives a description's value under the statistical model; for a trained learner, it gives the value that the learner realizes. To see how well learners use descriptions, we test how close each learner comes to the ESS of a Bayesian predictor given the same information.

1.1 Main Findings

Our main findings are as follows:

- **Specificity and dimension determine the value of reliable descriptions.** When little is known about the task without the description, a description that places each coordinate of the task within about τ of its prior mean is worth about $c + d + 1$ demonstrations before any are seen, where $c = \tau^{-2}$ is its precision: c because each demonstration adds one unit of precision, and $d + 1$ more because random inputs leave some directions of the task poorly constrained for a learner without the description. Demonstrations lower this value, to about c after many; a

description that constrains only part of the task can instead gain value from them (Chapter 4).

- **Unreliability caps zero-shot ESS.** Any fixed positive probability of irrelevance caps the ESS without demonstrations, the zero-shot ESS, at a finite value, however specific the description. Demonstrations can then show whether the description is relevant and raise its ESS above this ceiling (Chapter 5, Figure 4.1b).
- **Meta-trained Transformers approach the Bayes-optimal ESS when test descriptions are as reliable as training ones.** Without demonstrations, they weight a description by how often descriptions were relevant in training, whatever the test reliability (Chapter 6).
- **Language models use a precision stated in the prompt mainly with reasoning enabled.** Without reasoning, every model tested falls far short of a Bayesian predictor for what the prompt states when the description is highly specific; with reasoning, most closely match its error (Chapter 7).

1.2 Outline

Chapter 2 reviews in-context learning and its Bayesian interpretation, in-context linear regression and its squared error, the effective sample size of a prior, and robust mixture priors, together with related work. Chapter 3 sets up the problem: tasks and demonstrations, descriptions as uncertain task information, prediction error, and predictive ESS. Chapter 4 derives the value of a reliable description, with its proofs and the analysis of partial descriptions. Chapter 5 derives the cost of unreliability, with its proofs. Chapter 6 tests meta-trained Transformers, with their training details and further networks. Chapter 7 tests pretrained language models on numerical prediction tasks with explicit probabilistic descriptions, with the evaluation details and further experiments on stated reliability and on a message-sorting experiment, in which a note describes a sorting rule in words. Chapter 8 discusses the findings and their limitations, and outlines future work. Appendix A reports the language-model tables at every evaluated demonstration count.

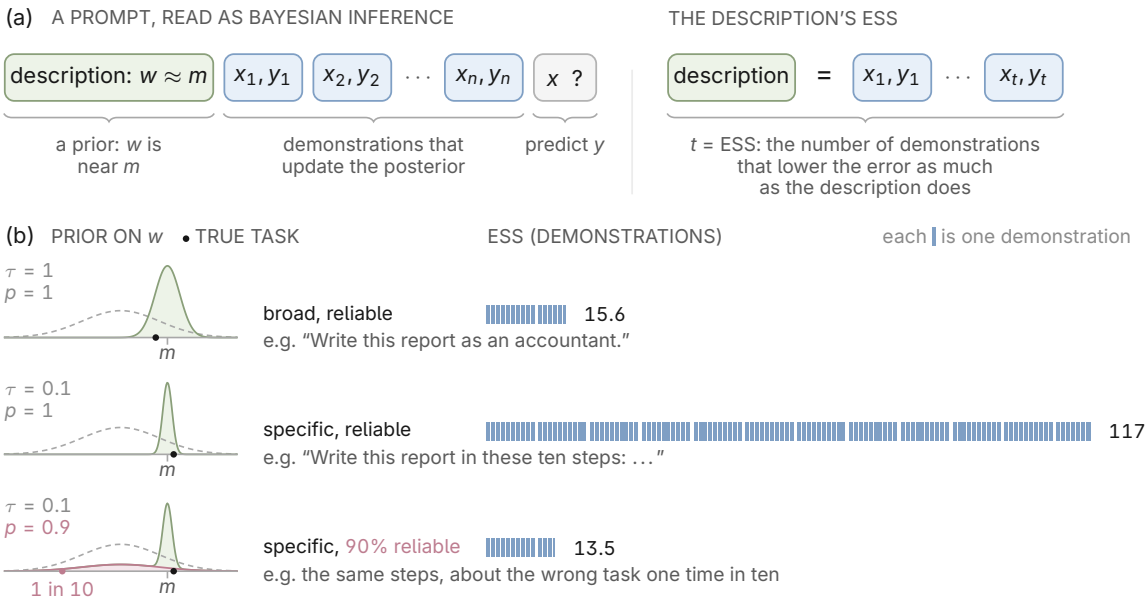


Figure 1.1: **Measuring a task description in demonstrations.** (a) A description provides information about a latent task w, and demonstrations supply additional evidence. The description's ESS is the number of demonstrations that lower the error as much as it does; the schematic on the right shows a prompt with no demonstrations ($n = 0$). (b) The prior on w under three descriptions, and each description's Bayes-optimal ESS at $n = 0$, $d = 16$, and baseline prior variance $\tau_0^2 = 10$; τ is the standard deviation of the task around the description's prior mean m. The curves are sketches, not drawn to scale. Dashed curves show the baseline prior; the pink curve is the irrelevant component, which has the baseline prior's shape. Dots mark the true task; for the unreliable description, it lies elsewhere one time in ten. The example instructions illustrate each kind of description; the ESS values are those of our Gaussian regression model. Increasing specificity greatly increases ESS when $p = 1$, whereas a 10% probability of irrelevance sharply limits this benefit.

Chapter 2

Background

In this chapter, we review the ideas that the rest of the thesis builds on. Sections 2.1 to 2.3 introduce in-context learning, its interpretation as Bayesian inference, and in-context linear regression, a setting in which that interpretation can be computed exactly. Section 2.4 defines the squared error by which we compare learners and gives the error of the Bayesian predictor. Sections 2.5 and 2.6 present two tools from Bayesian statistics, the effective sample size of a prior and the robust mixture prior. We later apply both to descriptions, and these sections relate our use of them to other work. Finally, Section 2.7 reviews theories of ICL that include descriptions, and Section 2.8 reviews studies of how pretrained language models use descriptions and stated probabilities.

2.1 In-Context Learning

A language model takes a sequence of tokens, the prompt, and predicts what comes next. Brown et al. (2020) showed that large language models can perform a task specified in the prompt, without any update to their weights. This ability is called in-context learning (ICL). Brown et al. (2020) distinguished three kinds of prompts. A zero-shot prompt gives only a natural-language description of the task. One-shot and few-shot prompts add one or several *demonstrations*, solved examples of the task, before the query that the model must answer. We follow their terminology: a demonstration is one input together with its label, and a description gives information about the task other than through demonstrations, whether as a statement about

the task or as a command. Weller et al. (2020) likewise formalize learning from task descriptions by distinguishing the latent task, its natural-language description, and individual input–output examples. Chapter 3 gives the model of a description that we analyze. In our experiments, the meta-trained Transformers receive a description as a numeric token, and the language models receive it as a probabilistic statement or a note about a sorting rule. A prompt can contain both a description and demonstrations, and our question is how much the description adds to the demonstrations.

How models use the demonstrations is itself an active question. Pretrained models can perform well even when the labels of the demonstrations are replaced by random ones (Min et al., 2022), which suggests that they use demonstrations partly to recognize a task they already know. Larger models, however, follow flipped in-context labels more readily (Wei et al., 2023). Lin & Lee (2024) explain both behaviors, retrieving a known task and learning a new one, within a single Bayesian model. A description raises the same question from the other side. Does a model treat it as information about the task, to be weighed against the demonstrations, or does it use it in some other way?

2.2 In-Context Learning as Bayesian Inference

Theoretical work studies ICL as Bayesian inference over latent tasks (Xie et al., 2022; Panwar et al., 2024), as a learning algorithm implemented in a forward pass (Akyürek et al., 2023; von Oswald et al., 2023), and as meta-learning across tasks (Garg et al., 2022; Müller et al., 2022). Meta-trained predictors can approximate Bayesian inference under suitable training conditions (Genewein et al., 2025), and the learning algorithms that Transformers implement in context approach the Bayes-optimal estimator as the networks grow (Akyürek et al., 2023). Agreement with the Bayesian predictor can depend on training-task diversity and on the function class (Raventós et al., 2023; Panwar et al., 2024), and language-model ICL can violate the martingale property that Bayesian prediction implies (Falck et al., 2024). We adopt the Bayesian view, because it extends naturally from demonstrations to descriptions (Chapter 3).

In the Bayesian view, as Xie et al. (2022) proposed it, pretraining text is generated from latent concepts. A model that predicts such text well must then infer the concept behind each document, and a prompt of demonstrations is evidence about the concept behind the prompt. The model's prior over concepts is acquired in pretraining, so

even a prompt without demonstrations draws on it, and a description refines it before any demonstration is seen. Formally, a task is a latent parameter w, and a prompt contains n demonstrations $D_n = \{(x_i, y_i)\}_{i=1}^n$ and a query x. The ideal learner predicts the query's label with the posterior predictive distribution

$$f(y \mid x, D_n) = \int f(y \mid x, w)\, f(w \mid D_n)\, dw. \tag{2.1}$$

This view becomes testable when the prior over tasks is known. Consider a Transformer trained on prompts whose tasks are sampled from a known prior. Training on such prompts is called meta-training: the network is trained on many tasks and must infer from the context which task a prompt comes from. The posterior predictive minimizes the expected log loss over the prior, and this expected log loss is exactly the training objective. A well-trained network therefore approximates the posterior predictive (Müller et al., 2022), whose mean is the Bayes-optimal prediction under squared error. Section 2.4 shows both properties of the posterior predictive: that it minimizes expected log loss, and that its mean minimizes squared error. For regression with a Gaussian task prior, Transformers meta-trained on enough distinct tasks behave like the Bayesian predictor, which is ridge regression (Akyürek et al., 2023; Raventós et al., 2023; Panwar et al., 2024); Section 2.3 describes these experiments. The Bayesian predictor is then a benchmark against which a network can be measured, and Chapter 6 uses it in this way.

2.3 In-Context Linear Regression

Garg et al. (2022) introduced in-context linear regression as a controlled test of ICL. Each prompt has its own task vector $w \sim \mathcal{N}(0, I_d)$, its inputs are drawn as $x_i \sim \mathcal{N}(0, I_d)$, and its labels are $y_i = w^\top x_i$. Garg et al. (2022) found that a Transformer trained from scratch on such prompts predicts the query's label about as well as least squares. Later work added label noise, $y_i = w^\top x_i + \epsilon_i$ (Akyürek et al., 2023; Raventós et al., 2023). Raventós et al. (2023) also varied the number of distinct tasks seen in training. With few tasks, the network behaves like the Bayesian predictor for a prior over those tasks alone. With enough tasks, it behaves like ridge regression, the Bayesian predictor for the Gaussian prior.

With Gaussian noise and a Gaussian prior, Bayes-optimal prediction has a closed form (Gelman et al., 2013, Section 14.8). We measure labels in units of their noise, so

that $y_i \sim \mathcal{N}(w^\top x_i, 1)$. Let the prior be $w \sim \mathcal{N}(m, \Lambda^{-1})$, with mean m and precision Λ. Let $X_n \in \mathbb{R}^{n \times d}$ contain the demonstration inputs as rows, and write $W_n = X_n^\top X_n$. The posterior over w is Gaussian, with covariance and mean

$$\Sigma = \big(\Lambda + W_n\big)^{-1}, \qquad \mu = \Sigma\big(\Lambda m + X_n^\top y_{1:n}\big), \tag{2.2}$$

and the posterior predictive for the query is

$$y \mid x, D_n \sim \mathcal{N}\big(\mu^\top x,\ 1 + x^\top \Sigma x\big). \tag{2.3}$$

The predictive variance has two parts. The first, 1, is the label noise. The second, $x^\top \Sigma x$, is the remaining uncertainty about the task in the direction of the query. Under the baseline prior $\mathcal{N}(0, \tau_0^2 I_d)$, the posterior mean $\mu = (W_n + \tau_0^{-2} I_d)^{-1} X_n^\top y_{1:n}$ is the ridge regression estimate with penalty τ_0^{-2}. Equation (2.2) also shows that a prior acts like additional data. Each demonstration adds $x_i x_i^\top$ to the posterior precision, and the prior adds Λ. Chapter 3 uses this fact to give a description a prior of its own, and in Chapter 4 a reliable description replaces the baseline precision $c_0 = \tau_0^{-2}$ with its own precision $c = \tau^{-2}$.

2.4 Squared Error

A predictor maps the prompt to a prediction $\hat{y}$ of the query's label. We score it by squared error. The label is $y = w^\top x + \epsilon$, where the noise ϵ has mean 0 and variance 1 and is independent of the prompt and the task. The expected squared error therefore splits into two parts:

$$\mathbb{E}\big[(\hat{y} - y)^2\big] = \mathbb{E}\big[(\hat{y} - w^\top x)^2\big] + 1. \tag{2.4}$$

The second part is the noise variance, which is the same for every predictor. Following Garg et al. (2022), we therefore score a predictor by the first part alone. We call it the predictor's *prediction error*, or simply *error*:

$$R = \mathbb{E}\big[(\hat{y} - w^\top x)^2\big].$$

It is the expected squared distance from the noise-free label $w^\top x$. A predictor that knows the task predicts $w^\top x$ and has zero error.

Given the prompt, the task is uncertain. For any prediction $\hat{y}$ computed from the prompt,

$$\mathbb{E}\big[(\hat{y} - w^\top x)^2 \mid x, D_n\big] = \big(\hat{y} - \mathbb{E}[w^\top x \mid x, D_n]\big)^2 + \mathrm{Var}(w^\top x \mid x, D_n).$$

The first term vanishes only when $\hat{y}$ is the posterior mean of $w^\top x$. When tasks are drawn from the prior, the posterior mean therefore has the smallest error of any predictor. We call it the Bayesian predictor and its error Bayes-optimal. It is also the mean of the posterior predictive (2.1). Its error is the expected posterior variance of $w^\top x$. In linear regression, the posterior is Gaussian with covariance Σ (Equation (2.2)), so the posterior variance of $w^\top x$ is $x^\top \Sigma x$, the part of the predictive variance (2.3) that comes from uncertainty about the task. The query input is independent of the demonstrations, with $\mathbb{E}[xx^\top] = I_d$, so the error of the Bayesian predictor is

$$R = \mathbb{E}\left[x^\top \Sigma x\right] = \mathbb{E}\,\mathrm{tr}\,\Sigma, \tag{2.5}$$

the expected trace of the posterior covariance. The error is zero when the posterior has concentrated on the task, and it grows with the uncertainty about the task that remains.

Two standard facts give this trace in the cases that later chapters need. Chapter 3 writes $R^{\mathrm{base}}(n)$ for the error with n demonstrations and no description. With fewer demonstrations than dimensions, $n < d$, the demonstrations carry no information about $d - n$ directions, so under the baseline prior $\mathcal{N}(0, \tau_0^2 I_d)$ we have $R^{\mathrm{base}}(n) \geq (d - n)\,\tau_0^2$. In the limit $\tau_0 \to \infty$ of a flat prior, and with $n \geq d + 2$ demonstrations, the posterior covariance is W_n^{-1}, the inverse of a Wishart matrix. The error is then $\mathbb{E}\,\mathrm{tr}\,W_n^{-1} = d/(n - d - 1)$ (Muirhead, 1982). Chapter 4 builds its limits for a reliable description (Proposition 4.1) from these two facts.

A predictor that gives a density $g(y \mid x, D_n)$ rather than a number can instead be scored by its log loss, $-\log g(y \mid x, D_n)$. The expected log loss is smallest for the true conditional density of y given the prompt (Cover & Thomas, 2006), which is the posterior predictive when tasks are drawn from the prior. Meta-trained networks are often trained with this loss (Section 2.2), and so are the networks of Chapter 6. The mean of the posterior predictive is the posterior mean, so a network that minimizes log loss and outputs a predictive mean can be scored by squared error. We use squared error throughout for two reasons. First, it requires only a point prediction, so it applies to the number a language model writes (Chapter 7). Second, the Bayes-optimal error (2.5) has the exact forms above, so the theory in Chapters 4 and 5 can work with exact errors.

Squared error also splits exactly over the components of a mixture. Suppose that a latent indicator Z selects the component that generates the task. By the law of total

variance, the posterior variance is the sum of two parts:

$$\begin{aligned}\operatorname{Var}(w^\top x \mid x, D_n) = \mathbb{E}\big[\operatorname{Var}(w^\top x \mid x, D_n, Z) \mid x, D_n\big] \\ + \operatorname{Var}\big(\mathbb{E}[w^\top x \mid x, D_n, Z] \mid x, D_n\big).\end{aligned} \tag{2.6}$$

The first part is the average variance within the components. The second is the variance of the prediction across the components. Chapter 5 applies this view to a description that may be irrelevant. There, a predictor that also observes Z has only the first part, and the second is the cost of not knowing whether the description is relevant.

2.5 The Effective Sample Size of a Prior

Bayesian statistics measures the information in a prior by its effective sample size (ESS), the number of observations that would carry the same information. In the conjugate normal model, this number is exact. Suppose that the observations are $y_i \sim \mathcal{N}(w, \sigma^2)$ and the prior is $w \sim \mathcal{N}(m, \sigma^2/c)$. After n observations with mean $\bar{y}$, the posterior is

$$w \mid y_{1:n} \sim \mathcal{N}\Big(\frac{c\,m + n\bar{y}}{c+n},\ \frac{\sigma^2}{c+n}\Big).$$

This is the posterior that $c+n$ observations would give under a flat prior if c of them averaged m. The prior is therefore worth c observations: c is its precision in units of the precision of one observation. Chapter 7 uses this model for its Bayesian reference. The precision c of a description (Chapter 3) is this ESS in each direction of the task, counted in direct observations of the task with unit noise.

Beyond the conjugate case, how many observations a prior is worth depends on what is matched, and the literature offers several choices (Morita et al., 2008; Neuenschwander et al., 2020). Morita et al. (2008) match information, which they measure by the curvature of the log density. Their ESS is the number of hypothetical observations that would turn a very vague prior into a posterior carrying about the same information as the prior. Neuenschwander et al. (2020) add a consistency requirement: on average, the ESS of the posterior after N observations should equal the prior's ESS plus N. Other definitions match performance at the current sample instead. Reimherr et al. (2021) count the observations that a baseline prior would need to match a given prior's expected posterior loss at a given sample size; their ESS can

be negative when the prior and the likelihood conflict. The effective current sample size of Wiesenfarth & Calderazzo (2020) starts from the accuracy of an analysis of the current sample with the prior. It counts the observations that must be added to the sample, or removed from it, for an analysis without the prior to be equally accurate. Reznik (2026) uses predictive log loss to choose the discount of a power prior, which raises the likelihood of historical data to a power between 0 and 1.

Our aim is to compare what a description is worth to different learners: the Bayes-optimal predictor, meta-trained networks, and pretrained language models. Matching information requires an explicit prior, which only the first of these has. A network or a language model also brings prior knowledge to a task, acquired in training, but only implicitly, in its parameters, and only its predictions can be observed. Matching performance requires only those predictions. Our predictive ESS (Definition 3.1) therefore follows the loss-based measures of Reimherr et al. (2021) and Wiesenfarth & Calderazzo (2020): it is the number of additional demonstrations that a learner needs, without the description, to reach the prediction error that it has with the description. It differs from those measures in three ways: it evaluates population prediction error, it allows a non-Bayesian learner, whose own curve without the description serves as the baseline, and it depends on the number of demonstrations already observed. The word *predictive* names what is matched; it does not assert the predictive-consistency property of Neuenschwander et al. (2020).

Matching performance rather than information changes the answer even for the Bayes-optimal predictor. By the measure of Morita et al. (2008), a prior of precision c adds c to the posterior precision regardless of the sample size, so a description that raises the precision from c_0 to c in every direction is worth $c - c_0$ demonstrations, regardless of how many have been observed. Prediction error behaves differently when the inputs are random. In d dimensions under a flat prior, the Bayesian predictor's error after n demonstrations is $d/(n - d - 1)$ (Section 2.4). The expected information of the demonstrations would suggest d/n; the error is larger because random inputs constrain some directions less than others. Chapter 4 shows the consequence. Without demonstrations and under a diffuse baseline prior, an ESS that matches error is about $c + d + 1$, more than the added precision by a term that grows with the dimension; after many demonstrations, it approaches $c - c_0$, the value of Morita et al. (2008) measured relative to the baseline prior.

Two predictive comparisons in machine learning are especially close to ours. Le Scao

& Rush (2021) measure the horizontal separation of learning curves in prompt-based fine-tuning, averaged over a range of accuracies; we evaluate a pointwise difference in demonstration counts. Selby et al. (2025) compare priors elicited from language models with data-based predictors; their measure is the number of observations needed to match predictive loss. We instead study descriptions supplied to a learner, and how their benefit changes after demonstrations. Beyond ICL, an instruction can replace up to 100 execution examples in the fine-tuning settings studied by Cook et al. (2026). Wenliang et al. (2026) treat a prompt as conditioning for a Bayesian sequence predictor and compare an optimized prompt against a set of demonstrations: the optimized prompt can be shorter than demonstrations that have the same effect.

2.6 Robust Mixture Priors

A prior built from historical data can conflict with the current data. Evans & Moshonov (2006) check for such prior–data conflict by asking whether the observed data are surprising under the prior predictive distribution. Heavy-tailed priors accommodate such conflict (O'Hagan & Pericchi, 2012), and clinical trials that borrow historical control data protect against it with a robust mixture prior (Schmidli et al., 2014). The informative prior f_I, derived from the historical data, is mixed with a vague prior f_V:

$$f(w) = p\, f_I(w) + (1 - p)\, f_V(w).$$

The posterior is then a mixture of the two component posteriors, and the weight on the informative component is updated in proportion to p times its marginal likelihood. When the data agree with the informative component, its weight grows, and the analysis borrows strength from the historical data. When they conflict, the weight moves to the vague component, and the analysis falls back on the current data.

A description is a prior of the same kind. It may describe the task at hand, but it may also not apply to it, for example because it was written for another task. Chapter 3 therefore models a description as a robust mixture prior whose weight is its reliability p, the probability that the description is relevant. Our mixture is specified directly, rather than derived from a historical-data meta-analysis. Chapter 5 calls the updated weight the posterior relevance probability π_n and shows how reliability affects the description's ESS. Averaging the component predictions with weights π_n is Bayesian model averaging (Hoeting et al., 1999). Chapter 7 also compares language-model

predictions with *fixed-weight averaging*, which keeps the weights at p and $1-p$ instead of updating them.

In our model, an irrelevant description is independent of the task and may happen to agree with it. Irrelevance is therefore a property of how the description was generated, and it differs from the prior–data conflict above, which concerns disagreement between prior information and observed data.

2.7 Descriptions in Theories of ICL

Theories of ICL have added task information to demonstrations in several forms: a token that carries the mean of the input distribution (Huang & Ge, 2025), tokens that indicate a class of candidate hypotheses (Lin et al., 2025), and related datasets supplied as context (Zhu et al., 2026). Other theories study a prior that mixes task groups (Lin & Lee, 2024). These theories generally do not measure the value of a description in units of demonstrations, and they do not model the possibility that it is irrelevant. For example, Huang & Ge (2025) show that a learner can use the description token to predict better, but they do not say how many demonstrations it is worth, or what it is worth when it may be irrelevant. The work closest to our setting is that of Zhu et al. (2026), who show that a Transformer given a prefix of related datasets performs amortized hierarchical Bayesian prediction. We use explicit Bayesian predictors to quantify how such conditioning information changes the number of demonstrations needed to reach a given prediction error, and we allow that information to be irrelevant.

The Bayesian theory of Xie et al. (2022) models the prompt as demonstrations alone. We keep its view of the prompt as evidence about a latent task and add a description, which induces a distribution over tasks before the demonstrations are observed. Our networks in Chapter 6 use the prefix embedding of Huang & Ge (2025), in which the description is a single token placed before the demonstrations.

2.8 Descriptions and Priors in Pretrained Language Models

Empirical studies differ on how much descriptions help language models. Task summaries distilled from demonstrations can replace the examples in context (Honda et al.,

2025), whereas other studies report weak sensitivity to instruction relevance (Webson & Pavlick, 2022) or limited benefits from repository-level context files (Gloaguen et al., 2026), and Tong et al. (2026) study how sensitivity to the prompt template varies with the number of demonstrations. When a description helps little, these studies cannot tell whether it carried little information or the model failed to use it. Our controlled experiments instead specify the joint distribution of task, description, and demonstrations, so the information a description carries is known and a model's use of it can be measured against a Bayesian reference.

Other work evaluates language-model predictions against Bayesian references. Gupta et al. (2025) compare explicitly stated coin biases with observations in this way, and find that in the models' predictions the observed flips outweigh the stated bias. The learning curves of pretrained models approach a Bayesian reference on synthetic in-context function-learning tasks with a known optimum (Akata et al., 2026), and Arora et al. (2025) model in-context learning curves with a Bayesian scaling law that includes a task prior. Qiu et al. (2026) also study how training can improve probabilistic reasoning; He et al. (2025) examine Bayesian consistency during iterative reasoning on open-ended questions; and concurrent work studies how reasoning and code execution affect language models' approximation of Bayesian posteriors (Filbry et al., 2026). Text supplied alongside numbers can also change numerical predictions (Requeima et al., 2024), and models estimate a density from numbers in context like a kernel density estimator with an adaptive bandwidth (Liu et al., 2025).

Our numerical experiments state uncertainty explicitly and vary stated precision and reliability separately, with conditions in which the stated precision differs from the generating one while the prior mean and demonstrations stay fixed, so that predictions can be compared with a specified Bayesian reference. We vary only reasoning; code execution, whose effect Filbry et al. (2026) study, is disabled in every run. The message-sorting experiment extends this comparison to constraints stated in words on a finite family of classification rules.

Our settings express specificity in three forms that share no numerical scale. Chapter 3 measures it by what a relevant description leaves unknown about the task: the smallest expected loss of a prediction made from the description's prior alone, which Grünwald & Dawid (2004) call the generalized entropy of that prior under the loss. Under squared error this is the prior's variance, τ^2 in each direction of the isotropic Gaussian model; a partial description constrains only k of the d directions (Chapter 4).

In the message-sorting experiment of Chapter 7, a note leaves a uniform prior over the rules compatible with it, and we describe its specificity by their number, 20 for a broad note and 2 for a specific one; the logarithm of that number is the prior's entropy, the corresponding measure under log loss. ESS puts these forms on one scale, the number of demonstrations a description is worth. To show how a model uses a description, Chapters 6 and 7 also summarize predictions by *description weights*. A description weight is the weight that a prediction places on the description's prior mean; we either fit one across prompts or compute one for each response. In Gaussian mean estimation, the Bayesian reference weights the prior mean and the sample mean in proportion to their precisions (Neuenschwander et al., 2020). A normalized description weight can be compared with posterior component probabilities, but need not itself be a probability.

Chapter 3

Problem Setup

This chapter applies the tools of Chapter 2 to task descriptions: a description becomes a robust mixture prior over the regression task, and its value is measured by a predictive effective sample size.

A description carries information about a task that does not come from demonstrations, so we model it as a prior over tasks. Let Q_0 be the *baseline prior*, the distribution of tasks when no description is given. A description comes with a prior of its own, Q_{desc}. If the description is about the right task, which we call *relevant*, the task is drawn from Q_{desc}. Otherwise the description is *irrelevant*, and the task is drawn from Q_0 whatever the description says. Given a description, the task therefore has the prior

$$p\,Q_{\text{desc}} + (1-p)\,Q_0, \tag{3.1}$$

where p, the description's *reliability*, is the probability that it is relevant. We call the description *reliable* when $p = 1$ and *unreliable* when $p < 1$. This two-component form is the robust mixture prior that Bayesian clinical trials use to borrow external information that may not apply (Section 2.6; Schmidli et al., 2014).

A description's *specificity* is how little it leaves unknown about the task when it is relevant. What a prior leaves unknown can be measured by the smallest expected loss that a prediction made from the prior alone attains, its generalized entropy (Grünwald & Dawid, 2004). We score predictions by squared error, as is standard for in-context regression (Section 2.4; Garg et al., 2022). The best prediction from a prior is then the one made with its mean, and for the isotropic inputs used here its expected squared error is the prior's variance, summed over the directions of the task. We therefore

measure a description's specificity by the variance of Q_{desc}: the lower the variance, the more specific the description. We study this model in Gaussian linear regression.

What counts as a description. A *description* is information about the task that a prompt gives other than through demonstrations, such as a numeric token given to a Transformer (Chapter 6) or a probabilistic statement or a note about a sorting rule given in text to a language model (Chapter 7). Anything that a prompt contains whether or not it has a description, such as instructions on the answer format, is not part of the description.

Tasks and demonstrations. We use the noisy in-context linear regression setting of Section 2.3 (Garg et al., 2022; Raventós et al., 2023). A task is a vector $w \in \mathbb{R}^d$, and the baseline prior is $Q_0 = \mathcal{N}(0, \tau_0^2 I_d)$, where $\tau_0 > 0$. A prompt contains n demonstrations $D_n = \{(x_i, y_i)\}_{i=1}^n$ and a query input x, with

$$x_i, x \overset{\text{iid}}{\sim} \mathcal{N}(0, I_d), \qquad y_i = w^\top x_i + \epsilon_i, \quad \epsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, 1).$$

Inputs and observation noise are independent of the task, of its description, and of one another. Numerical illustrations in Chapters 4–5 use $d = 16$ and $\tau_0^2 = 10$; other dimensions and baseline variances give the same pattern (Table 5.3 in Chapter 5).

Descriptions. A description supplies a prior mean $m \in \mathbb{R}^d$, and its prior is $Q_{\text{desc}} = \mathcal{N}(m, \tau^2 I_d)$ with $0 < \tau \leq \tau_0$. Its variance is τ^2 in each direction, so the smaller τ, the more specific the description: a relevant description places each coordinate of the task within about τ of that of m. So that learners with and without descriptions face the same tasks, the task must follow Q_0 once we average over descriptions. Under a relevant description, w is m plus independent noise of variance τ^2, so m must have variance $\tau_0^2 - \tau^2$, the part of the task's variance that the description accounts for. With a latent relevance indicator $Z \sim \text{Bernoulli}(p)$ independent of m, the joint model is

$$\begin{aligned} m &\sim \mathcal{N}(0, (\tau_0^2 - \tau^2) I_d), \\ w \mid m, Z = 1 &\sim \mathcal{N}(m, \tau^2 I_d), \\ w \mid m, Z = 0 &\sim \mathcal{N}(0, \tau_0^2 I_d). \end{aligned}$$

The description is relevant when $Z = 1$ and irrelevant when $Z = 0$, which gives the prior in (3.1), and integrating over m recovers $w \sim Q_0$ under either component. A description of the wrong task carries no information about this one, so when $Z = 0$ its mean is independent of the task. A description of the right task that is only somewhat inaccurate is modeled instead as a relevant description with a larger τ.

Relevance identifies the generating component, irrespective of the realized distance between m and w. At $\tau = \tau_0$, $m = 0$ and the two components coincide, so the description provides no task information. The model separates uncertainty within the described region from uncertainty about whether that region applies. It does not assume that a variance and a reliability can summarize every natural-language description.

To compare a description with demonstrations, we use precision, the inverse of variance, because the precisions of independent Gaussian sources of information add (Equation (2.2)). With unit noise and isotropic inputs, each demonstration adds expected precision 1 in every direction, since $\mathbb{E}W_n = nI_d$. We therefore write $c = \tau^{-2}$ for a description's precision and $c_0 = \tau_0^{-2}$ for the baseline prior's: a description carries the precision that c demonstrations add. What it is worth in prediction error, its ESS, need not equal c, as Chapters 4 and 5 show. A description of precision c thus carries the expected information of c demonstrations, the ESS that Morita et al. (2008) assign to a normal prior of that precision (Section 2.5); relative to the baseline prior, it adds precision $c - c_0$.

Prediction error. As in Section 2.4, for a prediction $\hat{y}$, we measure expected squared error against the noise-free query response:

$$R = \mathbb{E}\big[(\hat{y} - w^\top x)^2\big]. \tag{3.2}$$

We call this quantity *prediction error*, or simply error; it excludes irreducible query noise. The expectation is over tasks, descriptions, demonstrations, and queries. For a fixed learner, $R^{\text{desc}}(n)$ and $R^{\text{base}}(n)$ denote the error with and without a description. As a function of the demonstration count, R^{base} is the learner's *baseline curve*, its learning curve without the description. To express a description's value in demonstrations, we compare these two error curves, as the loss-based ESS of Section 2.5 does for a prior.

Definition 3.1 (Predictive effective sample size)**.** Given n demonstrations, let n^* be the smallest $t \geq 0$ satisfying $R^{\text{base}}(t) = R^{\text{desc}}(n)$. When such a t exists, we call n^* the

crossing, and the description's ESS is

$$\mathrm{ESS}_n = n^* - n.$$

We interpolate the baseline curve between integer counts, so n^* and the ESS need not be integers.

For example, $\mathrm{ESS}_n = 10$ means that n demonstrations plus the description match the error of $n + 10$ demonstrations alone. The ESS without demonstrations, ESS_0, is the *zero-shot ESS*. A learner without a description still has prior knowledge, from the baseline prior or from training, and ESS measures what the description adds to it, so we measure ESS against each learner's own baseline curve. For a learner whose baseline curve falls slowly, the same error reduction is worth more demonstrations, so a large ESS need not imply low absolute error. A negative ESS means that the description raises error. For the Bayes-optimal predictor under the correctly specified model, conditioning on the description cannot increase expected error, so $R^{\mathrm{desc}}(n) \leq R^{\mathrm{base}}(n)$ and ESS is nonnegative. Because R averages over descriptions, the ESS is a property of a class of descriptions, those with the same τ and p, rather than of a particular value of m.

Definition 3.1 follows the prior sample size of Reimherr et al. (2021), who match the mean squared error of an estimate under a prior with that under a baseline prior and additional observations (Section 2.5). We match prediction error, which can be measured for any learner, including a network or a language model, whose prior knowledge is only implicit in its parameters.

Empirical ESS interpolates a learner's measured error curve. The interpolation is linear in the demonstration count for the Bayes-optimal predictor, the meta-trained Transformers, and the message-sorting experiment. For the numerical language-model tasks, whose evaluation grid doubles n, it is linear in the logarithm of the count. It uses the first crossing in the evaluated range. When no crossing is observed, we report censoring at the corresponding boundary under the conventions in Section 7.5.2; an exact ESS is then unavailable.

Chapter 4

The Value of a Reliable Description

We first take a reliable description, $p = 1$. Its value without demonstrations depends on its specificity and also on the task's dimension. In this chapter and the next, error and ESS are those of the Bayes-optimal predictor. Section 4.1 treats a description that constrains every direction of the task, and Section 4.2 a partial description, which constrains only some directions and can gain value as demonstrations accumulate.

4.1 Complete Descriptions

We state the result for a diffuse baseline prior, the limit $\tau_0 \to \infty$, in which a learner knows little about the task without the description. The limit can be taken in two ways: a description may remove a fixed share of the baseline uncertainty, or its own variance τ^2 may stay fixed. For the first, let $s = 1 - \tau^2/\tau_0^2 \in [0, 1)$ be the share of the baseline variance that the description removes.

Proposition 4.1 (Reliable descriptions)**.** *Let $p = 1$.*

(i) With the share s fixed, $\mathrm{ESS}_0 \geq sd$, and $\mathrm{ESS}_0 \to sd$ as $\tau_0 \to \infty$.

(ii) With τ fixed and $c = \tau^{-2}$ a positive integer, $\mathrm{ESS}_0 \to c + d + 1$ as $\tau_0 \to \infty$.

(iii) With τ and τ_0 fixed, $\mathrm{ESS}_n \to c - c_0$ as $n \to \infty$.

Part (i) fixes the share s of the baseline variance that the description removes, so τ grows in proportion to τ_0. A learner with $t < d$ demonstrations and no description knows nothing about $d - t$ directions of w, and as $\tau_0 \to \infty$ its error is dominated by what those directions contribute, $(d - t)\tau_0^2$. The description alone leaves an error of $d\tau^2 = (1 - s)d\tau_0^2$, the variance of its prior summed over the d directions, which this

learner matches at $t = sd$: a description that removes a share of the uncertainty is worth that share of d demonstrations.

Part (ii) fixes the description's variance instead. The result follows by comparing the description's error, $d\tau^2 = d/c$, with the limiting baseline error after t demonstrations, which is $d/(t - d - 1)$ for integer $t > d + 1$ (Section 2.4; Muirhead, 1982): the two match at $t = c + d + 1$. If the demonstration inputs gave every direction of the task the same information, the baseline error would be d/t and the description would be worth c demonstrations, the count that an information-matching ESS such as that of Morita et al. (2008) assigns to it (Section 2.5). Random inputs constrain some directions less than that, so the error is $d/(t - d - 1)$, and matching the description takes $d + 1$ more demonstrations. The baseline curve is interpolated linearly between integer counts, so the limit is exactly $c + d + 1$ only for integer c, and we state the result for that case.

At $d = 16$ and $\tau_0^2 = 10$ (Figure 4.1a), a reliable description with $\tau = 0.1$, so that $c = 100$, is worth about 117 demonstrations before any are seen, the value $c + d + 1$ that Proposition 4.1(ii) gives for a diffuse baseline prior. A description with $\tau = 1$, so that $c = 1$, is worth 15.6, against the lower bound $sd = 14.4$ of part (i).

Demonstrations lower the value of a complete description. The descriptions modeled so far are *complete*: they constrain every direction of w. By part (iii), which is the case $k = d$ of Proposition 4.2 in Section 4.2, after many demonstrations the dimension no longer adds to such a description's value, which tends to $c - c_0$. This limit is the precision that the description adds to the baseline prior, its conventional prior ESS (Morita et al., 2008) measured relative to that prior. A complete description therefore loses value as demonstrations accumulate: a $c = 100$ description is worth 117 demonstrations before any are seen and 109 after 100 (Figure 4.1b), and a $c = 1$ description falls from 15.6 to 1.1, close to its added precision of 0.9 (Table 4.1).

Demonstrations can raise the value of a partial description. Descriptions often specify only part of a task. A *partial* description constrains only $k < d$ of the d directions (Section 4.2). Without demonstrations, its error comes almost entirely from the $d - k$ directions it leaves open, and a learner without the description reaches that error after about k demonstrations. After many demonstrations, the description is worth the precision it adds, averaged over all d directions: about $k(c - c_0)/d$ demonstrations, which exceeds k when $c - c_0 > d$. At $d = 16$ and $\tau_0^2 = 10$, a $c = 100$

partial description with $k = 8$ is worth 8.1 demonstrations before any are seen and 36.2 after 100 (Figure 4.1b). Over the same range, the complete description's ESS falls from 117 to 109, so demonstrations substitute for a complete description and complement a partial one.

The rest of this section proves Proposition 4.1, starting from the properties of W_n that the proof needs.

Notation and error. As in Chapter 3, $c_0 = 1/\tau_0^2$ is the baseline prior precision and $c = 1/\tau^2$ is the description's. With $W_n = X_n^\top X_n$ as in Section 2.3, define, for a prior precision $\kappa > 0$,

$$R^\kappa(n) = \mathbb{E}\,\mathrm{tr}(\kappa I_d + W_n)^{-1}.$$

Thus $R^{\text{base}} = R^{c_0}$ and, for a reliable description, $R^{\text{desc}} = R^c$. Both identities hold because the Bayes-optimal prediction is the posterior mean, whose error is the expected trace of the posterior covariance (Equation (2.5)).

Throughout the proofs, d is a positive integer, $0 < \tau \le \tau_0$, so that $c \ge c_0$, $s = 1 - \tau^2/\tau_0^2 \in [0, 1)$, and $p \in [0, 1]$. Noninteger demonstration counts refer to linear interpolation between the errors at adjacent integers.

Wishart identities. We use three standard properties of W_n. The first two give the facts of Section 2.4.

(W1) If $n < d$, then W_n has rank n almost surely. Consequently, $(\kappa I_d + W_n)^{-1}$ has $d - n$ eigenvalues equal to $1/\kappa$.

(W2) If $n \ge d$ and e is a fixed unit vector, then $(e^\top W_n^{-1} e)^{-1} \sim \chi^2_{n-d+1}$ (Muirhead, 1982, Theorem 3.2.12). Hence

$$\mathbb{E}\,\mathrm{tr}\,W_n^{-1} = \frac{d}{n-d-1} \quad (n \ge d+2),$$

whereas this expectation is infinite for $n = d$ and $n = d+1$.

(W3) For fixed d, as $n \to \infty$,

$$\mathbb{E}\,\mathrm{tr}\,W_n^{-2} = \frac{d}{n^2} + O(n^{-3}), \qquad \mathbb{E}\,\mathrm{tr}\,W_n^{-3} = O(n^{-3})$$

(von Rosen, 1988).

The Sherman–Morrison identity implies that an additional demonstration strictly decreases the expected trace of the posterior covariance. Thus $R^{\kappa}(n)$ is strictly decreasing, and (W2) implies that it tends to zero. Its linear interpolation is therefore continuous and strictly decreasing. In the correctly specified joint model, conditioning on the description cannot increase expected posterior variance, so $R^{\mathrm{desc}}(n) \le R^{\mathrm{base}}(n)$ and the Bayes-optimal ESS is nonnegative.

Diffuse baseline prior. Several results use the baseline error measured in units of the baseline variance τ_0^2. For every integer n,

$$\frac{R^{c_0}(n)}{\tau_0^2} = \mathbb{E}\,\mathrm{tr}(I_d + \tau_0^2 W_n)^{-1} \longrightarrow (d-n)^+ \qquad \text{as } \tau_0 \to \infty.$$

Indeed, an eigenvalue $1/(1+\tau_0^2\lambda)$ is bounded by one and converges to the indicator of $\lambda = 0$, so dominated convergence and the almost-sure rank of W_n apply. The same limit holds at every real $t \ge 0$ by interpolation. Each finite-τ_0 curve is strictly decreasing, and monotone functions that converge pointwise to a continuous limit converge locally uniformly. Hence, if an error level $\ell(\tau_0)$ satisfies $\ell(\tau_0)/\tau_0^2 \to \lambda$ with $0 < \lambda < d$, the count at which the baseline error equals $\ell(\tau_0)$ converges to $d - \lambda$, the only point where $(d-t)^+ = \lambda$.

Proof of Proposition 4.1. A reliable description without demonstrations has error $R^{\mathrm{desc}}(0) = d\tau^2 = (1-s)d\tau_0^2$, the trace of its prior covariance.

(i) Fixed share. By (W1), $R^{c_0}(n) \ge (d-n)\tau_0^2$ for $n < d$; the inequality also holds trivially for $n \ge d$. Linear interpolation preserves this bound, so

$$R^{\mathrm{base}}(t) \ge (d-t)\tau_0^2 \qquad (t \ge 0).$$

For $t < sd$, the right-hand side exceeds $(1-s)d\tau_0^2$. The crossing n^* of Definition 3.1 therefore satisfies $n^* \ge sd$, so $\mathrm{ESS}_0 \ge sd$. In units of τ_0^2, the description error is $(1-s)d$. For $s \in (0,1)$, this level lies strictly between 0 and d, so by the diffuse-baseline limit the crossing converges to $d-(1-s)d = sd$. For $s = 0$, the description is uninformative and $\mathrm{ESS}_0 = 0$ for every τ_0.

(ii) Fixed specificity. For fixed τ, the description error $d\tau^2 = d/c$ does not depend on τ_0. As $\tau_0 \to \infty$, monotone convergence, (W1), and (W2) give

$$R^{c_0}(n) \longrightarrow \begin{cases} +\infty, & n \le d+1, \\ \dfrac{d}{n-d-1}, & n \ge d+2. \end{cases}$$

The finite part of this limiting curve equals d/c at the integer $n = c + d + 1$. For every $t < c + d + 1$ the limiting error exceeds d/c, and for every $t > c + d + 1$ it is smaller, so the crossing of the interpolated finite-τ_0 error converges to $c + d + 1$. □

4.2 Partial Descriptions

Descriptions often specify only part of a task. In our model, a reliable description may then constrain only k of the d directions of w; Proposition 4.2 gives its ESS, and the case $k = d$ is the complete description of Section 4.1. Formally, in the k constrained coordinates, $m_i \sim \mathcal{N}(0, \tau_0^2 - \tau^2)$ and $w_i \mid m \sim \mathcal{N}(m_i, \tau^2)$, as in Chapter 3; in the remaining $d - k$ coordinates, $m_i = 0$ and $w_i \sim \mathcal{N}(0, \tau_0^2)$. The prior precision given the description is $c = 1/\tau^2$ in the constrained directions and $c_0 = 1/\tau_0^2$ elsewhere. Write

$$\Lambda = \operatorname{diag}(cI_k, c_0 I_{d-k}).$$

The prediction error is $\mathbb{E}\,\operatorname{tr}(\Lambda + W_n)^{-1}$. Rotational invariance of W_n means that the result depends on the number of constrained directions, not on their orientation.

Proposition 4.2 (ESS of a partial description)**.** *Let $p = 1$ and let $0 \le k \le d$ be an integer.*

(i) If $k < d$, then $\mathrm{ESS}_0 \to k$ as $\tau_0 \to \infty$ with τ fixed.

(ii) For fixed d, τ_0, and τ, as $n \to \infty$,

$$\mathrm{ESS}_n \longrightarrow \frac{k}{d}(c - c_0) = \frac{k}{d}\, sc.$$

In particular, a complete description has $\mathrm{ESS}_n \to c - c_0$.

As the baseline prior becomes diffuse with τ fixed, the unconstrained directions dominate the zero-shot error, and ESS approaches the number of constrained directions, k. With even one unconstrained direction, the error with the description stays of order τ_0^2, which a learner without the description matches after about k demonstrations; only a complete description brings the error down to order τ^2, which takes $c + d + 1$ demonstrations to match. After many demonstrations, ESS instead approaches the added precision averaged across all d directions. When $c - c_0 > d$, this large-sample ESS, $k(c - c_0)/d$, exceeds the diffuse-prior zero-shot ESS k, so for a diffuse enough baseline prior the description is worth more after many demonstrations than before

Table 4.1: **Demonstrations can increase the value of a specific partial description.** Bayes-optimal ESS for a reliable description of k of $d = 16$ directions, with $\tau_0^2 = 10$ and precision $c = \tau^{-2}$. The case $k = 16$ is a complete description. Each block reports ESS after n demonstrations and its large-sample limit $(c - c_0)k/d$. Estimates use eight batches of 2,000 prompts; standard errors are at most 0.05.

	$c = 1$					$c = 100$				
k	$n = 0$	$n = 10$	$n = 100$	$n = 400$	$n \to \infty$	$n = 0$	$n = 10$	$n = 100$	$n = 400$	$n \to \infty$
1	0.9	0.8	0.1	0.1	0.1	1.0	1.0	3.5	5.2	6.2
4	3.6	2.9	0.3	0.2	0.2	4.0	4.2	15.5	21.7	25.0
8	7.3	5.1	0.5	0.5	0.5	8.1	9.8	36.2	46.0	50.0
12	11.1	6.7	0.8	0.7	0.7	12.4	23.4	65.1	73.0	74.9
16	15.6	8.0	1.1	0.9	0.9	117	116	109	103	99.9

any. Table 4.1 illustrates this: at $c = 100$, a description of half of the $d = 16$ directions has ESS 8.1 without demonstrations and 36.2 with 100 demonstrations. At $c = 1$, ESS falls as demonstrations accumulate, whether the description is partial or complete.

Proof of Proposition 4.2. (i) No demonstrations. The description error is $\operatorname{tr} \Lambda^{-1} = (d - k)\tau_0^2 + k\tau^2$. In units of τ_0^2, it is

$$d - k + \frac{k\tau^2}{\tau_0^2} \longrightarrow d - k \qquad \text{as } \tau_0 \to \infty.$$

If $0 < k < d$, this limit lies strictly between 0 and d, so by the diffuse-baseline limit of Section 4.1 the crossing converges to $d - (d - k) = k$. If $k = 0$, the description leaves the prior unchanged and its ESS is zero.

(ii) Many demonstrations. Applying the resolvent identity twice gives

$$(\Lambda + W_n)^{-1} = W_n^{-1} - W_n^{-1}\Lambda W_n^{-1} + W_n^{-1}\Lambda W_n^{-1}\Lambda(\Lambda + W_n)^{-1}.$$

Rotational invariance implies

$$\mathbb{E} W_n^{-2} = \frac{\mathbb{E} \operatorname{tr} W_n^{-2}}{d} I_d.$$

The expected trace of the middle term is therefore $\operatorname{tr}(\Lambda)/n^2 + O(n^{-3})$ by (W3). To bound the remainder, use $|\operatorname{tr}(MB)| \le \|M\|_{\mathrm{op}} \operatorname{tr} B$ for $B \succeq 0$, together with $(\Lambda + W_n)^{-1} \preceq W_n^{-1}$. Its absolute trace is at most

$$\|\Lambda\|_{\mathrm{op}}^2 \|W_n^{-1}\|_{\mathrm{op}}^2 \operatorname{tr} W_n^{-1} \le d\|\Lambda\|_{\mathrm{op}}^2 \operatorname{tr} W_n^{-3},$$

whose expectation is $O(n^{-3})$. Consequently,

$$\mathbb{E}\,\mathrm{tr}(\Lambda + W_n)^{-1} = \frac{d}{n-d-1} - \frac{\mathrm{tr}\,\Lambda}{n^2} + O(n^{-3}).$$

The baseline error is the case $\Lambda = c_0 I_d$; if $k = 0$ or $c = c_0$, the two errors coincide and $\mathrm{ESS}_n = 0$. Otherwise, fix an integer $T > k(c - c_0)/d$. Since $d/(n + t - d - 1) = d/(n - d - 1) - dt/n^2 + O(n^{-3})$, uniformly over integers $0 \le t \le T$,

$$R^{\mathrm{base}}(n + t) - \mathbb{E}\,\mathrm{tr}(\Lambda + W_n)^{-1} = \frac{d}{n^2}\left[\frac{\mathrm{tr}\,\Lambda - dc_0}{d} - t + O(n^{-1})\right].$$

Linear interpolation gives the same expansion for real $t \in [0, T]$. The baseline error is strictly decreasing, so for large n the left-hand side vanishes at exactly one t in this range, which is ESS_n. Hence

$$\mathrm{ESS}_n = \frac{\mathrm{tr}\,\Lambda - dc_0}{d} + O(n^{-1}) \longrightarrow \frac{k(c - c_0)}{d}.$$

One more demonstration and one more unit of prior precision in every direction both lower the error by d/n^2 to leading order, which is why a description's large-sample ESS is the precision it adds, averaged across directions. □

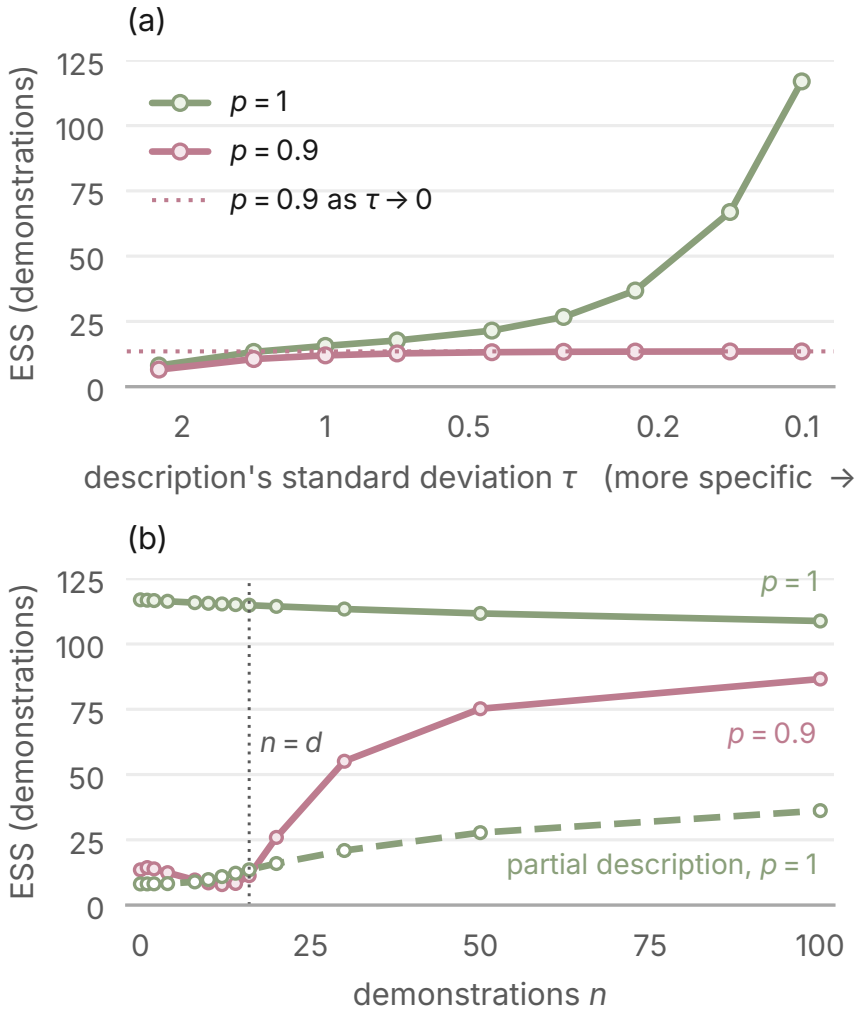


Figure 4.1: **Without demonstrations, reliability limits the benefit of specificity; demonstrations recover much of it.** Bayes-optimal ESS with $d = 16$ and $\tau_0^2 = 10$. (a) ESS_0 versus the description's standard deviation τ, for $p = 1$ and $p = 0.9$; a smaller τ means a more specific description. The dotted line is the finite ceiling as $\tau \to 0$ at $p = 0.9$, about 13.5 here; it tends to $p^2 d = 12.96$ as $\tau_0 \to \infty$. At $\tau = 0.1$, that is, $c = 100$, ESS is 117 for $p = 1$ and 13.5 for $p = 0.9$. (b) ESS_n versus the number of demonstrations n for three descriptions with $\tau = 0.1$: reliable, unreliable ($p = 0.9$), and a reliable partial description, which constrains $k = 8$ of the 16 directions. Estimates use eight batches of 4,000 prompts (2,000 for the partial description); standard errors are below 0.05 demonstrations.

Chapter 5

The Cost of Unreliability

We now allow the description to be irrelevant with probability $1-p$. The Bayes-optimal predictor must then weigh the two components by each one's probability given the demonstrations. Let $\pi_n = P(Z = 1 \mid m, D_n)$ be the *posterior relevance probability*. The predictor averages the two components' predictions with weights π_n and $1 - \pi_n$, which is Bayesian model averaging (Hoeting et al., 1999).

5.1 Unreliable Descriptions

Proposition 5.1 (Unreliable descriptions)**.** *Let* $0 < p < 1$.

(i) Without demonstrations, the error is

$$R^{\text{desc}}(0) = \left[(1 - p^2)\,\tau_0^2 + p^2\,\tau^2\right] d. \tag{5.1}$$

(ii) With τ *fixed,* $\text{ESS}_0 \to p^2 d$ *as* $\tau_0 \to \infty$.

(iii) With τ *and* τ_0 *fixed,* $\limsup_{n\to\infty} \text{ESS}_n \leq p\,(c - c_0)$.

Without demonstrations, unreliability imposes an error floor. The error in Equation (5.1) is the variance of the mixture prior (3.1), averaged over descriptions. It is a floor, $(1 - p^2)\tau_0^2 d$, plus p^2 times the error $d\tau^2$ of a reliable description. However specific the description, the error stays above the floor, so the zero-shot ESS stays below a finite ceiling: the number of demonstrations at which the baseline error equals the floor. For a diffuse baseline prior the ceiling is $p^2 d$, the limit in part (ii), which does not depend on τ. The floor arises because, without demonstrations, nothing shows whether the description is relevant: the posterior relevance probability is still

the prior one, $\pi_0 = p$, and the Bayes-optimal predictor shrinks the prior mean m by the factor p. A predictor that knew whether the description was relevant would have a ceiling that tends to pd in place of $p^2 d$; the further factor of p is the cost of not knowing (proved below). Figure 4.1a shows the effect at $p = 0.9$, the weight that the robust-prior example of Schmidli et al. (2014) gives its informative component, our relevant component: lowering τ from 1 to 0.1, which raises c from 1 to 100, multiplies a reliable description's ESS by more than seven, but raises an unreliable one's only from 12.0 to 13.5, already within 0.1 of the ceiling.

Demonstrations distinguish low specificity from unreliability. By Proposition 5.1, an unreliable description has, without demonstrations, the error of a less specific reliable one. For example, at $d = 16$ and $\tau_0^2 = 10$, a $c = 100$ description with $p = 0.9$ and a reliable one with $c \approx 0.52$ both have ESS about 13.5. After 100 demonstrations, their ESS values are 86.6 and 0.5. Once demonstrations show that the unreliable description is relevant, its high specificity takes effect; the reliable description has low specificity to begin with, so demonstrations cannot raise its value in this way. The unreliable description's ESS does not rise steadily: after a small rise at $n = 1$, it falls to 7.8 at $n = 12$, and it climbs steeply only as n approaches d, to 25.9 at $n = 20$ (Figure 4.1b). With fewer demonstrations than d, its error is dominated by the cases in which it is irrelevant. The baseline curve falls steeply while directions of the task remain undetermined, so it matches that error within a few demonstrations. The rise also levels off: in the limit of many demonstrations, an unreliable description is worth at most $p(c - c_0)$, which is p times the limit $c - c_0$ of a reliable one (Proposition 5.1(iii)). Table 5.1 collects the limits of this chapter and the last.

The rest of this section proves Proposition 5.1.

Component posteriors. The posterior relevance probability is $\pi_n = P(Z = 1 \mid m, D_n)$, whereas p remains the prior relevance probability. Conditional on $Z = 1$, the posterior parameters are

$$\Sigma_1 = (cI_d + W_n)^{-1}, \qquad \mu_1 = \Sigma_1(cm + X_n^\top y_{1:n}).$$

Conditional on $Z = 0$, the description's mean m is independent of w, and

$$\Sigma_0 = (c_0 I_d + W_n)^{-1}, \qquad \mu_0 = \Sigma_0 X_n^\top y_{1:n}.$$

Table 5.1: **The Bayes-optimal value of a description, in demonstrations.** Limits of the ESS of a description with precision $c = \tau^{-2}$: without demonstrations, for a diffuse baseline prior ($\tau_0 \to \infty$ with τ fixed), and after many demonstrations ($n \to \infty$ with τ_0 fixed, $c_0 = \tau_0^{-2}$). The unreliable description's entry after many demonstrations is an upper bound (Propositions 4.1–4.2).

	Demonstrations	
Description	none	many
Reliable	$c + d + 1$	$c - c_0$
Unreliable, $p < 1$	$p^2 d$	$\leq p\,(c - c_0)$
Partial, k of d directions	k	$k\,(c - c_0)/d$

Because m has the same distribution under both relevance states, its density cancels from the posterior odds, and the posterior relevance probability follows from the two Gaussian marginal likelihoods of the demonstrations,

$$\begin{aligned} L_1 &= \mathcal{N}\big(y_{1:n}; X_n m, I_n + \tau^2 X_n X_n^\top\big)\,, \\ L_0 &= \mathcal{N}\big(y_{1:n}; 0, I_n + \tau_0^2 X_n X_n^\top\big)\,, \\ \pi_n &= \frac{pL_1}{pL_1 + (1-p)L_0}. \end{aligned}$$

The query input does not enter this update because it is independent of (Z, m, D_n). Without demonstrations, both marginal likelihoods equal one, so $\pi_0 = p$. The Bayes-optimal prediction is the posterior mean, $\big(\pi_n \mu_1 + (1 - \pi_n)\mu_0\big)^\top x$ (Hoeting et al., 1999).

Proof of Proposition 5.1. (i) No demonstrations. At $n = 0$, $\mu_1 = m$, $\mu_0 = 0$, and $\pi_0 = p$, so the prediction is $p\, m^\top x$. For each coordinate,

$$\mathrm{Var}(w_i) = \tau_0^2, \qquad \mathrm{Var}(m_i) = \tau_0^2 - \tau^2 = s\tau_0^2, \qquad \mathrm{Cov}(w_i, m_i) = ps\tau_0^2.$$

The covariance follows because, conditional on relevance, w_i is m_i plus independent noise, and conditional on irrelevance, w_i is independent of m_i. It follows that

$$\mathbb{E}(w_i - pm_i)^2 = \tau_0^2 - 2p^2 s\tau_0^2 + p^2 s\tau_0^2 = (1 - p^2 s)\tau_0^2 = (1 - p^2)\tau_0^2 + p^2\tau^2.$$

Since $\mathbb{E}[xx^\top] = I_d$, the prediction error is $\mathbb{E}\|w - pm\|^2$. Summing over coordinates proves Equation (5.1). Because $p\, m$ is the mean of the mixture prior (3.1) given m, this error is the trace of that prior's covariance, averaged over m.

(ii) The ceiling and its limit. Since $R^{\text{base}}(0) = d\tau_0^2$ and $s < 1$, Equation (5.1) gives, for every $\tau > 0$,

$$R^{\text{desc}}(0) = (1 - p^2 s) R^{\text{base}}(0) > (1 - p^2) R^{\text{base}}(0).$$

The error on the left decreases to the right-hand side as $\tau \to 0$, equivalently $s \to 1$. The interpolated baseline error is continuous, strictly decreasing, and tends to zero. Let $n_p^\sharp$ be the count at which the baseline error equals $(1 - p^2)\tau_0^2 d$, the ceiling of the proposition; it is unique and finite. The crossing n^* is strictly below $n_p^\sharp$ and converges to it as $\tau \to 0$. Finally, the error level that defines $n_p^\sharp$ is $(1 - p^2)d$ in units of τ_0^2, which lies strictly between 0 and d. By the diffuse-baseline limit of Section 4.1, $n_p^\sharp \to d - (1 - p^2)d = p^2 d$ as $\tau_0 \to \infty$. With τ fixed instead of s, $s \to 1$ as $\tau_0 \to \infty$, and the same limit gives $\text{ESS}_0 \to p^2 d$.

For comparison, a predictor that also observes Z predicts $m^\top x$ when $Z = 1$ and 0 when $Z = 0$, with zero-shot error

$$p\, d\tau^2 + (1 - p)\, d\tau_0^2 = (1 - ps)\, d\tau_0^2.$$

As $\tau \to 0$ this error tends to $(1 - p)d\tau_0^2$, so by the same limit its zero-shot ESS ceiling approaches pd as $\tau_0 \to \infty$, against $p^2 d$ for the Bayes-optimal predictor, which does not observe Z. The difference, $p(1-p)d$, is the cost of not knowing whether the description is relevant. In error, the two predictors differ by $p(1-p)s\, d\tau_0^2$, the between-component term of the law of total variance (Equation (2.6)) at $n = 0$.

(iii) Many demonstrations. The predictor that observes Z also bounds the ESS after many demonstrations. Given Z, it is the Bayes-optimal predictor for the corresponding component, so its error with n demonstrations is $pR^c(n) + (1 - p)R^{c_0}(n)$. It predicts with the posterior mean given (Z, m, D_n), and the Bayes-optimal predictor with the posterior mean given (m, D_n) alone. Conditioning on less cannot lower the expected squared error, so

$$R^{\text{desc}}(n) \geq pR^c(n) + (1 - p)R^{c_0}(n).$$

The expansion in the proof of Proposition 4.2(ii), applied with $\Lambda = cI_d$ and with $\Lambda = c_0 I_d$, gives for every fixed real $t \geq 0$

$$R^{\text{base}}(n + t) - \left[pR^c(n) + (1 - p)R^{c_0}(n)\right] = \frac{d}{n^2}\Big[p\,(c - c_0) - t + O(n^{-1})\Big].$$

Fix $t > p(c - c_0)$. For all large n the right-hand side is negative, so $R^{\text{base}}(n + t) < R^{\text{desc}}(n)$. The crossing n^* exists, because $R^{\text{desc}}(n) \leq R^{\text{base}}(n)$ and the interpolated

Table 5.2: **The bound of Proposition 5.1 after many demonstrations is nearly attained.** The bound $p(c-c_0)$ on an unreliable description's ESS in the limit $n \to \infty$, and a Monte Carlo estimate of how far the ESS stays below it, at $\tau_0^2 = 10$. The gap is estimated from the difference in error between the Bayes-optimal predictor and the predictor that observes Z, with two million draws.

d	c	p	bound $p(c-c_0)$	gap
16	100	0.9	89.91	< 0.0001
16	1	0.9	0.81	< 0.0001
5	10	0.9	8.91	0.002
5	1	0.9	0.81	0.010
5	1	0.5	0.45	0.023
1	36	0.9	32.31	1.30
1	1	0.9	0.81	0.105
1	1	0.5	0.45	0.199

baseline error is continuous and tends to zero (Section 4.1); since that error is strictly decreasing, $n^* < n + t$, that is, $\mathrm{ESS}_n < t$. Since $t > p(c-c_0)$ is arbitrary, $\limsup_{n\to\infty} \mathrm{ESS}_n \le p(c-c_0)$: even a predictor that knew whether the description was relevant would gain the added precision $c - c_0$ only with probability p. □

Part (i) also relates the two propositions. Equation (5.1) equals $(1 - p^2 s)\, d\tau_0^2$, the zero-shot error of a reliable description that removes the share $p^2 s$ of the baseline variance. The proof of Proposition 4.1(i) uses the description only through this error, so it applies with $p^2 s$ in place of s: $\mathrm{ESS}_0 \ge p^2 s d$, and $\mathrm{ESS}_0 \to p^2 s d$ as $\tau_0 \to \infty$ with s fixed.

Simulations indicate that the bound is nearly attained in the settings we study. The ESS can stay below it in the limit because the Bayes-optimal predictor remains unsure whether the description is relevant, which the predictor that observes Z is not. Table 5.2 gives Monte Carlo estimates of this gap (`results/theory-check/vn_limit.py`): it is below 10^{-4} demonstrations at $d = 16$ and grows as the dimension falls. The bound concerns the limit, not each finite n: at $d = 16$, $\tau_0^2 = 10$, $c = 100$, and $p = 0.9$, where the bound is 89.91, the ESS is 86.6 at $n = 100$, 90.6 at $n = 400$, and 90.1 at $n = 3{,}000$, so it approaches the limit from above.

Other dimensions and baseline variances. The numerical illustrations use $d = 16$ and $\tau_0^2 = 10$. Table 5.3 repeats the comparison of a reliable and an unreliable

Table 5.3: **The cost of unreliability at other dimensions and baseline variances.** Bayes-optimal zero-shot ESS of a $c = 100$ description at $p = 1$ and $p = 0.9$, with the limits of Propositions 4.1 and 5.1. The baseline curve is estimated from 4,000 sets of demonstration inputs per count (1,500 at $d = 50$).

d	τ_0^2	ESS_0, $p = 1$	$c + d + 1$	ESS_0, $p = 0.9$	$p^2 d$
5	1	105.0	106	8.7	4.05
5	10	105.8	106	4.7	4.05
5	100	105.9	106	4.2	4.05
16	1	115.8	117	17.3	12.96
16	10	116.9	117	13.5	12.96
16	100	117.0	117	13.0	12.96
50	1	149.5	151	44.4	40.50
50	10	150.9	151	40.9	40.50
50	100	151.0	151	40.5	40.50

$c = 100$ description at other values. In every case the reliable description's zero-shot ESS is within 1.6 of $c + d + 1$. A one-in-ten chance of irrelevance cuts it to between 4.2 and 44.4, and it approaches the diffuse-prior limit $p^2 d$ of Proposition 5.1 as τ_0^2 grows.

Chapter 6

Meta-Trained Transformers

Transformers meta-trained on demonstrations can approach Bayes-optimal prediction (Garg et al., 2022; Genewein et al., 2025). We first test whether they also attain the Bayes-optimal ESS of descriptions under the correctly specified model. Following Garg et al. (2022), we train them from scratch, with $\tau_0^2 = 10$ as before. The dimension is $d = 5$ instead of 16: this is the dimension used by Huang & Ge (2025), and it keeps training short. Prompts contain an optional description token carrying m, up to 20 demonstration tokens (40 at $d = 10$), and a query token (Huang & Ge, 2025). The description token does not state τ or p; the network must learn these properties across training tasks. Half the prompts omit the description; a network's predictions on such prompts give its own baseline curve. We train separate networks for $c \in \{1, 10\}$ and $p \in \{1, 0.99, 0.9\}$, with two or three seeds per configuration. We choose values a factor of ten apart, so that a few networks span a wide range: the two precisions, and the two nonzero probabilities of irrelevance, $1 - p = 0.01$ and $1 - p = 0.1$. We do not train networks at $c = 100$, the precision used in Chapters 4 and 5: at $d = 5$, its Bayes-optimal ESS, about 106, lies far beyond the 20 demonstrations over which a network's baseline curve is measured. We train further networks with $c \in \{0.2, 2\}$, with $d = 10$, and with a partial description. Section 6.1 gives the training and evaluation details. Section 6.2 compares the networks with the Bayes-optimal predictor when training and test reliability match, and Section 6.3 tests them when the two differ.

6.1 Training and Evaluation

Architecture and optimization. The Transformer has 12 layers, 8 attention heads, and width 256, matching the model size of Garg et al. (2022). Layers use pre-normalization, GELU activations, feed-forward width 1,024, and no dropout. At $d = 5$, the model has 9.5 million parameters. Unlike the model of Garg et al. (2022), ours uses bidirectional attention and no positional encoding; two indicator coordinates distinguish token types. Each demonstration token contains an input and its label. When a description is absent, its token is masked out of attention. The prediction is read from the query token.

Following Zhu et al. (2026), the model predicts a mean and a log variance for the query label and minimizes log loss. The optimal predicted mean is then the posterior mean, which also minimizes squared error (Section 2.4), so we evaluate the predicted mean by the prediction error of Equation (3.2).

We train with Adam, the optimizer of Garg et al. (2022), at learning rate 10^{-4} for 40,000 steps, using batches of 1,024 newly sampled prompts and no curriculum. Adam uses the PyTorch defaults $\beta_1 = 0.9$ and $\beta_2 = 0.999$ and stability constant 10^{-8}, with no weight decay and a constant learning rate. Labels and description means are divided by τ_0, gradients are clipped to norm 1, and training uses bf16 autocast. The demonstration count is sampled uniformly from 0 to 20, except for the $c = 0.2$ and $c = 2$ experiments, which use 0 to 15, and the $d = 10$ experiments, which use 0 to 40. We train three seeds for $c = 10$, for $d = 10$, and for the partial description; two for $c = 1$; and one each for $c = 0.2$ and $c = 2$. A $d = 5$ network takes approximately 80 minutes to train on one NVIDIA L4 GPU, and a $d = 10$ network takes approximately 2.8 hours. Training all 25 networks takes approximately 38 GPU-hours.

Evaluation. We evaluate each network on 20,000 fresh prompts. The networks share these evaluation prompts, so standard deviations across training seeds exclude prompt-sampling variation. Each network with a complete description and $c \in \{1, 10\}$ is tested at its training reliability and at $p = 1$ and $p = 0.9$; all but one seed of each configuration are also tested at $p = 0.99$. At $d = 5$, we evaluate the baseline curve at every integer from 0 to 20, except for $c = 0.2$ and $c = 2$, which use counts from 0 to 15. We compute ESS at the first crossing of this curve with the prediction error obtained using the description, as in Definition 3.1. Figures 6.1a and 6.2 compare each

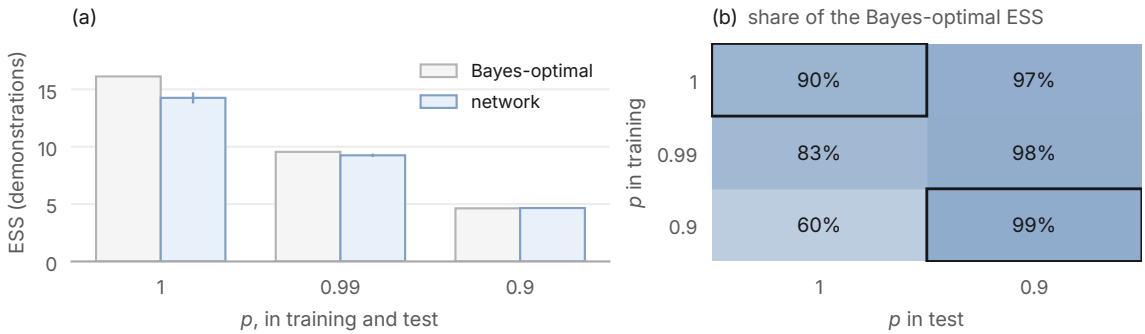


Figure 6.1: **Meta-trained Transformers approach Bayes-optimal ESS and weight descriptions by how often descriptions were relevant in training.** Results use $c = 10$, $d = 5$, $\tau_0^2 = 10$, and no demonstrations. (a) Training and test reliability match. Network ESS uses each network's own baseline curve; bars show the mean and one standard deviation across three training seeds. The Bayes-optimal comparison uses the same prompts. (b) Networks trained at each reliability, evaluated at each test reliability. Cells show network ESS as a fraction of the Bayes-optimal ESS under the test distribution, estimated on 200,000 prompts; outlined cells have matching reliabilities. Table 6.1 includes $c = 1$.

network with the Bayes-optimal predictor on the same prompts. Table 6.1 reports results at $n = 0$, including test reliabilities that differ from the training reliability, with Bayes-optimal values estimated on 200,000 prompts.

6.2 Matched Training and Test Reliability

Because ESS is measured against each network's own baseline curve, we first check that the networks learn the task from demonstrations alone. Without a description, every $d = 5$ network's error is within 0.21 of the Bayes-optimal error of approximately 50 at $n = 0$, and within 0.32 on average over its training demonstration counts. At $d = 10$, where the Bayes-optimal error at $n = 0$ is approximately 100, the gap is at most 0.50 at $n = 0$ and at most 0.41 on average. With a description at $c = 10$, $p = 1$, and $n = 0$, network error ranges from 0.67 to 0.71, compared with 0.49 for the Bayes-optimal predictor. The networks therefore remove 98.6% of the baseline error, compared with 99.0% for the Bayes-optimal predictor.

ESS expresses this remaining gap in units of demonstrations. With no demonstra-

tions and $c = 10$, network ESS is 14.3 ± 0.5 at $p = 1$, 9.3 ± 0.2 at $p = 0.99$, and 4.66 ± 0.01 at $p = 0.9$, compared with Bayes-optimal values of 16.1, 9.6, and 4.63 on the same prompts (Figure 6.1a). A network's ESS can exceed the Bayes-optimal ESS on the same prompts, as it does at $p = 0.9$, because the network's ESS is measured against its own baseline curve, not the Bayes-optimal curve. On 200,000 prompts, the Bayes-optimal value at $p = 1$ is 15.9, close to the limiting expression $c + d + 1 = 16$ of Proposition 4.1. At $p = 0.99$, the Bayes-optimal value is 8.96 on 200,000 prompts, below the networks' mean ESS of 9.3, and 9.6 on the networks' evaluation prompts, above that mean. For $c = 1$, network ESS is within 0.1 of the Bayes-optimal ESS at every reliability (Table 6.1); at $p = 1$, for example, it is 5.85 ± 0.03, compared with 5.89.

To check that this agreement does not depend on the two precisions above, we train four further networks, with $c = 0.2$ and $c = 2$, each trained and tested at $p = 1$ and at $p = 0.9$. At $n = 0$, each comes within 0.03 of the Bayes-optimal ESS on the same prompts. At $c = 0.2$, the network and Bayes-optimal values are 2.64 and 2.66 for $p = 1$, and 2.16 and 2.15 for $p = 0.9$. At $c = 2$, they are 7.32 and 7.35 for $p = 1$, and 4.38 and 4.37 for $p = 0.9$.

To test whether the networks also show the $d + 1$ term of Proposition 4.1, we train three seeds at $d = 10$, with $c = 10$ and $p = 1$, and extend training and evaluation to 40 demonstrations so that the baseline curve reaches the ESS crossing. Network ESS is 17.8, with standard deviation 0.3 across seeds, compared with a Bayes-optimal value of 20.8 and the limiting expression $c + d + 1 = 21$. As at $d = 5$ with $c = 10$ and $p = 1$, the networks capture most of the description's predictive value but fall short of the Bayes-optimal ESS: by 3.0 demonstrations here, compared with 1.8 at $d = 5$.

Figure 6.2 follows ESS as demonstrations accumulate. For $c = 1$, network ESS remains within 0.6 of the Bayes-optimal value at every demonstration count whose crossing lies within the evaluation grid. For $c = 10$, the largest shortfalls are 2.4, 2.9, and 1.4 demonstrations at $p = 1$, 0.99, and 0.9, respectively; the largest excess is 0.3. Demonstrations can raise the ESS of an unreliable description (Chapter 5): for the three $c = 10$ networks trained at $p = 0.9$, ESS rises from 4.66 at $n = 0$ to between 4.77 and 4.83 at $n = 5$, and the Bayes-optimal ESS on the same prompts rises from 4.63 to 5.20. The shortfall is consistent across seeds: at $n = 5$, all three networks are below the Bayes-optimal value.

Networks also learn that a partial description and demonstrations complement

Table 6.1: **Transformer ESS and fitted description weights under matched and shifted reliability.** Results use $d = 5$, $\tau_0^2 = 10$, and no demonstrations, with 20,000 test prompts per network. Rows are grouped by training reliability. *Network* reports mean ESS and standard deviation across the indicated training seeds; for a single seed, the uncertainty is the standard error over prompts. The two Bayesian predictors use p_{train} and p_{test}, respectively; the latter is Bayes-optimal under the test distribution. We show one reference value when they coincide. At $n = 0$, a network's prediction depends only on m and x, whose distribution does not change with the test reliability, so its fitted weight is the same at every p_{test}; the $p_{\text{test}} = 0.99$ rows for training reliabilities 1 and 0.9 differ only because they average one seed fewer. The values using p_{test}, and those using p_{train} where the two reliabilities coincide, use 200,000 prompts, with standard errors at most 0.1. The other values using p_{train} use the networks' 20,000 test prompts and can differ from a large-sample value by up to 0.6, which is why some exceed the Bayes-optimal value beside them. *Weight* is the fitted description weight q in $\hat{y} \approx q\, m^\top x$ (Section 6.3), averaged across the available seeds. This coefficient is unrestricted and need not be a relevance probability.

				ESS			
c	p_{train}	p_{test}	Seeds	Network	Bayes p_{train}	Bayes p_{test}	Weight q
1	1	1	2	5.85 ± 0.03	5.89	—	0.964
1	1	0.99	1	5.65 ± 0.06	5.66	5.61	0.991
1	1	0.9	2	4.02 ± 0.03	3.98	4.10	0.964
10	1	1	3	14.26 ± 0.48	15.91	—	1.008
10	1	0.99	2	9.38 ± 0.16	9.54	8.96	1.015
10	1	0.9	3	4.58 ± 0.03	4.54	4.70	1.008
1	0.99	1	2	5.87 ± 0.01	5.88	5.89	0.963
1	0.99	0.99	2	5.65 ± 0.01	5.61	—	0.963
1	0.99	0.9	2	4.02 ± 0.03	3.99	4.10	0.963
10	0.99	1	3	13.28 ± 0.57	16.01	15.91	0.982
10	0.99	0.99	3	9.25 ± 0.15	8.96	—	0.982
10	0.99	0.9	3	4.60 ± 0.04	4.56	4.70	0.982
1	0.9	1	2	5.67 ± 0.06	5.75	5.89	0.883
1	0.9	0.99	1	5.51 ± 0.06	5.55	5.61	0.901
1	0.9	0.9	2	4.03 ± 0.01	4.10	—	0.883
10	0.9	1	3	9.49 ± 0.44	10.79	15.91	0.876
10	0.9	0.99	2	8.08 ± 0.28	8.55	8.96	0.880
10	0.9	0.9	3	4.66 ± 0.01	4.70	—	0.876

each other. The partial-description networks receive a reliable partial description (Section 4.2) that constrains $k = 3$ of the $d = 5$ coordinates with precision $c = 10$.

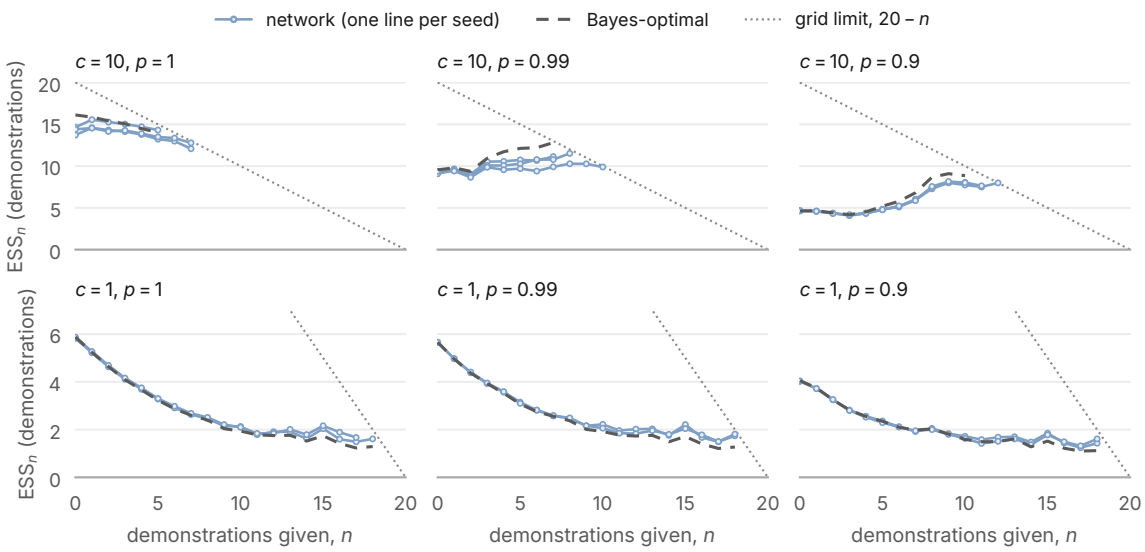


Figure 6.2: **Transformer ESS as a function of demonstration count.** Each line represents a training seed; the Bayes-optimal predictor is evaluated on the same prompts. Training and test reliability are matched. Results cover $c = 10$ and $c = 1$ at $d = 5$, using 20,000 prompts. A curve ends when its crossing exceeds the maximum of 20 demonstrations. The dotted line, $20 - n$, marks the largest ESS identifiable on this grid.

The description token contains m in the first three coordinates and zeros elsewhere, and the other two task coordinates follow the baseline prior. Their mean ESS rises from 3.27 at $n = 0$ to 5.55 at $n = 10$, compared with Bayes-optimal values 3.22 and 5.64; the standard deviation across the three seeds at $n = 0$ is 0.03. This rise agrees with Proposition 4.2, whose large-sample limit here is $k(c - c_0)/d = 5.94$.

6.3 Reliability Shifts

Each network learns how much to weight a description from how often descriptions were relevant in its training data, but descriptions at test time may be more or less reliable than that. We therefore evaluate each network at test reliabilities that differ from its training reliability, and compare it with two Bayesian predictors (Raventós et al., 2023): one that uses the training reliability p_{train}, and one that uses the test reliability p_{test} and is therefore Bayes-optimal. At $n = 0$, a Bayesian predictor with

reliability p predicts $p\,m^\top x$. We therefore summarize a network's predictions by the *fitted description weight* q, the coefficient from regressing them on $m^\top x$. This coefficient describes prediction behavior and need not represent a probability inside the network.

Without demonstrations. For $c = 10$, the networks trained at $p = 0.9$ have fitted description weights of 0.87 to 0.89 across the three seeds, and keep them even when every test description is relevant (Figure 6.1b and Table 6.1). Their ESS on reliable descriptions is then 9.5 ± 0.4. The Bayesian predictor that uses p_{train} reaches 10.8 on the same prompts, and the Bayes-optimal predictor under the test distribution reaches 15.9 on 200,000 prompts. This is consistent with the information available: m alone does not reveal the changed test reliability. The networks trained at $p = 1$ or $p = 0.99$ have fitted weights between 0.95 and 1.02, so the two training reliabilities cannot be distinguished without demonstrations. At test reliability $p = 0.9$, assuming relevance costs the networks trained at $p = 1$ little at $n = 0$: their ESS is 4.58 ± 0.03, compared with 4.70 for the Bayes-optimal predictor under the test distribution on 200,000 prompts. The two shifts differ because the Bayes-optimal error differs in size. When one description in ten is irrelevant, that error is already large, and a weight of 1 instead of 0.9 adds only about 5% to it. When every description is relevant, the error is small, and a weight of 0.9 instead of 1 roughly doubles it.

After demonstrations are observed. With demonstrations, the networks trained at $p = 1$ and at $p = 0.99$ respond differently to a test reliability of $p = 0.9$, under which 10% of test descriptions are irrelevant. For the networks trained at $p = 0.99$, the description reduces error at every evaluated n, as it does for the Bayes-optimal predictor under the test distribution. Across seeds, the networks' reduction is 3.5 to 3.7 with six demonstrations and 0.08 to 0.13 with 20. For the networks trained at $p = 1$, the description increases error from six demonstrations onward, in every seed: by 0.77 to 1.06 at $n = 6$ and by 2.74 to 2.94 at $n = 9$, where the excess peaks. With eight demonstrations, for example, one of these networks has error 4.6 with the description and 2.3 without it. A Bayesian predictor that assumes relevance ($p = 1$) also first has higher error than the baseline predictor at $n = 6$, with excess error 1.12 there and 2.96 at $n = 9$; this predictor returns the relevant component's prediction, $\mu_1^\top x$, whatever the demonstrations show. The networks trained at $p = 1$ thus behave

as if every description were relevant.

Chapter 7

Pretrained Language Models

Pretrained language models are not trained on our task distribution, so they cannot learn a description's precision and reliability as the networks do. We therefore state both in the prompt, and ask whether the models use what it states. To isolate the use of a description from the ability to carry out linear regression in context, we study numerical mean estimation, which is the scalar case of our regression model with fixed input $x = 1$. A message-sorting experiment, in which a note describes a sorting rule in words, follows in Section 7.4. The scalar task tests the use of stated precision and relevance probabilities without the $d + 1$ term of Proposition 4.1, which arises from random inputs.

7.1 Task and Evaluation

Each numerical task has a hidden mean $w \sim \mathcal{N}(500, 100^2)$. Its demonstrations are drawn from $\mathcal{N}(w, \sigma^2)$, with $\sigma = 60$, and rounded to integers; the prompt states σ. Language models predict short unsigned numbers best (Gruver et al., 2023; Requeima et al., 2024), so we keep every number to three digits: a task is redrawn if any of its numbers falls outside 100 to 999. A description gives a prior mean m and a standard deviation τ: "Its mean was drawn from a normal distribution with mean 510 and standard deviation 30." Before rounding and redrawing, the relevant component satisfies $w \mid m \sim \mathcal{N}(m, \tau^2)$. With $x = 1$, each demonstration is a direct observation of the task and adds precision σ^{-2}, so, as in Chapter 3, the description carries the precision of $c = \sigma^2/\tau^2$ demonstrations. Stated standard deviations 60, 30, and 10

correspond to $c = 1$, 4, and 36. Over the demonstration counts we evaluate, $n = 1$, 2, and 4, the Bayesian weight on the description, $c/(c+n)$, ranges from 0.2 to 0.97. With $\sigma = 60$, these precisions correspond to standard deviations that suit three-digit numbers: the smallest, 10, is still large relative to rounding, and the largest, 60, is small relative to the range 100 to 999.

We generate descriptions at $c = 1$, 4, and 36, and the prompt states the generating precision, except in the one condition below. Because a more precise description also has a more accurate prior mean, a model could do well with $c = 36$ descriptions without using the stated precision. To separate the two, we also give the $c = 36$ descriptions with a stated precision of $c = 4$, keeping each task's prior mean and demonstrations. We also add a sentence, the *warning*, that states a 10% or 50% chance that the statement is wrong, the probability $1 - p$ that the description is irrelevant. Section 7.3 uses two designs for it. In the *mixed design*, descriptions are irrelevant with the stated probability, and the warning says what holds then: how the mean is distributed. In the *open-alternative design*, the warning says only that the mean is then "unknown," and every description remains relevant. We also evaluate irrelevant descriptions without a warning (Section 7.5.4).

Table 7.1 lists the evaluated configurations. Using a stated precision requires weighting the description and the demonstrations by their precisions. Reasoning may help a model carry out this computation, so we select models that can answer with reasoning enabled and with it disabled under the same model identifier; both runs then use the same model. They include open-weight and proprietary models from several developers. The main comparison uses the same 200 tasks for five such models: gpt-oss-20b (OpenAI, 2025), Qwen3.5 9B (Qwen Team, 2026), Gemma 4 31B (Gemma Team, 2026), GPT-6 Sol (OpenAI, 2026), and Claude Opus 4.6 (Anthropic, 2026b). Two further models are evaluated only with reasoning. Claude Haiku 5.5 (Anthropic, 2026a) could not have its reasoning disabled in our setup; we run it at medium effort and, in a second run, at low effort. Olmo 3 7B Think (Team Olmo, 2025) is a small open-weight model that always reasons. To measure behavior without reasoning more precisely and on more models, the 2,000-*task run* evaluates six open-weight models without reasoning: gpt-oss-20b, Qwen3.5 9B, Gemma 4 31B, Phi-4 (14B; Abdin et al., 2024), OLMo 2 32B Instruct (Team OLMo et al., 2025), and Gemma 4 12B. It uses 2,000 tasks, the first 200 of which are the tasks of the main comparison. "Reasoning enabled" denotes the configured reasoning mode; individual

Table 7.1: **Evaluated model configurations.** "2,000 tasks" denotes the 2,000-task run without reasoning, and "200 tasks" the first 200 of its tasks; the open-weight results without reasoning on these 200 tasks are taken from the 2,000-task run. Effort is the provider's reasoning-effort setting; "on" marks reasoning without such a setting. GPT-6 Sol at low effort and Haiku 5.5 at medium effort also run the mixed design (Section 7.3) and the message-sorting experiment, and GPT-6 Sol runs both without reasoning as well; Opus 4.6 also runs the mixed design in both modes; Haiku 5.5 also runs the wording test with reasoning (Section 7.5.7). Section 7.5.5 gives the serving and decoding settings.

Model	Without reasoning	With reasoning	Run through
gpt-oss-20b	2,000 tasks	200 tasks, low	Transformers; vLLM
Qwen3.5 9B	2,000 tasks	200 tasks, on	Transformers; OpenRouter
Gemma 4 31B	2,000 tasks	200 tasks, on	Transformers; Gemini API
Phi-4	2,000 tasks	–	Transformers
OLMo 2 32B Instruct	2,000 tasks	–	Transformers
Gemma 4 12B	2,000 tasks	–	Transformers
Olmo 3 7B Think	–	200 tasks, on	vLLM
GPT-6 Sol	200 tasks	200 tasks, low	Codex command line
Claude Opus 4.6	200 tasks, high	200 tasks, high	Claude Code command line
Claude Haiku 5.5	–	200 tasks, medium and low	Claude Code command line

replies may still contain no recorded reasoning output. Paired runs use the same model identifier; for the open-weight models, they also differ in serving software and answer extraction (Section 7.5.5). GPT-6 Sol and the two Claude models are served through their developers' command-line agents, with code execution disabled and no user instructions. Section 7.5 gives the evaluation details and the further results.

Bayesian reference. Our *Bayesian reference* uses the observation noise, description uncertainty, and chance of irrelevance stated in the prompt, even when they differ from the generating values. Agreement with it therefore measures how a model uses the prompt's statistical information; it need not imply optimal prediction under the evaluation distribution. Without a description, the prompt says only that the mean is "unknown": a stated distribution for it would itself be a description, and the baseline should reflect only what a model already knows (Chapter 3). A warning in the open-alternative design likewise says only that the mean is then "unknown." In both cases, the reference therefore needs a baseline prior of its own. It takes a uniform one over the integers 100 to 999, the range of every number shown in a prompt, so its prediction without a description is approximately the sample mean of the demonstrations. When the prompt asserts relevance, its prediction is approximately the posterior mean of the conjugate normal model of Section 2.5 (Neuenschwander et al., 2020),

$$\hat{w}_{\text{ref}} = \frac{c\,m + n\bar{y}}{c + n}, \tag{7.1}$$

where $\bar{y}$ is the sample mean; rounding and boundary effects on this expression are negligible. The description is equivalent to c additional demonstrations at m. In this idealized continuous model with a flat baseline prior, the squared error is therefore $\sigma^2/(c+n)$ with the description and σ^2/n without it, so a correctly specified description has exact ESS c. When the prompt states a chance of irrelevance, the reference takes it as the prior probability of irrelevance and updates that probability from the demonstrations (Section 7.3).

We compute ESS at $n = 1$, 2, and 4 from the error averaged over tasks, against each model's own baseline curve, which we evaluate from $n = 1$ to 64 and interpolate in $\log n$ (Section 7.5.2). We report the *median ESS* of these three values. We exclude $n = 0$ because models rarely answer with a number when given neither a description nor demonstrations, and $n > 4$ because there ESS mainly reflects how slowly a model's own baseline error falls rather than how it uses the description. Applied to

the reference, this estimator departs from the idealized value because of finite-task sampling and interpolation: on the first 200 tasks, the reference's median ESS is 1.0, 3.5, and 34.6 at $c = 1$, 4, and 36; on the 2,000-task run it gives the idealized value for the $c = 36$ description; and its values under a stated chance of irrelevance in the open-alternative design vary more between task sets (Section 7.5.2). Each model is therefore compared with the reference on the same tasks. We also regress predictions on m, $\bar{y}$, and an intercept. The coefficient on m is the *fitted description weight*; the reference weight is $c/(c+n)$ when relevance is asserted.

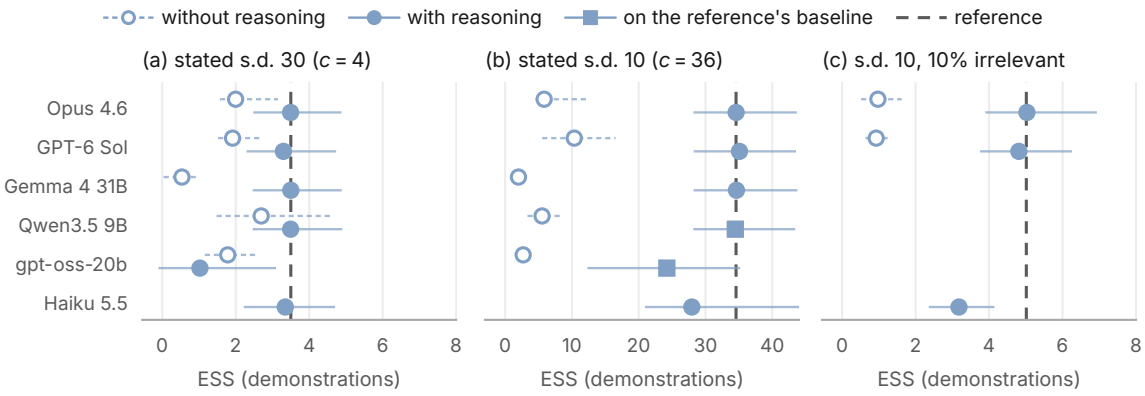


Figure 7.1: **Language models use the precision and the reliability that a prompt states mainly with reasoning enabled.** Median ESS over $n = 1$, 2, and 4, from error averaged over 200 tasks at $\sigma = 60$; 95% bootstrap intervals resample tasks. Open points indicate reasoning disabled; filled points indicate reasoning enabled; vertical dashed lines show the Bayesian reference. (a) Descriptions generated and stated with standard deviation 30, so $c = 4$. (b) With standard deviation 10, so $c = 36$. (c) GPT-6 Sol, Opus 4.6, and Haiku 5.5 on the $c = 36$ descriptions of (b), when the prompt states a 10% chance that the description is irrelevant; each task is presented with a relevant description and with an irrelevant one. Squares in (b) measure ESS on the reference's baseline curve, not the model's own, and so are not comparable with the other points. Intervals are clipped to the display.

7.2 Reasoning and Stated Precision

Without reasoning, models make too little use of precise descriptions. At $c = 36$, every model without reasoning falls far short of the reference: in the main comparison, median ESS ranges from 2.0 to 10.3, against 34.6 for the reference (Figure 7.1b). Counted in demonstrations, the shortfall is much smaller at $c = 4$ and $c = 1$, where the reference's own ESS is small: at $c = 4$, the models reach 0.5 to 2.7, against 3.5 for the reference (Figure 7.1a), and at $c = 1$ they reach -0.2 to 0.9, against 1.0 (Table 7.6).

Without reasoning, the models differ in how much they respond to the stated precision itself. When the prompt states $c = 4$ for descriptions generated at $c = 36$, the reference's ESS falls from 34.6 to 13.7. It stays above its value for descriptions generated at $c = 4$, because the prior mean is as accurate as before. GPT-6 Sol's ESS falls by a larger fraction than the reference's, from 10.3 to 2.4, and that of Opus 4.6 and Gemma 4 31B by about half, somewhat less than the reference's. The ESS of Qwen3.5 9B and gpt-oss-20b falls by a quarter or less, against three fifths for the reference, so these two models respond little to it (Table 7.6). Fitted weights fall short of the reference as well. With four demonstrations, the reference weights are 0.50 at stated $c = 4$ and 0.90 at stated $c = 36$; GPT-6 Sol's are 0.19 and 0.47, and Opus 4.6's are 0.20 and 0.40.

The 2,000-task run shows the same pattern for six open-weight models (Section 7.5.3): no model's median ESS for the $c = 36$ description exceeds 7.6, and the weights of four of the six barely change with c. The six models' ESS rises with c mainly because a more precise description's prior mean lies closer to the task. Rewording the description changes how much weight these models give it, but no tested wording yields the reference's sensitivity to precision.

Reasoning improves the use of stated precision. With reasoning enabled, four of the six models in Figure 7.1 come close to the reference's absolute error at $c = 4$ and $c = 36$: Opus 4.6, Gemma 4 31B, and Qwen3.5 9B are within 0.1 of it at every demonstration count, and GPT-6 Sol within 0.3 at all but one (Table 7.7). Opus 4.6, GPT-6 Sol, and Gemma 4 31B also attain approximately the reference ESS at both precisions (Figure 7.1a,b), and their ESS falls as the reference's does when only the stated precision is lowered (Table 7.6). Haiku 5.5 at medium effort matches the

reference's median ESS at $c = 4$; at $c = 36$ its median ESS is 28, below the reference's 34.6 though its interval includes it. At $n = 1$ its ESS is only 3.4, and it skips reasoning in more than a quarter of its replies; at $n = 2$ its ESS matches the reference, and at $n = 4$ it is 28 against 33. Even so, most individual replies of Opus 4.6, GPT-6 Sol, Gemma 4 31B, and Haiku 5.5 follow the reference's rule. A reply implies a description weight, $(\hat{w} - \bar{y})/(m - \bar{y})$, which is well determined on tasks with $|m - \bar{y}| \geq 30$; on those tasks, at $n = 4$ and $c \in \{1, 36\}$, between 89% and 100% of each model's replies have a weight within 0.05 of the reference's (Section 7.5.6). Reasoning alone is not enough, however: Olmo 3 7B Think, a small model that always reasons and is not shown in Figure 7.1, averages demonstrations well but reaches a median ESS of only 6.2 at $c = 36$ (Table 7.6); Section 7.5.6 examines how it combines the description with the demonstrations.

With reasoning, the fitted description weights of Qwen3.5 9B and gpt-oss-20b also come close to the reference's: at $c = 36$ and $n = 4$, they are 0.90 and 0.89, against 0.90. Without a description, however, their error stops falling at around 32 demonstrations, above the error they attain with a $c = 36$ description at most demonstration counts. Their median ESS on their own baseline curves is therefore not defined (> 60 in Table 7.6), and the squares in Figure 7.1b use the reference's baseline curve instead: they report $n^*_{\text{ref}} - n$, where n^*_{ref} is the demonstration count at which that curve reaches the model's error. This comparison uses another predictor's baseline, so it is not the model's own ESS. The remaining points use each model's own baseline. gpt-oss-20b's ESS stays below the reference's, partly because about 2% of its replies contain arithmetic errors (Section 7.5.5).

The gains from reasoning also depend on how the description states its uncertainty. The informal wordings "give or take" and "±" do not define a prior standard deviation, and under them Haiku 5.5 often repeats m even with reasoning (Section 7.5.7).

7.3 Stated Reliability

Chapter 5 shows that unreliability sharply lowers a description's value, and that demonstrations can then show whether the description is relevant. To test whether language models discount an unreliable description and use the demonstrations to judge its relevance, we make the $c = 36$ description unreliable and say so. The prompt adds: "There is a 10% chance that this statement is wrong, in which case the mean

was instead drawn uniformly at random from the whole numbers 100 to 999." The stated chance is the true one, whereas the stated alternative is the reference's baseline prior, not the distribution that generates the tasks. Each task is presented twice: once with its relevant description, and once with an irrelevant one, whose mean is drawn independently of the task from the same distribution as the means of relevant descriptions. We average the two errors with weights 0.9 and 0.1; this is the mixed design. For cost reasons, we run it on the three models that our subscriptions cover, GPT-6 Sol, Opus 4.6, and Haiku 5.5. We also run it with a 50% chance.

For the reference, a one-in-ten chance of irrelevance reduces ESS from 34.6 to 5.0 (Figure 7.1c). Most of the remaining value comes from using the demonstrations to judge whether the description is relevant. Consider *fixed-weight averaging*, a predictor that keeps the chance of irrelevance at the stated 10%, whatever the demonstrations show. Its ESS is only 0.3, because it cannot discount a description that the demonstrations contradict. With reasoning, GPT-6 Sol reaches 4.8, and its replies are within 2 of the reference's prediction in 89% of cases with a relevant description and 92% with an irrelevant one. Opus 4.6 reaches 5.0, as the reference does, and 100% of its replies are within 2 of the reference's prediction with a relevant description and 96% with an irrelevant one. Haiku 5.5 reaches 3.2; 85% of its replies are within 2 of the reference with a relevant description and 60% with an irrelevant one. Without reasoning, GPT-6 Sol and Opus 4.6 gain little from the description: their ESS is 0.9 and 1.0, and at most 19% of their replies are within 2 of the reference. At a 50% chance, the reference's ESS is 0.9, and a predictor that keeps the chance fixed or ignores the warning does worse with the description than without it; with reasoning, GPT-6 Sol is again within 0.2 of the reference and Opus 4.6 within 0.1 (Table 7.2). Thus, whereas the networks of Chapter 6 can learn a description's reliability only from their training data, GPT-6 Sol and Opus 4.6 with reasoning take the chance of irrelevance from the prompt and, like the reference, revise it in light of the demonstrations; Haiku 5.5 does so less consistently.

The prompt says how the mean is distributed when the description is irrelevant, because the posterior relevance probability is not determined without it. The open-alternative design shows what models do when it does not: the warning said only that the mean was then "unknown," and every description remained relevant. There, a stated 50% chance of irrelevance reduces reference ESS from 34.6 to 5.7 under the uniform baseline prior. With reasoning, the ESS of five of the six models in

Table 7.2: **Unreliable descriptions with the stated chance of irrelevance.** Median ESS in the mixed design, with 95% bootstrap intervals over tasks, and the share of replies within 2 of the reference's prediction when the description is relevant (rel.) and irrelevant (irr.), pooled over $n = 1$, 2, and 4. With stated chance $1 - p$, the error with the description is p times the mean squared error with the relevant description plus $1-p$ times that with the irrelevant one. A value of ≤ -1.0 is censored: at $n = 2$ the description does worse than one demonstration alone.

	10% irrelevant			50% irrelevant		
	ESS	rel.	irr.	ESS	rel.	irr.
Reference	5.0 [3.9, 6.2]			0.9 [0.5, 1.2]		
Fixed weights	0.3 [−0.2, 1.0]			−0.7 [−1.0, −0.4]		
Ignores the warning	−0.1 [−0.6, 0.6]			≤ -1.0		
GPT-6 Sol, low effort	4.8 [3.8, 6.3]	89%	92%	0.8 [0.5, 1.2]	90%	92%
Opus 4.6, high effort	5.0 [3.9, 6.9]	100%	96%	0.9 [0.5, 1.3]	98%	96%
Haiku 5.5, medium effort	3.2 [2.4, 4.1]	85%	60%	0.6 [0.3, 1.0]	58%	52%
GPT-6 Sol, without reasoning	0.9 [0.6, 1.2]	19%	12%	−0.1 [−0.2, 0.1]	10%	12%
Opus 4.6, without reasoning	1.0 [0.5, 1.6]	15%	11%	0.0 [0.0, 0.0]	14%	13%

Figure 7.1 falls with the reference's, to between 3.8 and 4.2. The sixth, gpt-oss-20b, largely ignores the stated chance, weighting the description almost as if there were no warning. Because every description in these runs is relevant, its ESS of 8.8 exceeds the reference's 5.7. Without reasoning, the ESS of Opus 4.6, gpt-oss-20b, Qwen3.5 9B, and Gemma 4 31B falls by about a third to three fifths, against 84% for the reference, and GPT-6 Sol's falls from 10.3 to 0.9. In the 2,000-task run, no model's ESS falls in proportion to the stated probability (Section 7.5.3).

The five models' ESS is below the reference's because the reference also updates the stated chance on the demonstrations: demonstrations close to the stated prior mean make relevance more probable, and distant ones less. The models apply the stated chance as given. They closely match fixed-weight averaging. Under it, each component's prediction conditions on the demonstrations, while the components keep their prior weights, p and $1 - p$. In the scalar approximation, the resulting weight on m is $pc/(c + n)$ (Table 7.8). Fixed-weight averaging has ESS 4.4, and 80 to 100% of the replies of Opus 4.6, GPT-6 Sol, Gemma 4 31B, and Qwen3.5 9B are within 2 of its prediction (Section 7.5.6). The gap to the reference widens with demonstrations.

At $n = 4$, the reference's ESS reaches 16.5; fixed-weight averaging reaches 8.6, and the five models reach between 3.8 and 8.6.

Updating the chance is Bayesian model averaging, which uses posterior component probabilities. These depend on both component marginal likelihoods and hence on the baseline prior under irrelevance. The open-alternative design's warning leaves the baseline prior unspecified, saying only that the mean is otherwise "unknown," so the prompt does not determine a unique posterior relevance probability: a model must supply the missing prior from its pretrained knowledge, and the reference uses its uniform baseline prior. The relevant-description half of the mixed design supplies the comparison. There, GPT-6 Sol and Haiku 5.5, in their main reasoning configurations, answer the same tasks with that prior stated explicitly. Opus 4.6, at high effort with reasoning and without it, answers the same prompts, which supply the relevant half of its mixed-design results.

With the baseline prior stated and a 50% probability of irrelevance, both models update the chance as the reference does. On tasks where the reference and fixed-weight predictions differ by at least 5, the share of replies closer to the reference rises from 0–5% to 92–97% for GPT-6 Sol and from 21–41% to 87–95% for Haiku 5.5, where each range is over $n = 1$, 2, and 4. The share of all replies within 2 of the reference prediction rises from 10% to 90% for GPT-6 Sol and from 23% to 58% for Haiku 5.5. GPT-6 Sol's ESS rises from 4.2 to 5.4. Haiku 5.5's rises less, from 3.8 to 4.2, because a minority of its replies lie far from the reference (Section 7.5.6).

Replies could move closer to the reference merely by giving the description a larger constant weight, because the reference's posterior relevance probability exceeds the stated 50% on most tasks. To test whether the weights instead vary with the evidence, we divide each reply's description weight by $c/(c + n)$, which gives its *normalized description weight*. Under the scalar approximation in Equation (7.1), the reference's normalized weight equals its posterior relevance probability, π_n, whereas fixed-weight averaging keeps it constant at p. Regressing the models' normalized weights on π_n gives slopes 0.96 [0.84, 1.04] and 0.98 [0.82, 1.15] for GPT-6 Sol and Haiku 5.5, respectively, compared with 0.01 [−0.09, 0.11] and 0.22 [−0.01, 0.45] under the open-alternative design's warning (Section 7.5.6).

At a stated 10% chance of irrelevance, the two weightings rarely differ by much, so this comparison is uninformative. In that condition, stating the prior lowers Haiku 5.5's ESS from 32.6 to 15.1, against 27.5 for the reference, because a minority of its replies

lie far from the reference prediction (Section 7.5.6).

Apart from the mixed design, the experiments above keep descriptions relevant and vary only what the prompt states. The 2,000-task run and the main comparison also include irrelevant descriptions that the prompt presents as relevant (Section 7.5.4; Table 7.6). In the 2,000-task run, the models give an irrelevant description less weight than a relevant one, and less than the reference, which treats it as relevant.

7.4 Descriptions in Words: Sorting Messages

The numerical descriptions state a distribution over the task, whereas descriptions in practice usually state constraints in words that the model must interpret. We therefore design a message-sorting experiment with such descriptions that still admits an exact Bayes-optimal predictor. As in Bayesian models of rule-based concept learning (Goodman et al., 2008), the hidden task is one of a finite set of rules, here 60 sorting rules under a uniform prior. Each message has six yes-or-no properties, such as whether it requests a payment, and a hidden rule sorts it into one of two folders according to two of these properties. The prompt shows messages already sorted and a note describing the rule, and asks for the probability that each new message goes into A; we score it by squared error. A broad note, such as "whether it concerns travel is one of the two properties the policy uses," leaves 20 compatible rules, and a specific note, which names both properties and the value one must take, leaves 2 (Section 7.5.8); a note leaves the rules compatible with it equally likely, so it is more specific the fewer rules it leaves. Under log loss on the rule, the logarithm of that number is the generalized entropy by which Chapter 3 measures specificity; under the squared error used here, a broad note leaves the error without demonstrations unchanged. We evaluate two of the models of the mixed design, GPT-6 Sol and Haiku 5.5.

With reasoning, Haiku 5.5 and GPT-6 Sol obtain ESS 9.9 and 9.5 for the specific note at $n = 2$, compared with 11.4 for the reference. For unreliable notes, the prompt adds that the note has a 50% chance of being wrong; the generator then draws the note independently of the task with probability 0.5, and the prompt says that a wrong note says nothing about the rule. We compare Bayesian predictions with fixed-weight averaging, which in this experiment keeps the stated weight on the note unless the sorted messages rule the note out. Among queries where these predictions differ by

at least 0.05, 77–99% of the two models' predictions with reasoning are closer to the Bayesian reference. GPT-6 Sol and Haiku 5.5 with reasoning therefore also revise a stated chance in light of the demonstrations when a description states a rule.

7.5 Additional Results and Evaluation Details

This section gives the evaluation details and the full results that Sections 7.2–7.4 summarize. Sections 7.5.1 and 7.5.2 give the prompts, answer extraction, and ESS estimation, and Sections 7.5.3 and 7.5.4 report the 2,000-task run; unless stated otherwise, these four sections use the 2,000-task run, which Appendix A reports at every evaluated demonstration count. Sections 7.5.5–7.5.7 cover the 200-task runs with and without reasoning, and Section 7.5.8 the message-sorting experiment, which uses a separate set of 200 tasks.

7.5.1 Prompts, Tasks, and Answer Extraction

Prompt. The following text is supplied as the user turn in the model's chat template:

```
The numbers below are drawn from a normal distribution with standard deviation 60.
Its mean was drawn from a normal distribution with mean 510 and standard deviation 30.

548
431

Predict the next number.  Reply with the number only.
```

Without a description, the second sentence is "Its mean is unknown." To state unreliability, a third sentence is added. In the mixed design of Section 7.3, it reads "There is a 10% chance that this statement is wrong, in which case the mean was instead drawn uniformly at random from the whole numbers 100 to 999." In the open-alternative design, its final clause is "in which case the mean is unknown." With no demonstrations, the final question is "Predict the first number."

Task generation. A relevant description's prior mean is sampled conditional on the task so that the joint distribution satisfies $m \sim \mathcal{N}(500, 100^2 - \tau^2)$ and $w \mid m \sim \mathcal{N}(m, \tau^2)$. A task is redrawn if a displayed number or prior mean falls outside 100 to 999; this occurs for less than 1% of tasks. The 2,000-task run also includes $\sigma = 30$ conditions. Both standard deviations stated in the prompt are then halved, preserving $c \in \{1, 4, 36\}$. The qualitative pattern is unchanged (Table A.1).

Execution. The 2,000-task run uses Hugging Face Transformers at each model's native precision, with checkpoints `openai/gpt-oss-20b`, `microsoft/phi-4`, `allenai/OLMo-2-0325-32B-Instruct`, `Qwen/Qwen3.5-9B`, `google/gemma-4-12B-it`, and `google/gemma-4-31B-it`. Tasks are generated with a fixed random seed, so every model sees the same tasks. Reasoning is disabled where the chat template supports this option. gpt-oss-20b writes its reasoning in a separate channel before its answer; we start generation directly in the answer channel.

Distribution and written readouts. For gpt-oss-20b, Phi-4, and OLMo 2 32B without reasoning, the prediction is the mean of the first-token distribution, renormalized over the 900 three-digit number tokens. We call this the *distribution readout*. It is possible for these three models because their tokenizers encode each number from 100 to 999 as a single token; the other models' tokenizers split numbers into digits. Other models, and all reasoning runs, use the number in the final answer, called the *written readout*. We also evaluate written readouts for the three distribution-readout models as a robustness check (Table 7.3 for gpt-oss-20b, and Table 7.5). OLMo 2 32B assigns 41% of first-token probability to numbers without a description at $n = 1$, and 75% at $n = 2$. Its distribution readout at $n = 1$ therefore summarizes a minority of the first-token probability mass.

Numeric response rates. For the distribution readout, the numeric response rate is the probability mass assigned to number tokens at the first position. For the written readout, it is the fraction of replies containing a three-digit number. Tables 7.5, A.1, A.2, and A.3 report this quantity as "number". Tables 7.5 and A.3 italicize cells whose rate is below three quarters.

Model inclusion. ESS requires a baseline curve built from numeric answers, so we retain a model only if it meets two conditions. It must produce a number in at least three quarters of its replies overall. It must also do so in at least half of its replies without a description, averaged over $n = 1$, 2, and 4 and both noise levels, because these replies determine the baseline curve used for median ESS. This criterion excludes MiniCPM5 2B (MiniCPM Team, 2026), OLMo 2 7B Instruct, and OLMo 2 13B Instruct, whose replies begin with a number in 62%, 20%, and 10% of cases, respectively. These models often begin with explanations such as "To predict…" or

Table 7.3: **Description ESS and weights in the 2,000-task run without reasoning.** Left: median ESS over $n = 1$, 2, and 4, for relevant descriptions with $c = 1$, 4, and 36, and for the $c = 36$ description when irrelevant or with a stated 10% or 50% chance of irrelevance ("10% wrong" and "50% wrong", after the prompt's wording). The stated-chance columns come from the open-alternative design, in which every description is relevant and the prompt says only that the mean is otherwise unknown; they are not the mixed design of Section 7.3. Brackets give 95% bootstrap intervals. The marker $\leq$ shows that a median is only an upper bound, because at one of its three counts a single demonstration already gives lower error than the description (Section 7.5.2); such a median is rounded up so that the printed bound holds, and so can be 0.1 above the same value as rounded in Tables A.1 and A.2. Right: the fitted description weight on the prior mean at $n = 1$; the Bayesian reference weight is $c/(c + n)$. All results use $\sigma = 60$. "Written" denotes the written readout. Reference values are computed on the same 2,000 tasks and differ from those on the first 200 tasks (Section 7.5.2).

	ESS, median over $n = 1$, 2, 4						weight, $n = 1$		
	relevant			$c = 36$			relevant		
Model	$c = 1$	$c = 4$	$c = 36$	irrelevant	10% wrong	50% wrong	$c = 1$	$c = 4$	$c = 36$
Reference	1.3	4.4	36	≤ -1.0	22	5.8	0.50	0.80	0.97
gpt-oss-20b	0.9 [0.7, 1.0]	2.0 [1.7, 2.3]	2.8 [2.6, 3.1]	≤ -0.4	2.4 [1.9, 2.6]	2.3 [1.9, 2.7]	0.53	0.48	0.49
Phi-4	0.7 [0.4, 0.9]	4.1 [3.5, 4.6]	7.6 [6.5, 8.6]	≤ -1.0	8.7 [7.8, 9.8]	9.7 [8.7, 11]	0.82	0.76	0.74
OLMo 2 32B	1.9 [1.7, 2.1]	3.4 [3.2, 3.9]	5.3 [4.5, 6.3]	≤ -0.6	6.0 [5.6, 7.1]	5.4 [5.0, 6.0]	0.72	0.70	0.70
gpt-oss-20b (written)	0.2 [0.1, 0.4]	2.0 [1.9, 2.6]	3.7 [3.0, 4.4]	≤ -1.0	2.4 [2.1, 2.6]	2.3 [2.1, 2.6]	0.64	0.52	0.52
Qwen3.5 9B (written)	0.2 [−0.1, 0.4]	2.7 [2.1, 3.2]	4.0 [3.5, 4.5]	≤ -1.0	0.6 [0.3, 1.0]	1.0 [0.6, 1.5]	0.81	0.76	0.80
Gemma 4 12B (written)	0.0 [−0.1, 0.1]	0.5 [0.3, 0.7]	1.8 [1.4, 2.2]	≤ -0.7	1.3 [1.1, 1.6]	2.7 [2.5, 3.1]	0.34	0.24	0.59
Gemma 4 31B (written)	0.0 [0.0, 0.0]	0.2 [0.1, 0.3]	1.4 [1.2, 1.6]	≤ -0.7	0.6 [0.5, 0.7]	0.5 [0.4, 0.6]	0.48	0.49	0.72

"Given…" despite the numeric-only instruction. For example, OLMo 2 7B's median ESS for the $c = 36$ description is 21, but this value comes from a readout that covers only a minority of the model's first-token probability mass.

Written-readout implementation and checks. The written readout extracts the number in the last `\boxed{}` expression in the final answer, if present. Otherwise, it takes the last number with three digits before an optional decimal part, retaining the decimals. A digit group after a thousands separator, as in 14,945, does not count as a qualifying number. In the 2,000-task run, decoding is greedy with a limit of eight new tokens, so the reply is the number itself; for these replies, the readout takes the first qualifying number and keeps only its integer part. Replies without a qualifying number are excluded from both error curves and weight regressions; the reported response rates identify the extent of this exclusion.

For gpt-oss-20b, written-readout weights differ from distribution-readout weights by at most 0.21 across all conditions and demonstration counts. For $c = 1$, 4, and 36, written-readout median ESS is 0.2, 2.0, and 3.7, compared with 0.9, 2.0, and 2.8 from the distribution readout. A single greedy response is a noisier summary than the distribution mean. Phi-4 produces a number in only 58% of written replies across all conditions, so Table 7.3 reports only its distribution readout. In the wording experiment of Section 7.5.4, which uses written replies, Phi-4's numeric response rate nevertheless exceeds three quarters for the original wording, so Table 7.5 includes its written replies. OLMo 2 32B produces a number in only 5%, 36%, and 48% of written replies without a description at $n = 1$, 2, and 4, respectively, with $\sigma = 60$; we therefore do not report its written-readout ESS. Its written-readout weights for the three relevant descriptions at $n = 1$ are 0.83, 0.81, and 0.83, compared with distribution-readout weights of 0.72, 0.70, and 0.70.

7.5.2 ESS Estimation and the Bayesian Reference

Measurement. We apply Definition 3.1 to squared error against the hidden mean. In the 2,000-task run, the baseline curve is evaluated at $n = 1, 2, 4, \ldots, 64$. Because this grid doubles n, we interpolate squared error, rather than its square root, linearly in $\log n$, which is less biased here than interpolation in n (see *Sensitivity to interpolation* below). The ESS at n is the crossing n^*, the first interpolated demonstration count

whose baseline error matches the error with the description, minus n. Figure 7.2 shows the curves of the 2,000-task run.

If the error with the description lies below the entire baseline curve, which ends at $t_{\max} = 64$, we report $\mathrm{ESS}_n > t_{\max} - n$. Because the curve starts at one demonstration, we use the marker $\mathrm{ESS}_n \leq 1 - n$ when one demonstration alone already gives lower error. These markers record censoring at the evaluation boundaries; they do not define an extrapolated matching count or an exact ESS when the crossing does not exist. Median ESS excludes $n = 0$ because models rarely return a number without a description or demonstrations. At large n, ESS is strongly affected by slow baseline learning and excessive weight on the $c = 1$ and $c = 4$ descriptions. At $\sigma = 60$ and $n = 64$, model root-mean-square error without a description ranges from 9 to 20, compared with 7.5 for the reference. A learner with a flatter baseline curve can assign a larger ESS to the same error reduction. Description-weight estimates complement ESS because they do not use that curve.

Bootstrap replicates resample tasks jointly for the curves with and without the description and recompute the complete ESS statistic. The intervals therefore measure variation across tasks; they do not include interpolation bias, quantified below. We use 500 replicates with a fixed seed and the same resampled tasks for every model.

The reference's baseline prior and its exact ESS. Integer rounding, finite-range truncation, and the task-rejection rule are not part of the exact ESS c of Section 7.1, so we compute the reported reference values on the same generated tasks as the models.

Applying the empirical ESS procedure to the reference on the 2,000-task run gives 1.3, 4.4, and 36 for $c = 1$, 4, and 36 at $\sigma = 60$, and 1.2, 4.7, and 43 at $\sigma = 30$. Interpolation in $\log n$ induces an upward bias, while finite-task sampling can shift the estimate in either direction. We use the same estimator for the reference and the models. On the first 200 tasks, the grid additionally includes $n = 40$ and 48 near the $c = 36$ crossing, and the reference values are 1.0, 3.5, and 34.6. In the open-alternative design, in which every description is relevant and the prompt says only that the mean is otherwise unknown, the reference's median ESS varies more across task samples: on 20,000 tasks it is 7.3 at a stated 50% and 23.4 at 10%, compared with 5.8 and 22 on all 2,000 tasks and 5.7 and 27.5 on the first 200. Each model is compared with the reference on the same tasks.

The actual task prior, $\mathcal{N}(500, 100^2)$, is not stated in the prompt. A reference using

that prior would have baseline precision equivalent to 0.36 demonstrations at $\sigma = 60$ and 0.09 at $\sigma = 30$. At $\sigma = 60$, the corresponding idealized continuous Gaussian model would give a relevant description exact ESS $c - 0.36$, namely 0.64, 3.64, and 35.64. This is the scalar analogue of $c - c_0$ in Chapter 4. We use the Bayesian reference because the models are not given the actual task prior. This distinction matters under misspecification: unlike the correctly specified Bayesian predictor in Chapters 4–5, a reference that incorrectly assumes an irrelevant description to be relevant can have negative ESS, indicating increased prediction error (Reimherr et al., 2021).

Sensitivity to interpolation. The grid of the 2,000-task run doubles n, so interpolation can affect ESS. The reference's idealized errors are σ^2/n without a description and $\sigma^2/(c+n)$ with one. On these errors, interpolation in $\log n$ overestimates ESS by 0.2, 0.3, and 1.8 for $c = 1$, 4, and 36. Linear interpolation in n gives overestimates of 0.3, 0.6, and 4.1.

For the models without reasoning, every crossing is unique, because every baseline curve decreases over the counts where crossings occur. Among the twelve baseline curves across the two noise levels, the only increase is Phi-4's at $\sigma = 60$, whose root-mean-square error rises slightly from 20.3 at $n = 32$ to 20.4 at $n = 64$. Table 7.4 compares four estimators: interpolation in $\log n$; linear interpolation in n; a least-squares non-increasing fit using the pool-adjacent-violators algorithm; and a relative-least-squares fit of $A/(n+n_0)+B$ to the baseline errors. The fourth estimator's curve, $A/(n+n_0)+B$, is a Bayesian learning curve plus an error floor, and it recovers exact ESS when fit to the reference's idealized errors.

Across these choices, no model's median ESS changes by more than 1.4 demonstrations. For the $c = 36$ description, no model exceeds 8.1, while the reference ranges from 35 to 38. Differences become larger at higher demonstration counts, where a flat learning curve turns a small error difference into a large change in ESS. For example, gpt-oss-20b at $n = 8$ and $c = 36$ has ESS 17 under the primary log-n estimator (Table A.1), but 9.2 under the fitted curve.

7.5.3 Results of the 2,000-Task Run

Median ESS. Across the six models, the $c = 36$ description has median ESS 1.4 to 7.6, and its ESS is at most 15 at any of $n = 1$, 2, and 4; the reference's ESS on

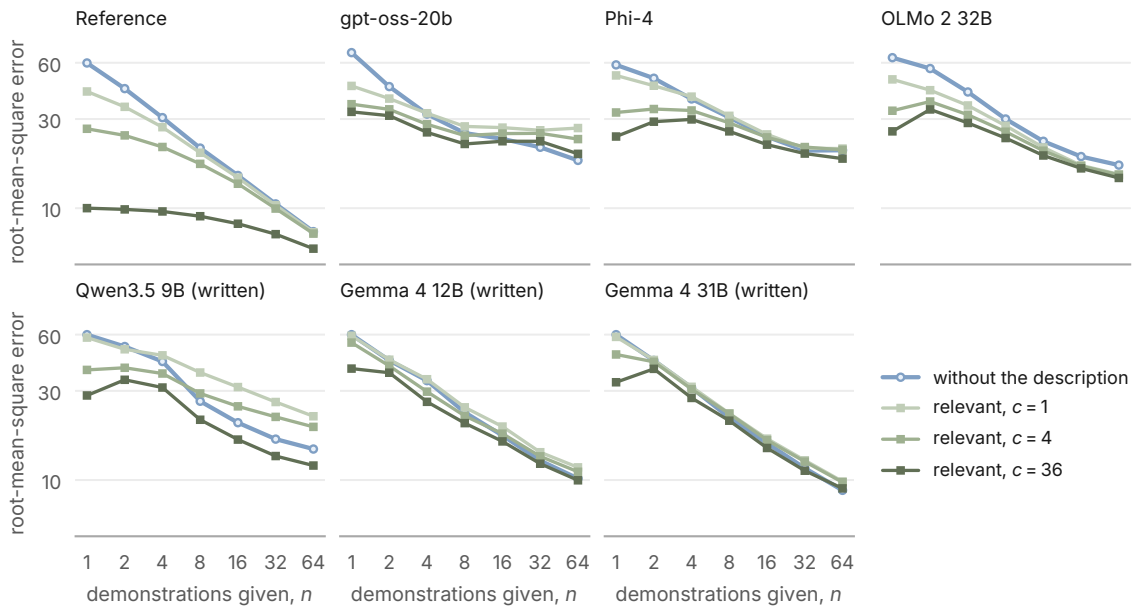


Figure 7.2: **Learning curves used to estimate language-model ESS.** Root-mean-square error versus demonstration count for the reference and six models in the 2,000-task run without reasoning, at $\sigma = 60$. Blue curves omit the description; green curves use relevant descriptions, with darker shades indicating higher c. Replies without a number are excluded. ESS is the horizontal distance from the point with the description at n to the baseline curve at the same error. For Phi-4, OLMo 2 32B, Qwen3.5 9B, and Gemma 4 31B, error with $c = 36$ increases between one and two demonstrations because the models shift toward the observations more than the reference does; Table A.3 shows this for the first two.

Table 7.4: **Sensitivity of median ESS to baseline-curve estimation.** Median ESS over $n = 1$, 2, and 4 for relevant descriptions, using the 2,000-task run without reasoning at $\sigma = 60$. "Log n" is the primary estimator; "linear" interpolates in n; "monotone" applies a non-increasing fit; and "fit" uses $A/(n + n_0) + B$. The reference's idealized ESS values are 1, 4, and 36.

	$c = 1$				$c = 4$				$c = 36$			
Predictor	log n	linear	monotone	fit	log n	linear	monotone	fit	log n	linear	monotone	fit
Reference	1.3	1.4	1.3	1.0	4.4	4.7	4.4	4.0	36	38	36	35
gpt-oss-20b	0.9	1.0	0.9	0.9	2.0	2.4	2.0	2.4	2.8	2.9	2.8	3.4
Phi-4	0.7	0.8	0.7	0.4	4.1	4.4	4.1	3.6	7.6	8.1	7.6	7.1
OLMo 2 32B	1.9	2.1	1.9	1.4	3.4	3.7	3.4	3.3	5.3	5.8	5.3	5.5
Qwen3.5 9B (written)	0.2	0.3	0.2	−0.1	2.7	2.9	2.7	1.3	4.0	4.3	4.0	2.8
Gemma 4 12B (written)	0.0	0.0	0.0	0.0	0.5	0.6	0.5	0.4	1.8	1.9	1.8	1.6
Gemma 4 31B (written)	0.0	0.0	0.0	−0.1	0.2	0.3	0.2	0.0	1.4	1.7	1.4	1.0

these tasks is 36 (Table 7.3). The $c = 1$ and $c = 4$ descriptions have median ESS 0.0 to 1.9 and 0.2 to 4.1, compared with 1.3 and 4.4. Median ESS thus rises with c. The description weights below indicate that the rise comes mainly from the prior mean, which lies closer to the task at higher c, rather than from the stated precision.

Description weights. At $\sigma = 60$, with no demonstrations and no stated reliability, the models return the prior mean (weight 0.92 to 1.10), although Phi-4 and OLMo 2 32B then begin only 20 to 23% and 50 to 59% of their replies with a number. At $n = 1$, the weight of gpt-oss-20b, Phi-4, and OLMo 2 32B changes by at most 0.08 from $c = 1$ to $c = 36$, and if anything falls, whereas the reference weight rises from 0.50 to 0.80 to 0.97 (Table 7.3). At every n from 1 to 64 and at both noise levels, the spread of these three models' weights across c is at most 0.16, compared with a reference spread of 0.34 to 0.71 (Table A.3). The weights of the two Gemma models do rise with c, by about 0.25 between $c = 1$ and $c = 36$ at $n = 1$, half the reference rise. The weight of Qwen3.5 9B does not. Models therefore underweight a precise description and, with many demonstrations, overweight the least precise one. At $n = 64$, gpt-oss-20b still assigns weight 0.23 to the $c = 1$ prior mean, compared with reference weight 0.02, and that description raises its error (ESS −57).

Stated unreliability. In the open-alternative design, a stated 10% or 50% chance of irrelevance barely changes the ESS of gpt-oss-20b and OLMo 2 32B, and Phi-4's

ESS even rises, from 7.6 to 8.7 and 9.7. Qwen3.5 9B and Gemma 4 31B have lower ESS, but by about as much for 50% as for 10%. Gemma 4 12B's ESS falls at 10% and rises at 50% (Table 7.3). No model's ESS falls in proportion to the stated probability, whereas reference ESS falls to 22 and 5.8.

7.5.4 Irrelevant Descriptions and Wording

Apart from the mixed design, the experiments above keep every description relevant, and all of them state its precision in one wording. This section relaxes each of these choices in the 2,000-task run.

Irrelevant descriptions. An irrelevant description states a prior mean drawn independently of the task, from the same distribution as relevant prior means, with the same stated precision. At $c = 36$, these descriptions have negative ESS for every model and for the Bayesian reference, which treats them as relevant. The models assign less weight to the description than the reference does and consequently incur lower error in this condition: at $n = 4$, their root-mean-square errors range from 35 to 64, compared with 124 for the reference. They also discount it relative to a relevant description: at $n = 4$, their weight on an irrelevant $c = 36$ prior mean is 0.01 to 0.33, compared with 0.18 to 0.44 on a relevant one.

Wording without reasoning. Descriptions in practice express uncertainty in many ways. To test whether the results depend on the original wording, we repeat the three relevant-description conditions and the no-description baseline at $n = 1$, 2, and 4 using three alternative description sentences: "Its mean is about m, give or take τ.", "Mean: $m \pm \tau$", and "An earlier estimate put its mean at m, with a standard error of τ." Table 7.5 uses the tasks of the 2,000-task run and written readouts. In that table, a fitted coefficient above 1 means that a unit change in the prior mean corresponds to more than a unit change in the prediction, with the sample mean held fixed in the regression.

Wording substantially changes the weight assigned to the description. For Gemma 4 12B with $c = 1$ and $n = 1$, this weight is 0.34 under the original wording and 1.00 under each alternative. For gpt-oss-20b, it ranges from 0.17 to 0.67. Nevertheless, none of the tested wordings yields the reference's sensitivity to prior precision: across

models and wordings, the weight difference between $c = 36$ and $c = 1$ is at most 0.25 at $n = 1$ and 0.44 at $n = 4$, compared with reference differences of 0.47 and 0.70. The informal descriptions do not uniquely specify a probability distribution, so this comparison changes how explicitly the prompt specifies the prior, not only its wording.

7.5.5 Reasoning Settings and Inference Details

Comparisons between modes. As in Section 7.1, "with reasoning" and "without reasoning" refer to generation settings. When we say that a reply contains reasoning, we mean that the provider recorded reasoning output for it; with reasoning enabled, some replies have none. Paired runs share the model identifier and prompts, but reasoning settings, serving software, readout, and retry policy vary across the runs; code execution is disabled in every run. The comparisons therefore concern the evaluated configurations and do not identify a unique internal mechanism.

Shared evaluation prompts. Most configurations in Figure 7.1 are evaluated on the same 7,800 prompts from the first 200 tasks. These include no-description prompts at $n = 1$, 2, 4, ..., 64, with additional points at $n = 40$ and 48 near the $c = 36$ crossing, and every description condition at $n = 1$, 2, and 4. The gpt-oss-20b and Olmo 3 7B Think runs omit the two additional baseline points and contain 7,400 prompts. Results without reasoning for the open-weight models are taken from the 2,000-task run restricted to these tasks. At least 99.7% of replies in each run contain a number.

Open-weight reasoning models. With reasoning, gpt-oss-20b runs in vLLM using greedy decoding, low reasoning effort, and at most 4,096 new tokens. It reaches the final channel in 99.9% of replies and uses 322 tokens on average, and at least 97% of replies reach it in every cell. An earlier run with a limit of 384 tokens finished only 28 to 54% of the replies in the stated-reliability conditions. That run seemed to show a difference between a stated 10% and a stated 50%, but the difference came from which replies finished. Qwen3.5 9B runs through OpenRouter at temperature 0, with reasoning enabled and a limit of 24,576 new tokens. We pin requests to DeepInfra, which serves the model in bf16, because quantization varies across providers. Of its replies, 2,305 repeat themselves until they reach the token limit, a failure its model

Table 7.5: **Description weights under four wordings without reasoning.** Weights are reported for the three relevant descriptions at $n = 1$ and 4, using written readouts on the tasks of the 2,000-task run. "Number" denotes the numeric response rate averaged over demonstration counts; a weight is italicized when the response rate at that count is below three quarters. The reference weights are 0.50, 0.80, and 0.97 at $n = 1$, and 0.20, 0.50, and 0.90 at $n = 4$. Wording changes overall reliance on the description, but the observed change with stated precision remains smaller than the reference change.

			weight, $n = 1$			weight, $n = 4$		
Model	Wording of the description	number	$c = 1$	$c = 4$	$c = 36$	$c = 1$	$c = 4$	$c = 36$
Gemma 4 12B	normal, mean m, sd τ (ours)	1.00	0.34	0.24	0.59	0.19	0.24	0.31
	about m, give or take τ	1.00	1.00	1.00	1.00	0.87	0.76	0.82
	Mean: $m \pm \tau$	1.00	1.00	1.00	1.00	0.60	0.66	0.74
	earlier estimate m, s.e. τ	1.00	1.00	1.00	1.00	0.45	0.54	0.67
Gemma 4 31B	normal, mean m, sd τ (ours)	1.00	0.48	0.49	0.72	0.02	0.04	0.18
	about m, give or take τ	1.00	1.00	1.00	1.00	1.01	1.22	1.45
	Mean: $m \pm \tau$	1.00	1.00	1.00	1.00	0.90	1.03	1.15
	earlier estimate m, s.e. τ	1.00	0.94	0.94	0.96	0.09	0.23	0.41
gpt-oss-20b	normal, mean m, sd τ (ours)	1.00	0.64	0.52	0.52	0.23	0.29	0.34
	about m, give or take τ	1.00	0.24	0.17	0.38	0.28	0.29	0.38
	Mean: $m \pm \tau$	1.00	0.17	0.26	0.40	0.23	0.24	0.30
	earlier estimate m, s.e. τ	1.00	0.67	0.63	0.67	0.28	0.37	0.42
OLMo 2 32B	normal, mean m, sd τ (ours)	0.61	0.83	0.81	0.83	*0.20*	*0.29*	0.38
	about m, give or take τ	0.82	0.93	0.92	0.96	0.60	0.57	0.65
	Mean: $m \pm \tau$	0.82	0.88	0.83	0.88	0.44	0.46	0.55
	earlier estimate m, s.e. τ	0.82	0.95	0.94	0.95	0.49	0.57	0.65
Phi-4	normal, mean m, sd τ (ours)	0.77	0.96	0.78	0.76	*0.38*	*0.37*	*0.38*
	about m, give or take τ	0.97	1.02	1.01	1.02	0.65	0.57	0.62
	Mean: $m \pm \tau$	0.83	*0.99*	*0.87*	*0.87*	0.45	0.42	0.50
	earlier estimate m, s.e. τ	0.81	0.88	0.76	0.73	*0.47*	*0.49*	0.54
Qwen3.5 9B	normal, mean m, sd τ (ours)	1.00	0.81	0.76	0.80	0.26	0.27	0.44
	about m, give or take τ	1.00	0.99	1.00	0.99	0.35	0.43	0.57
	Mean: $m \pm \tau$	1.00	1.08	1.25	1.14	0.19	0.27	0.38
	earlier estimate m, s.e. τ	1.00	0.87	0.86	0.86	0.61	0.75	0.82

card warns of. Each is rerun once with the model card's recommended reasoning sampling settings: temperature 1.0, top-p 0.95, top-k 20, and presence penalty 1.5. Eleven remain unfinished and one request fails.

Gemma 4 31B (`gemma-4-31b-it`) runs through the Gemini API at temperature 0. Reasoning is enabled by default, and the model uses 2,613 reasoning tokens on average, with the same 24,576-token limit. Of its replies, 435 reach the limit; all finish after one rerun with the model card's recommended sampling settings: temperature 1.0, top-p 0.95, and top-k 64.

Olmo 3 7B Think (`allenai/Olmo-3-7B-Think`), a small open-weight model that always reasons and so has no run without reasoning, runs in vLLM with greedy decoding and at most 32,768 new tokens, the limit its model card recommends; at 8,192 tokens, only 12 of 20 pilot replies finished. In the run reported here, 169 replies reach the 32,768-token limit. We rerun each of them once, with temperature 0.6 and top-p 0.95; 17 remain unfinished. The model generates 9,215 tokens on average and writes a worked solution before most answers. Its long greedy outputs depend on the hardware: an earlier run of the same conditions on a different GPU gave the same final number on only half the prompts. We report a single run and treat it as one sample from this variation; Section 7.5.6 analyzes its prediction rules.

Models served through command-line agents. GPT-6 Sol (`gpt-6-sol`, the model identifier served by Codex during the runs) runs through the Codex command-line agent in a read-only sandbox, with an empty configuration directory so that no user instructions or settings load, and with every tool that can run code disabled. We evaluate the model with reasoning disabled (no reasoning tokens) and enabled at low effort, at which, in a pilot, its answers were already within 1 of the reference's predictions.

Claude Opus 4.6 (`claude-opus-4-6`) and Haiku 5.5 (`claude-haiku-5-5`) run through the Claude Code command line, with one prompt per call, no tools, no project or user instructions, and the system prompt "You are a helpful assistant." Opus 4.6 uses high effort in both evaluated modes, with reasoning either enabled or disabled; the disabled mode emits no reasoning tokens. With reasoning enabled, at least 99% of replies in each description condition include reasoning output, with an average of 599 reasoning tokens across all replies. Without a description, only 1%, 24%, and 79% of replies include reasoning output at $n = 1$, 2, and 4, respectively, so its

baseline curve at small n mostly reflects replies without reasoning; using the reference's baseline curve instead changes its ESS by at most 1 (see below). Without reasoning, Opus 4.6 still writes out intermediate steps in 64 of its 7,800 replies; excluding these replies changes no median ESS by more than 0.1. Haiku 5.5 is evaluated mainly at medium effort, because at low effort it often answers without reasoning; we also report low effort for comparison. Both Claude models use the command line's default sampling settings, which the command line does not report, and we collect one reply per prompt.

At medium effort, Haiku 5.5 includes reasoning output in 37 to 88% of replies at $n = 1$ across conditions and in nearly all replies from $n = 2$ onward, with 270 reasoning tokens on average. Its no-description error stays close to the reference through $n = 48$; at $n = 16$, for example, it is 15.8 against 14.8, with two replies far from the sample mean. Its error rises at $n = 64$, where it divides the sum by 60 instead of 64 in 9 of 200 replies.

Run dates. Because hosted models can change over time, we record when each was evaluated. The 2,000-task run took place on 2–3 October 2026. Its conditions in which the stated precision differs from the generating one, and gpt-oss-20b's reasoning run, followed on 20 October. Haiku 5.5 was evaluated on 20 October for the runs of Section 7.2 and 4 November for wording, the relevant descriptions of the mixed design, and the message-sorting experiment. Opus 4.6, Qwen3.5 9B, Gemma 4 31B, and Olmo 3 7B Think were evaluated on 3–4 November. GPT-6 Sol was evaluated on 4 November. The irrelevant-description runs of the mixed design took place on 9–10 November, when Opus 4.6's runs with the baseline prior stated were also completed. The irrelevant descriptions of the mixed design, and GPT-6 Sol's run of that design without reasoning, were evaluated on 9 November. The GPT-6 Sol logs do not record a model version beyond the identifier above.

ESS when a model's baseline curve stops falling. With reasoning, gpt-oss-20b and Qwen3.5 9B nearly match the reference description weight: at $n = 4$ and $c = 36$, their fitted weights are 0.89 and 0.90, compared with 0.90 for the reference. Their errors without descriptions, however, stop falling at larger n. gpt-oss-20b makes errors when averaging long sequences: its no-description root-mean-square error is 19.0 at $n = 32$ and 24.7 at $n = 64$, compared with reference errors 10.0 and 7.4. At these

counts, 37 and 62 replies, respectively, differ from the sample mean by more than 20. Qwen3.5 9B's error levels off near 11 from $n = 32$ onward (11.2 at $n = 64$); in each cell, at most eleven of its replies differ from the sample mean by more than 20.

For these models, the error with a description sometimes lies below the whole baseline curve, so the evaluated grid contains no crossing. Table 7.6 therefore also reports, in parentheses, the comparison on the reference's baseline, $n^*_{\text{ref}} - n$, defined in Section 7.2. gpt-oss-20b also makes arithmetic errors with descriptions: about 2% of its replies are more than 40 from the reference prediction, from errors such as exchanging the description's and the demonstrations' precisions. These errors reduce gpt-oss-20b's ESS. When the prompt states a 10% chance of irrelevance in the open-alternative design, for example, its median ESS is 1.3. At a stated 50%, by contrast, it largely ignores the stated chance (Table 7.8), and because every description in those runs is relevant, its ESS of 8.8 exceeds the reference's 5.7. On the reference's baseline, its $c = 36$ description is worth 24 demonstrations, well below the reference's 34.6, although its fitted weight matches the reference's. A fitted weight is insensitive to such errors, whereas ESS reflects them.

Opus 4.6 and GPT-6 Sol do not exhibit these arithmetic failures. Opus 4.6 has root-mean-square error 7.4 at $n = 64$, matching the reference on these tasks. For these two models, changing between the model and reference baselines changes ESS by at most 1.

7.5.6 Prediction Rules in Individual Replies

Per-response description weights. At $n = 4$, we compute the per-response description weight $(\hat{w} - \bar{y})/(m - \bar{y})$. This quantity describes one response and differs from the fitted coefficient obtained by regressing predictions across tasks: a fitted coefficient averages over replies, whereas per-response weights show which prediction rule each reply follows. We use the tasks with $|m - \bar{y}| \geq 30$, where the weight is well determined: 61 tasks at $c = 36$, whose more accurate prior means lie closer to the sample mean, and 124 at $c = 1$. We exclude replies without a number. A reply is classified as consistent with a rule when its per-response weight is within 0.05 of that rule's weight. The candidate weights are the Bayesian reference (0.90 for $c = 36$ and 0.20 for $c = 1$); the reference with the description's and the demonstrations' precisions exchanged, $n/(c + n)$ (0.10 and 0.80); the description counted as one additional

Table 7.6: **Description ESS across reasoning settings.** Results use the first 200 tasks at $\sigma = 60$. Median ESS uses each run's own baseline curve; parentheses report the value computed using the reference baseline where it differs. "Stated irrelevance" gives the stated probability that a description generated at $c = 36$ is irrelevant, in the open-alternative design: every description is in fact relevant, and the prompt says only that the mean is otherwise unknown. These columns therefore differ from the mixed design of Section 7.3, whose results are in Table 7.2. The last column fixes the generating precision at 36 and states it as 4. Each run names the mode and, in parentheses, the provider's effort setting. An upper bound, $\leq$, indicates that one demonstration alone gives lower error; > 60 means that the error with the description lies below the model's entire baseline curve, through $n = 64$, so no crossing exists on the grid. Reference values use the same tasks and estimator and therefore differ from those in Table 7.3 (Section 7.5.2). The open-weight rows without reasoning take these 200 tasks from the 2,000-task run, whose results on all its tasks are in Table 7.3. The runs also include three further conditions in which the stated precision differs from the generating one, not shown.

					Stated irrelevance		
Run	$c = 1$	$c = 4$	$c = 36$	Irrelevant	10%	50%	Stated $c = 4$ Generated $c = 36$
Reference	1.0	3.5	34.6	≤ -1.0	27.5	5.7	13.7
GPT-6 Sol, without reasoning	0.9	1.9	10	≤ -1.0	3.3	0.9	2.4
GPT-6 Sol, with reasoning (low)	0.9	3.3	35 (34)	≤ -1.0	30 (29)	4.2	11
Opus 4.6, without reasoning	0.9	2.0	5.8 (5.9)	≤ -1.0	6.1 (6.0)	3.8	3.0
Opus 4.6, with reasoning (high)	1.0	3.5	35	≤ -1.0	29	3.8	13
Gemma 4 31B, without reasoning	0.0 (−0.2)	0.5 (0.2)	2.0 (1.6)	≤ -0.3	1.0 (0.7)	0.8 (0.4)	1.0 (0.6)
Gemma 4 31B, with reasoning	1.0	3.5	35	≤ -1.0	21	4.0	14
Qwen3.5 9B, without reasoning	−0.2 (−0.8)	2.7 (0.4)	5.5 (1.1)	≤ -1.0	1.6 (−0.7)	2.3 (−0.6)	4.1 (1.1)
Qwen3.5 9B, with reasoning	0.9	3.5	> 60 (34)	≤ -1.0	29 (27)	3.9	15 (14)
gpt-oss-20b, without reasoning	0.5 (0.3)	1.8 (1.3)	2.7 (1.7)	≤ -0.5	1.8 (1.5)	1.7 (1.5)	2.5 (1.6)
gpt-oss-20b, with reasoning (low)	0.3	1.0	> 60 (24)	≤ -1.0	1.3 (1.2)	8.8	8.6 (8.4)
Haiku 5.5, with reasoning (medium)	1.1	3.3 (3.5)	28	≤ -1.0	33	3.8 (3.9)	8.5 (8.8)
Haiku 5.5, with reasoning (low)	0.8 (0.7)	3.2 (3.3)	24	≤ -1.0	27 (26)	5.5 (5.6)	8.6 (8.4)
Olmo 3 7B Think, with reasoning	−0.5	0.0	6.2 (6.0)	≤ -1.0	5.9 (5.6)	5.5 (5.3)	15 (13)

Table 7.7: **Absolute error with a description.** Root-mean-square error against the hidden mean, for descriptions generated and stated at $c = 4$ and $c = 36$, on the first 200 tasks at $\sigma = 60$. Rows are the runs of Table 7.6, whose ESS compares each run with its own baseline curve; this table reports the error itself. Replies without a number are excluded.

	$c = 4$			$c = 36$		
Run	$n = 1$	2	4	1	2	4
Reference	26.7	24.0	19.9	9.5	9.3	9.3
GPT-6 Sol, without reasoning	32.1	28.6	24.2	15.2	16.3	18.4
GPT-6 Sol, with reasoning (low)	26.7	26.7	20.1	9.5	9.4	9.6
Opus 4.6, without reasoning	29.4	28.1	23.7	13.6	18.9	18.1
Opus 4.6, with reasoning (high)	26.7	24.0	19.8	9.5	9.3	9.3
Gemma 4 31B, without reasoning	43.4	41.6	27.6	32.6	39.1	24.0
Gemma 4 31B, with reasoning	26.7	24.0	19.9	9.5	9.3	9.3
Qwen3.5 9B, without reasoning	38.0	38.7	34.5	28.5	33.6	26.0
Qwen3.5 9B, with reasoning	26.7	24.0	19.9	9.6	9.3	9.3
gpt-oss-20b, without reasoning	34.5	32.6	26.5	31.5	30.5	23.7
gpt-oss-20b, with reasoning (low)	28.0	34.5	52.4	11.9	32.2	11.0
Haiku 5.5, with reasoning (medium)	33.7	24.0	19.8	27.0	9.3	10.1
Haiku 5.5, with reasoning (low)	46.1	24.0	20.2	41.9	9.7	11.1
Olmo 3 7B Think, with reasoning	28.7	42.1	32.1	20.8	16.5	28.8

observation, $1/(n+1)$; the description alone, with weight 1; and the demonstrations alone, with weight 0. The one-additional-observation rule coincides with the reference at $c = 1$.

With reasoning, Opus 4.6, GPT-6 Sol, Gemma 4 31B, and Haiku 5.5 follow the reference rule (Section 7.2); without reasoning, 77% of Opus 4.6 replies at $c = 36$ match none of these candidate rules, as do more than half of GPT-6 Sol replies. Olmo 3 7B Think averages demonstrations well but places more weight than the reference on the $c = 1$ and $c = 4$ descriptions. At $n = 1$, it repeats the prior mean in 57 to 80% of replies, more often at higher stated precision. At $n = 4$, about half of its eligible replies return the prior mean alone and a third match the reference. Without a description, its root-mean-square error falls from 56 at $n = 1$ to 10.8 at $n = 32$, compared with 10.0 for the reference. Its shortfall in ESS (Section 7.2) therefore lies in combining the description with demonstrations.

Within each cell, Haiku 5.5's replies split by whether they include reasoning output. At $n = 1$ and at either effort, replies with reasoning output assign weights 0.50, 0.80, and 0.97 to the relevant $c = 1$, 4, and 36 descriptions; these are the reference weights to two decimals. Replies without reasoning output assign weight 0.00 to 0.05 and repeat the single demonstration value. Its fitted weight at $n = 1$ mixes the two kinds of reply: 0.44, 0.66, and 0.74 at medium effort. On these three descriptions, 34 to 53% of its replies include reasoning output at low effort, against 72 to 82% at medium effort. Its fitted weights at $n = 1$ are therefore lower at low effort: 0.36, 0.34, and 0.41, nearly the same for all three precisions. This holds even at $c = 1$ and $c = 4$, where its median ESS is close to the reference's (Table 7.6). With more demonstrations, more replies include reasoning output, and its weight approaches the reference again: at $n = 4$ and low effort, it is 0.17, 0.48, and 0.86, compared with 0.20, 0.50, and 0.90. Whether a reply includes reasoning output is the model's own choice, so this split is observational rather than experimental.

Fixed-weight averaging. Table 7.8 reports the fitted description weights behind Section 7.3, for $c = 36$ under the open-alternative design's warning, which leaves the baseline prior unspecified, together with the weight $p\,36/(36+n)$ that fixed-weight averaging gives when the uniform baseline prior is treated as flat. GPT-6 Sol, Gemma 4 31B, and Qwen3.5 9B track fixed-weight averaging at the stated 50%. Opus 4.6 starts near the reference at $n = 1$ but falls below fixed-weight averaging by $n = 4$, and

Haiku 5.5 at medium effort lies between the two at $n = 1$ and 2. At the stated 50% and $n = 2$ and 4, Haiku 5.5 at low effort comes closest to the reference; at $n = 1$, where its replies often contain no reasoning output, its weight equals that of fixed-weight averaging. gpt-oss-20b assigns nearly the weight of a relevant description regardless of the stated probability. The individual replies show the same pattern. Pooling $n = 1$, 2, and 4 at the stated 50%, the share of replies within 2 of the fixed-weight prediction is 100% for Gemma 4 31B, 97% for Qwen3.5 9B, 89% for GPT-6 Sol, 80% for Opus 4.6, 58% for Haiku 5.5 at medium effort, and 21% for gpt-oss-20b. The share within 2 of the reference prediction is at most 24% for every model. Regressing the replies' normalized description weights (per-response weights divided by $c/(c+n)$, as described below) on the reference's π_n gives slopes of 0.01 for GPT-6 Sol, 0.02 for Gemma 4 31B, 0.06 for Qwen3.5 9B, 0.22 for Haiku 5.5, 0.30 for Opus 4.6, and 0.00 for gpt-oss-20b, where updating as the reference does gives 1.

Table 7.8: **Description weights under stated unreliability.** Fitted coefficients on the $c = 36$ prior mean for the first 200 tasks, with stated irrelevance probability $1 - p$. All model rows use reasoning. "Fixed weights" uses weight $p\,36/(36+n)$; the reference updates the relevance probability using a uniform baseline prior over 100 to 999. Bootstrap standard errors are at most 0.03 at $n = 4$ in the stated-50% condition and at most 0.06 overall, except for gpt-oss-20b, whose maximum is 0.09. At the stated 50%, several reasoning configurations closely track fixed-weight averaging.

	Stated irrelevance 50%			Stated irrelevance 10%		
	$n = 1$	2	4	1	2	4
Reference	0.58	0.61	0.64	0.88	0.87	0.84
Fixed weights	0.49	0.47	0.45	0.88	0.85	0.81
GPT-6 Sol	0.49	0.46	0.40	0.90	0.86	0.71
Opus 4.6	0.56	0.43	0.40	0.93	0.86	0.77
Gemma 4 31B	0.49	0.46	0.45	0.89	0.68	0.76
Qwen3.5 9B	0.49	0.46	0.43	0.88	0.84	0.80
Haiku 5.5, medium	0.55	0.56	0.46	0.79	0.93	0.89
Haiku 5.5, low	0.49	0.62	0.58	0.42	0.94	0.88
gpt-oss-20b	0.89	0.87	0.82	0.95	0.87	0.90

Stating the baseline prior. The mixed design of Section 7.3 uses the same 200 tasks, $c = 36$, stated irrelevance probabilities of 10% and 50%, and $n = 1$, 2, and

4. Each model answers 1,200 prompts with the task's relevant description and 1,200 with an irrelevant one. This subsection concerns the relevant descriptions, which differ from the open-alternative design only in stating the baseline prior. Stating it is needed because the reference's assumption matters under the open-alternative design's warning: with its baseline curve held fixed, its median ESS at the stated 50% is 5.7 under our uniform prior over 100 to 999, 4.6 under the generating distribution $\mathcal{N}(500, 100^2)$, and 13.4 under a uniform prior over 0 to 2000. Table 7.9 compares the fitted description weights with those from the open-alternative design.

At the stated 50%, the reference and fixed-weight predictions differ by at least 5 on 123 to 155 tasks per cell; these are the tasks on which Section 7.3 reports the share of replies closer to the reference. At the stated 10%, at most 7 tasks per cell separate the two rules by this amount, so the comparison is uninformative there.

The normalized description weight of Section 7.3 is computed from each reply's per-response weight, on tasks with $|m - \bar{y}| \geq 30$. For the scalar reference, it equals π_n up to the negligible boundary effects of Section 7.1. For model predictions, the ratio can lie outside $[0, 1]$ and does not identify an internal probability. Here π_n ranges from 0.08 to 0.88 as the demonstrations provide different evidence for relevance. The regression pools $n = 1$, 2, and 4 at the stated 50%, with 269 replies per model, and its 95% intervals resample the 153 tasks. With the prior stated, the intercepts are 0.01 and 0.03, and R^2 is 0.57 and 0.33.

The agreement also holds when the demonstrations count against the description. Among the 23 replies with π_n below 0.5 (mean 0.33), the mean normalized weights are 0.30 and 0.38 for the two models with the baseline prior stated, compared with 0.48 and 0.50 under the open-alternative design's warning. Both models therefore vary their description weights across tasks in approximately the amount predicted by Bayesian model averaging.

Median ESS uses each model's baseline curve from the runs of Section 7.2. At the stated 50%, fixed-weight averaging gives 4.4 and the Bayesian reference gives 5.7. Stating the baseline prior raises ESS from 4.2 to 5.4 for GPT-6 Sol and from 3.8 to 4.2 for Haiku 5.5. At the stated 10%, it changes GPT-6 Sol's ESS from 29.7 to 25.0, and lowers Haiku 5.5's to 15.1, against 27.5 for the reference (Section 7.3). The two models' fitted weights agree within 0.01 in this condition (Table 7.9). Haiku 5.5's lower ESS therefore does not reflect lower average reliance on the description; it comes from a minority of replies far from the reference, some of which repeat a demonstration

value. We also ran two further configurations on the relevant descriptions with the baseline prior stated. Haiku 5.5 at low effort also updates (slope 0.93), with median ESS 2.7 at the stated 50%, against 5.5 under the open-alternative design's warning. Opus 4.6, at high effort and without reasoning, does not (slope −0.18), with median ESS 3.8 under both warnings; its replies to these prompts supply the relevant half of its mixed-design results.

Table 7.9: **Stating the baseline prior changes description weights.** Fitted coefficients on the $c = 36$ prior mean for the first 200 tasks, as in Table 7.8. "Unstated" denotes the open-alternative design's warning, "the mean is unknown"; "stated" specifies that the mean is otherwise uniform over 100 to 999, as the reference assumes. GPT-6 Sol uses low effort and Haiku 5.5 uses medium effort.

	Stated irrelevance 50%			Stated irrelevance 10%		
	$n = 1$	2	4	1	2	4
Reference	0.58	0.61	0.64	0.88	0.87	0.84
Fixed weights	0.49	0.47	0.45	0.88	0.85	0.81
GPT-6 Sol, unstated	0.49	0.46	0.40	0.90	0.86	0.71
GPT-6 Sol, stated	0.58	0.59	0.59	0.88	0.83	0.80
Haiku 5.5, unstated	0.55	0.56	0.46	0.79	0.93	0.89
Haiku 5.5, stated	0.62	0.60	0.63	0.89	0.84	0.80

7.5.7 Wording with Reasoning Enabled

Because reasoning brings description weights close to the reference under the original wording, we test whether it does so under the alternative wordings as well. We evaluate the three alternative wordings with Haiku 5.5 at medium effort, for the three relevant descriptions at $n = 1$, 2, and 4 on the first 200 tasks; all 5,400 replies contain a number. With "An earlier estimate put its mean at m, with a standard error of τ", at least 71% of replies in each cell include reasoning output; 99% of these lie within 2 of the reference prediction, the tolerance of Section 7.3. Under "Its mean is about m, give or take τ" and "Mean: $m \pm \tau$", reasoning output is absent from 4 to 36% and 2 to 74% of replies per cell, respectively. Even among replies with reasoning output, 71% and 91% repeat the prior mean, and 43% and 32% lie within 2 of the reference prediction; the two groups overlap, because repeating the prior mean can lie within 2 of the reference

when the description is precise. At $c = 1$ and $n = 4$, Haiku 5.5's description weights under these two wordings are 0.72 and 0.93, compared with reference weight 0.20.

All wordings provide a value m and a scale τ, but "give or take" and "±" do not identify τ as a prior standard deviation or specify a distribution for the mean. As without reasoning (Section 7.5.4), the comparison therefore varies how explicit the probabilistic specification is, as well as how it is worded. A model that combines a description with demonstrations correctly when the prompt states a distribution therefore need not do so when the prompt does not.

7.5.8 Details of the Message-Sorting Experiment

Task. A message has six binary properties: its sender is outside the organization; it requests a payment; it concerns travel; it requests computer access; it has a deadline within a day; and it contains personal contact information. Each message states every property as present or absent, using one of three phrasings for each value, such as "I'm a contractor from another company." or "I'm on the sales team here." Properties are sampled independently with probability one half, phrasings are sampled uniformly, and sentences appear in random order.

A rule uses exactly two properties and requires each to be present or absent. A message goes into folder A when both requirements hold, and into B otherwise. There are $\binom{6}{2}2^2 = 60$ such rules. The prompt describes this family, calling the rule the company's filing policy, and states that it was chosen uniformly from the family. It then provides a note, n previously sorted messages with their folders, and four new messages. The model is asked for each new message's probability of going into A, as a number between 0 and 1, on four lines from "1: <probability>" to "4: <probability>".

Notes. A broad note reads, for example, "Policy note: whether it concerns travel is one of the two properties the policy uses," leaving 20 possible rules. A specific note reads "Policy note: to go into A, a message must request a payment. The other property the policy uses is whether it concerns travel," leaving 2. These notes state constraints in words; the model must determine which rules satisfy them.

The unreliable note is a specific note with an added sentence: "This note has a 50% chance of being wrong, in which case it says nothing about the policy, which is then any of the 60, equally likely." In the generator, the note is relevant with probability

one half and otherwise describes an independently drawn rule. The note is irrelevant in 92 of the 200 tasks. An independently drawn note can agree with the actual rule by chance, so we distinguish *relevance*, the latent component that generated the note, from *compatibility*, whether a rule satisfies it.

Reference and ESS. For a note asserted to be relevant, the reference takes the rules that are consistent with the note and the sorted messages, and predicts the fraction of them that assign the new message to A. For an unreliable note, it averages this prediction with the prediction from rules consistent with the sorted messages alone, weighting the two by the posterior relevance probability. The prior relevance probability is 0.5; the posterior probability is obtained using the likelihood of the demonstrations under each component. As a check on our implementation, on 3,000 simulated tasks the reference's predictive probabilities are calibrated in every condition, and its posterior probability that the unreliable note is relevant matches the frequency of relevance: where it exceeds 0.7, it averages 0.80, and the note is relevant in 80% of cases.

The loss is the Brier score, the squared error between the predicted probability of A and the true folder indicator, averaged over the four query messages. Without demonstrations, a broad note leaves the prediction unchanged: for any message, exactly a quarter of the 20 rules compatible with the note send it to A, the same fraction as among all 60, so the broad note's ESS at $n = 0$ is zero. ESS follows Definition 3.1. Each learner's baseline curve is its loss without a note, evaluated at $n = 0$, 2, 4, 6, 8, 12, 16, 20, 24, and 32 and interpolated linearly in n. On simulated tasks, this grid changes the reference ESS by at most 0.6 relative to evaluation at every integer demonstration count.

Runs. We use 200 tasks, each defined by a sampled rule, and 19 conditions: no note at the ten demonstration counts above, and each note at $n = 0$, 2, and 4. This gives 3,800 prompts per configuration. Haiku 5.5 runs at medium effort; GPT-6 Sol runs without reasoning and at low effort. We leave out Opus 4.6, because in a pilot it still wrote out its working on this task with reasoning disabled, so its two settings would not differ only in reasoning. The command-line agents are set up as in Section 7.5.5, with no tools, and no reply calls a tool. To check that GPT-6 Sol without reasoning reads the messages correctly, we asked it to state each property of 96 messages; it was

correct in 97 to 100% of cases for each property; all three errors concern one phrasing, "I'm not traveling for this." Fourteen replies from GPT-6 Sol at low effort contain no probability and are excluded; one Haiku 5.5 reply omits one of its four probabilities, and we score its other three. Every Haiku 5.5 reply includes reasoning output, with a median of 3,008 reasoning tokens. For GPT-6 Sol at low effort, 91% of replies include reasoning output, with a median of 142 reasoning tokens.

Results. Table 7.10 reports each note's ESS. With reasoning, both models are within 2 demonstrations of the reference in every condition except GPT-6 Sol's specific note at $n = 4$: its ESS is 26.1, compared with 14.8 for the reference. Its loss with this note, 0.008, is close to the reference's 0.007, but its baseline curve flattens beyond $n = 16$, with loss 0.022 at $n = 20$, compared with 0.004 for the reference. It consequently needs more demonstrations on its own baseline curve to match the note's error reduction.

For the unreliable note, fixed-weight averaging (Section 7.4) averages the two component predictions with the stated 50% weights. The one exception is when the sorted messages rule out every rule compatible with the note: it then assigns zero weight to the relevant component. We use query messages on which this predictor and the reference differ by at least 0.05: about 150 at $n = 2$ and 200 at $n = 4$, out of 800. The share of predictions closer to the reference is 90% at $n = 2$ and 99% at $n = 4$ for Haiku 5.5; 77% and 87% for GPT-6 Sol at low effort; and 45% at both counts for GPT-6 Sol without reasoning.

Without reasoning, GPT-6 Sol learns slowly from demonstrations. Its loss without a note is still 0.078 at $n = 32$, compared with less than 0.0005 for the reference. Measured against this weaker baseline, a note can have an ESS as large as the reference's or larger. For example, the specific note has ESS 13.5 at $n = 2$, although the model's loss with that note is 0.130, compared with 0.027 for the reference. This illustrates why learner-specific ESS must be interpreted alongside absolute prediction error.

Table 7.10: **ESS of notes in the message-sorting experiment.** Results use 200 tasks and each learner's own baseline curve, measured without a note. "≤ 0" means that the note did not lower loss; "> 28" means that the baseline curve does not reach the loss with the note by $n = 32$. On 6,000 simulated tasks, the reference ESS is 9.6 and 13.2 for the specific note at $n = 2$ and 4, and 2.5 and 2.9 for the unreliable note. Models are compared with the reference on the same 200 tasks. Paired 95% bootstrap intervals over tasks (1,000 replicates) for the specific note at $n = 2$ are: reference [9.7, 12.9], Haiku 5.5 [8.7, 11.5], GPT-6 Sol at low effort [7.8, 12.7], and GPT-6 Sol without reasoning [3.7, 21.9].

Note	n	Reference	Haiku 5.5 medium	GPT-6 Sol low	GPT-6 Sol no reasoning
Broad	0	0.0	0.0	≤ 0	≤ 0
	2	1.4	1.3	1.3	0.7
	4	1.8	1.6	1.7	8.6
Specific	0	3.7	3.6	3.3	3.7
	2	11.4	9.9	9.5	13.5
	4	14.8	14.5	26.1	> 28
Unreliable	0	1.3	1.3	1.6	1.7
	2	3.0	2.6	3.2	1.7
	4	5.0	4.9	5.1	2.7

Chapter 8

Discussion

We asked how natural-language descriptions can be characterized and how well learners use them in practice. For the first question, we treated a description as a prior with a specificity and a reliability, and we measured it by its predictive ESS. A description's value then depends on its specificity, its reliability, and the demonstrations already available. In our Gaussian model, a reliable description can replace many demonstrations by constraining the task across directions. Unreliability imposes a finite ESS ceiling before demonstrations are observed. Demonstrations can then raise a description's ESS, by supplying evidence about its relevance or, for a partial description whose added precision exceeds the dimension, by resolving the directions it leaves open so that its precision takes full effect.

For the second question, meta-trained Transformers approach the Bayes-optimal ESS and, without demonstrations, weight a description by how often descriptions in their training data were relevant. Without reasoning, the language models we test make too little use of a precise description; with reasoning, most of them use a stated precision as the Bayesian reference does. When descriptions are unreliable and the prompt says so, the three models we tested this way revise the stated chance from the demonstrations with reasoning, Haiku 5.5 less consistently than the other two, and for GPT-6 Sol and Haiku 5.5, the two models also tested on rules stated in words, the agreement extends from numerical predictions to those rules. When the prompt does not say how the task is distributed if the description is irrelevant, or expresses uncertainty informally, the models must supply what is missing from their pretrained knowledge, and their predictions depart from the reference.

Our controlled analysis suggests why a description can substitute for many examples

(Honda et al., 2025) or have limited influence (Webson & Pavlick, 2022; Gupta et al., 2025; Gloaguen et al., 2026). A description has little influence when it says little about the task, when it may be about the wrong task, when the demonstrations already supply what it says, or when the learner does not use it. The Bayes-optimal ESS reflects the first three, and a learner's shortfall from it reflects the fourth. For prompt design, the results suggest two practices. First, an instruction that may be wrong should come with demonstrations, which let the learner check it: without them, its value is capped however specific it is (Chapter 5). Second, a prompt that warns that an instruction may be wrong should also state the alternative. When our warning said only that the mean was otherwise "unknown", the models we tested made little use of the demonstrations to judge whether the description was right; when it stated the alternative, a mean drawn uniformly from 100 to 999, GPT-6 Sol and Haiku 5.5 used them as the Bayesian reference does (Section 7.3). More broadly, because a description enters as a prior, the Bayesian analysis of ICL extends from demonstrations to descriptions.

8.1 Scope and Limitations

The theory assumes Gaussian linear regression, squared error, and a particular mixture model of irrelevance. Within this setting, we model a relevant description as a single Gaussian prior around one prior mean: a description that points to one task and is more or less specific about it. Some descriptions take other shapes; one that names several candidate tasks, for example, is not a single Gaussian but a mixture, as in the task prior of Lin & Lee (2024). Future work could cover other forms of description with a mixture of Gaussians, which can approximate a wide class of priors (Dalal & Hall, 1983). Our main analysis also gives a description the same precision in every direction; Section 4.2 treats one that constrains only some directions. Learner-specific ESS measures improvement relative to a learner's own baseline curve and should be read alongside absolute error. The language-model tasks are small and controlled; the main numerical comparison uses 200 tasks, the test of unreliable descriptions covers three models, and the message-sorting experiment two. Some reasoning comparisons also differ in serving, decoding, or readout settings, and the frontier models are served through command-line agents (Section 7.5.5). Agreement with a Bayesian reference concerns a model's predictions: it does not show that the model computes them as

the reference does.

The networks are trained only at $d = 5$ and, for the higher-dimension experiment, at $d = 10$. Because the numerical language-model task is scalar, it has no $d + 1$ term and no partial description. The language models we test are post-trained chat models, and the wording test with reasoning enabled covers only one of them, Haiku 5.5. The warnings of the open-alternative design leave the baseline prior unspecified, so the Bayesian reference for them rests on an assumed prior (Section 7.3). Models without reasoning are evaluated through either the distribution readout or the written readout, so comparisons across these models mix differences between models with differences between readouts. Our language-model prompts state a description's precision and reliability explicitly, as numbers, so that every prediction can be compared with a Bayesian reference. In practice, these properties are implicit: a more specific instruction is typically a longer one that says more about the task, as a step-by-step workflow does, and nothing in it states a precision. How well models infer specificity and reliability from the wording alone is the next question, and predictive ESS applies to it directly, since it requires only a learner's errors with and without the description. Descriptions that leave their uncertainty unstated, more structured tasks, and richer or heavy-tailed priors (Dalal & Hall, 1983; O'Hagan & Pericchi, 2012) would test how far these findings extend.

8.2 Future Work

We outline four directions for future work. The first two follow from the limitations and findings above, the third from the descriptions studied here, and the fourth from the networks of Chapter 6. None of them is established in this thesis.

Partial descriptions and the $d + 1$ term in language models. Under a diffuse enough baseline prior, a partial description whose added precision exceeds the dimension is worth more after many demonstrations than before any (Proposition 4.2, Table 4.1), whereas the ESS of a reliable complete description falls toward $c - c_0$ (Proposition 4.2 with $k = d$). The networks of Chapter 6 capture this increase with $k = 3$ of $d = 5$ directions described, and their ESS at $c = 10$ is larger at $d = 10$ than at $d = 5$. The numerical language-model task, however, has a single unknown mean. A task with several unknown quantities, of which the description states only some,

would test whether a language model's ESS has a $d + 1$ term, and whether a partial description becomes more valuable to it as demonstrations accumulate.

Updating reliability. With reasoning, GPT-6 Sol, Opus 4.6, and Haiku 5.5 revise a stated chance of irrelevance in light of the demonstrations once the prompt says how the mean is distributed when the description is irrelevant (Section 7.3). In the mixed design, where descriptions are in fact irrelevant at the stated rate, GPT-6 Sol's and Opus 4.6's ESS are within 0.2 and 0.1 of the reference at both stated chances. Haiku 5.5 falls short: at a 10% chance, its ESS is 3.2 against 5.0, and only 60% of its replies to an irrelevant description are within 2 of the reference. On the relevant descriptions alone, its ESS is 15.1, against 27.5 for the reference. Its fitted weights there agree with GPT-6 Sol's within 0.01, so the shortfall does not come from lower average reliance on the description: a minority of its replies fall far from the reference, some repeating a demonstration value (Section 7.5.6). The mixed design covers three models, one precision ($c = 36$), and at most four demonstrations. Two questions remain open. Do the other reasoning models revise the chance once the prompt states the irrelevant case? And why do some of Haiku 5.5's replies depart so far from the reference?

Structural descriptions. Apart from the message-sorting experiment, the descriptions studied here state where the task lies, through a prior mean m with a stated precision. Many instructions instead state what kind of task it is, for example that only three features matter, or that the answer is monotone in an input. Such a description narrows the hypothesis class rather than pointing at one task. Our analysis already covers one case of this kind. A description that confines w to a known j-dimensional subspace states the other $k = d - j$ directions exactly, so it is a partial description of those directions in the limit $c \to \infty$. A heuristic calculation suggests that such a description keeps its value as demonstrations accumulate. Under a flat baseline prior, it would lower the Bayes-optimal error from about d/n to about j/n, so its ESS would be about $n(d/j - 1)$, which grows with the number of demonstrations. If a structural description is irrelevant, however, the Bayesian predictor that assumes relevance would keep a bias that no number of demonstrations removes. Our analysis does not cover structures that fix no direction, such as sparsity with unknown relevant features, or monotonicity. The definition of ESS and the network and language-model

tests apply to such structures as well. The message-sorting experiment of Section 7.4 gives a first test in a language model. Its broad note names a property that the rule uses without stating the value that the property must take. With reasoning, the broad note is worth 1.3 demonstrations to Haiku 5.5 and to GPT-6 Sol at $n = 2$, against 1.4 for the reference (Table 7.10).

Where a network stores its description weight. Without demonstrations, the meta-trained networks weight a description by how often descriptions in their training data were relevant (Chapter 6). A natural next step is to locate the components of a network that set this weight, and to test by intervention whether changing them changes the weight without retraining. Such a study would connect the behavioral measure of this thesis to the mechanisms by which networks learn in context.

Bibliography

Abdin, M., Aneja, J., Behl, H., et al. Phi-4 technical report. Technical report, Microsoft, 2024. arXiv:2412.08905.

Akata, E., Voudouris, K., Fortuin, V., and Schulz, E. In-context function learning in large language models. In *AISTATS*, 2026.

Akyürek, E., Schuurmans, D., Andreas, J., Ma, T., and Zhou, D. What learning algorithm is in-context learning? Investigations with linear models. In *ICLR*, 2023.

Anthropic. System card: Claude Haiku 5.5. https://www.anthropic.com/claude-haiku-5-5-system-card, 2026a.

Anthropic. System card: Claude Opus 4.6. https://anthropic.com/claude-opus-4-6-system-card, 2026b.

Arora, A., Jurafsky, D., Potts, C., and Goodman, N. D. Bayesian scaling laws for in-context learning. In *COLM*, 2025.

Brown, T. B. et al. Language models are few-shot learners. In *NeurIPS*, 2020.

Cook, J., Sapora, S., Ahmadian, A., Khan, A., Rocktäschel, T., Foerster, J., and Ruis, L. Programming by backprop: an instruction is worth 100 examples when finetuning LLMs. In *ICLR*, 2026.

Cover, T. M. and Thomas, J. A. *Elements of Information Theory*. Wiley, second edition, 2006.

Dalal, S. R. and Hall, W. J. Approximating priors by mixtures of natural conjugate priors. *Journal of the Royal Statistical Society: Series B (Methodological)*, 45(2): 278–286, 1983.

Evans, M. and Moshonov, H. Checking for prior-data conflict. *Bayesian Analysis*, 1 (4):893–914, 2006.

Falck, F., Wang, Z., and Holmes, C. Is in-context learning in large language models Bayesian? A martingale perspective. In *ICML*, 2024.

Filbry, L., Melo, L. C., and Gal, Y. From prompts to posteriors: Can LLM agents perform Bayesian inference in context? *arXiv:2610.29331*, 2026.

Garg, S., Tsipras, D., Liang, P., and Valiant, G. What can Transformers learn in-context? A case study of simple function classes. In *NeurIPS*, 2022.

Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., and Rubin, D. B. *Bayesian Data Analysis*. CRC Press, third edition, 2013.

Gemma Team. Gemma 4 technical report. *arXiv:2607.02770*, 2026.

Genewein, T., Wenliang, L. K., Grau-Moya, J., Ruoss, A., Orseau, L., and Hutter, M. Understanding prompt tuning and in-context learning via meta-learning. In *NeurIPS*, 2025.

Gloaguen, T., Mündler-Sasahara, N., Müller, M. N., Raychev, V., and Vechev, M. Evaluating AGENTS.md: are repository-level context files helpful for coding agents? *arXiv:2602.11988*, 2026.

Goodman, N. D., Tenenbaum, J. B., Feldman, J., and Griffiths, T. L. A rational analysis of rule-based concept learning. *Cognitive Science*, 32(1):108–154, 2008.

Grünwald, P. D. and Dawid, A. P. Game theory, maximum entropy, minimum discrepancy and robust Bayesian decision theory. *The Annals of Statistics*, 32(4): 1367–1433, 2004.

Gruver, N., Finzi, M., Qiu, S., and Wilson, A. G. Large language models are zero-shot time series forecasters. In *NeurIPS*, 2023.

Gupta, R., Corona, R., Ge, J., Wang, E., Klein, D., Darrell, T., and Chan, D. M. Enough coin flips can make LLMs act Bayesian. In *ACL*, 2025.

He, Z., Qiu, T., Shirado, H., and Sap, M. Martingale Score: An unsupervised metric for Bayesian rationality in LLM reasoning. In *NeurIPS*, 2025.

Hoeting, J. A., Madigan, D., Raftery, A. E., and Volinsky, C. T. Bayesian model averaging: A tutorial. *Statistical Science*, 14(4):382–417, 1999. doi: 10.1214/ss/1009212519.

Honda, U., Murakami, S., and Zhang, P. Distilling many-shot in-context learning into a cheat sheet. In *Findings of EMNLP*, 2025.

Huang, R. and Ge, R. Task descriptors help Transformers learn linear models in-context. In *ICLR*, 2025.

Kwa, T., West, B., Becker, J., Deng, A., Garcia, K., Hasin, M., Jawhar, S., Kinniment, M., Rush, N., Von Arx, S., et al. Measuring AI ability to complete long software tasks. *arXiv:2503.14499*, 2025.

Le Scao, T. and Rush, A. M. How many data points is a prompt worth? In *NAACL*, 2021.

Lin, Z. and Lee, K. Dual operating modes of in-context learning. In *ICML*, 2024.

Lin, Z., Bharti, S. K., and Lee, K. In-context learning with hypothesis-class guidance. *arXiv:2502.19787*, 2025.

Liu, T. J. B., Boullé, N., Sarfati, R., and Earls, C. J. Density estimation with LLMs: a geometric investigation of in-context learning trajectories. In *ICLR*, 2025.

Mao, Y., He, J., and Chen, C. From prompts to templates: A systematic prompt template analysis for real-world LLMapps. In *FSE Industry Track*, 2025. arXiv:2504.02052.

Min, S., Lyu, X., Holtzman, A., Artetxe, M., Lewis, M., Hajishirzi, H., and Zettlemoyer, L. Rethinking the role of demonstrations: What makes in-context learning work? In *EMNLP*, 2022.

MiniCPM Team. MiniCPM5-2B. https://huggingface.co/openbmb/MiniCPM5-2B, 2026. Model card; the series is described in *MiniCPM4: Ultra-Efficient LLMs on End Devices*, arXiv:2506.07900.

Morita, S., Thall, P. F., and Müller, P. Determining the effective sample size of a parametric prior. *Biometrics*, 64(2):595–602, 2008.

Muirhead, R. J. *Aspects of Multivariate Statistical Theory*. Wiley, 1982.

Müller, S., Hollmann, N., Pineda Arango, S., Grabocka, J., and Hutter, F. Transformers can do Bayesian inference. In *ICLR*, 2022.

Neuenschwander, B., Weber, S., Schmidli, H., and O'Hagan, A. Predictively consistent prior effective sample sizes. *Biometrics*, 76(2):578–587, 2020.

O'Hagan, A. and Pericchi, L. Bayesian heavy-tailed models and conflict resolution: A review. *Brazilian Journal of Probability and Statistics*, 26(4):372–401, 2012.

OpenAI. gpt-oss-120b & gpt-oss-20b model card. Technical report, OpenAI, 2025. arXiv:2508.10925.

OpenAI. GPT-6 Sol and GPT-6 Luna: October 2026 update. https://deployment safety.openai.com/gpt-6-october, 2026.

Panwar, M., Ahuja, K., and Goyal, N. In-context learning through the Bayesian prism. In *ICLR*, 2024.

Patsakis, C., Argyropoulos, V., and Alepis, E. Configuration, not conscience: A large-scale empirical study of LLM system prompts. In *ICTAI*, 2026. arXiv:2609.31575.

Qiu, L., Sha, F., Allen, K., Kim, Y., Linzen, T., and van Steenkiste, S. Bayesian teaching enables probabilistic reasoning in large language models. *Nature Communications*, 17:1238, 2026.

Qwen Team. Qwen3.5-9B. https://huggingface.co/Qwen/Qwen3.5-9B, 2026. Model card.

Raventós, A., Paul, M., Chen, F., and Ganguli, S. Pretraining task diversity and the emergence of non-Bayesian in-context learning for regression. In *NeurIPS*, 2023.

Reimherr, M., Meng, X.-L., and Nicolae, D. L. Prior sample size extensions for assessing prior impact and prior-likelihood discordance. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 83(3):413–437, 2021.

Requeima, J., Bronskill, J., Choi, D., Turner, R. E., and Duvenaud, D. LLM processes: numerical predictive distributions conditioned on natural language. In *NeurIPS*, 2024.

Reznik, Y. A. The optimal discounting parameter of the power prior under predictive log-loss. *arXiv:2608.12159*, 2026.

Schmidli, H., Gsteiger, S., Roychoudhury, S., O'Hagan, A., Spiegelhalter, D., and Neuenschwander, B. Robust meta-analytic-predictive priors in clinical trials with historical control information. *Biometrics*, 70(4):1023–1032, 2014.

Selby, D., Iwashita, Y., Spriestersbach, K., Saad, M., Bappert, D., Warrier, A., Mukherjee, S., Kise, K., and Vollmer, S. Had enough of experts? Quantitative knowledge retrieval from large language models. *Stat*, 14:e70054, 2025.

Team Olmo. Olmo 3. *arXiv preprint arXiv:2512.13961*, 2025.

Team OLMo, Walsh, P., Soldaini, L., Groeneveld, D., Lo, K., et al. 2 OLMo 2 Furious. In *COLM*, 2025. arXiv:2501.00656.

Tong, X., Zeng, Y., and Zhang, J. Demonstrations, CoT, and prompting: a theoretical analysis of ICL. *arXiv:2603.19611*, 2026.

von Oswald, J., Niklasson, E., Randazzo, E., Sacramento, J., Mordvintsev, A., Zhmoginov, A., and Vladymyrov, M. Transformers learn in-context by gradient descent. In *ICML*, 2023.

von Rosen, D. Moments for the inverted Wishart distribution. *Scandinavian Journal of Statistics*, 15(2):97–109, 1988.

Webson, A. and Pavlick, E. Do prompt-based models really understand the meaning of their prompts? In *NAACL*, 2022.

Wei, J., Wei, J., Tay, Y., Tran, D., Webson, A., Lu, Y., Chen, X., Liu, H., Huang, D., Zhou, D., and Ma, T. Larger language models do in-context learning differently. *arXiv:2303.03846*, 2023.

Weller, O., Lourie, N., Gardner, M., and Peters, M. E. Learning from task descriptions. In *EMNLP*, pp. 1361–1375, 2020. doi: 10.18653/v1/2020.emnlp-main.105.

Wenliang, L. K., Ruoss, A., Grau-Moya, J., Hutter, M., and Genewein, T. Why is prompting hard? Understanding prompts on binary sequence predictors. In *AISTATS*, 2026. PMLR 300:2134–2142.

Wiesenfarth, M. and Calderazzo, S. Quantification of prior impact in terms of effective current sample size. *Biometrics*, 76(1):326–336, 2020.

Xie, S. M., Raghunathan, A., Liang, P., and Ma, T. An explanation of in-context learning as implicit Bayesian inference. In *ICLR*, 2022.

Zhu, Q., Oermann, E. K., and Cho, K. Multi-task Bayesian in-context learning. In *ICML*, 2026. arXiv:2606.20538.

Appendix A

Language-Model Results at Every Number of Demonstrations

This appendix reports the 2,000-task run of Chapter 7, without reasoning, at every evaluated demonstration count. Tables A.1 and A.2 report ESS for the distribution-readout and written-readout models, and Table A.3 reports description weights for the distribution-readout models. Tables A.1 and A.3 each continue onto a second page.

Table A.1: **ESS by description condition and demonstration count: distribution readout.** Results cover the Bayesian reference and the three distribution-readout models in the 2,000-task run without reasoning. "Number" is the numeric response rate averaged over n. "10% wrong" and "50% wrong" denote stated irrelevance probabilities. The censoring marker $\leq$ marks error above the first baseline point; $> 64 - n$ means that the error with the description lies below the entire baseline curve through 64 demonstrations.

Predictor	Description	number	$n = 1$	$n = 2$	$n = 4$	$n = 8$	$n = 16$	$n = 32$	$n = 64$
Reference	$\sigma = 60$, $c = 1$, relevant		1.2	1.3	1.3	1.3	1.2	1.5	> 0
	$\sigma = 60$, $c = 1$, irrelevant		≤ 0.0	-0.5	-0.8	-1.2	-1.7	-1.5	-1.7
	$\sigma = 60$, $c = 4$, relevant		4.5	4.4	3.9	4.6	4.6	5.2	> 0
	$\sigma = 60$, $c = 4$, irrelevant		≤ 0.0	≤ -1.0	≤ -3.0	-6.2	-12	-20	-32
	$\sigma = 60$, $c = 36$, relevant		36	36	37	38	39	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant		≤ 0.0	≤ -1.0	≤ -3.0	≤ -7.0	≤ -15	≤ -31	-62
	$\sigma = 30$, $c = 1$, relevant		1.0	1.2	1.4	1.6	1.4	1.5	> 0
	$\sigma = 30$, $c = 4$, relevant		4.5	4.7	5.0	5.1	4.9	5.6	> 0
	$\sigma = 30$, $c = 36$, relevant		42	43	43	43	42	> 32	> 0
	$\sigma = 60$, $c = 4$, relevant, 10% wrong		3.8	3.9	3.6	4.4	4.5	5.1	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong		≤ 0.0	-0.7	-1.2	-1.6	-2.2	-1.9	-2.2
	$\sigma = 60$, $c = 36$, relevant, 10% wrong		13	22	26	35	38	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong		≤ 0.0	-0.9	-1.7	-2.8	-5.4	-6.9	-10
	$\sigma = 60$, $c = 4$, relevant, 50% wrong		1.9	2.2	2.9	3.7	4.1	4.5	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong		≤ 0.0	-0.2	-0.4	-0.5	-0.8	-0.6	-0.7
	$\sigma = 60$, $c = 36$, relevant, 50% wrong		3.1	5.8	12	21	31	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong		≤ 0.0	-0.3	-0.7	-1.2	-2.4	-2.8	-4.5
gpt-oss-20b	$\sigma = 60$, $c = 1$, relevant	0.98	1.0	0.9	-0.1	-1.5	-9.2	-25	-57
	$\sigma = 60$, $c = 1$, irrelevant	0.98	≤ 0.0	-0.3	-1.5	-3.6	-10	-25	-59
	$\sigma = 60$, $c = 4$, relevant	0.98	2.3	1.7	2.0	2.9	-7.5	-24	-48
	$\sigma = 60$, $c = 4$, irrelevant	0.97	≤ 0.0	-0.3	-1.4	-3.7	-11	-26	-59
	$\sigma = 60$, $c = 36$, relevant	0.98	2.8	2.3	3.8	17	3.7	-12	-17
	$\sigma = 60$, $c = 36$, irrelevant	0.98	≤ 0.0	-0.5	-1.5	-3.7	-11	-26	-60
	$\sigma = 30$, $c = 1$, relevant	0.99	0.8	0.4	-0.3	-1.8	-9.9	-26	-59
	$\sigma = 30$, $c = 4$, relevant	0.98	1.2	1.3	1.9	3.4	-8.1	-24	-44
	$\sigma = 30$, $c = 36$, relevant	0.99	1.6	2.3	4.7	31	3.0	-13	> 0
	$\sigma = 60$, $c = 4$, relevant, 10% wrong	0.96	2.2	1.8	0.0	-0.2	-8.1	-24	-52
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.6	-1.1	-2.2	-9.6	-25	-58
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	0.97	2.4	2.4	0.5	2.8	-3.6	-20	-33
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.7	-1.1	-2.3	-9.7	-25	-59
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	0.95	2.3	1.7	0.1	-0.5	-8.3	-24	-53
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	0.94	≤ 0.0	-0.5	-1.1	-2.3	-9.7	-25	-58
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	0.95	2.8	2.3	0.7	1.2	-4.5	-21	-37
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	0.95	≤ 0.0	-0.6	-1.1	-2.3	-9.7	-25	-59
Phi-4	$\sigma = 60$, $c = 1$, relevant	0.87	0.8	0.7	-0.2	-0.4	-1.0	-3.1	-35
	$\sigma = 60$, $c = 1$, irrelevant	0.86	≤ 0.0	≤ -1.0	≤ -3.0	-4.9	-6.1	-8.8	-40
	$\sigma = 60$, $c = 4$, relevant	0.85	5.8	4.1	2.4	2.1	0.7	-4.6	-33
	$\sigma = 60$, $c = 4$, irrelevant	0.84	≤ 0.0	≤ -1.0	≤ -3.0	-4.7	-6.2	-13	-42
	$\sigma = 60$, $c = 36$, relevant	0.86	15	7.6	4.7	5.6	8.5	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant	0.84	≤ 0.0	≤ -1.0	≤ -3.0	-4.5	-6.4	-12	-42
	$\sigma = 30$, $c = 1$, relevant	0.87	1.1	0.8	-0.3	-0.3	-1.0	-1.8	-15
	$\sigma = 30$, $c = 4$, relevant	0.86	4.8	3.5	2.0	1.6	0.6	-1.9	-2.2
	$\sigma = 30$, $c = 36$, relevant	0.88	9.7	6.0	3.8	4.4	8.4	17	> 0
	$\sigma = 60$, $c = 4$, relevant, 10% wrong	0.87	6.0	5.5	5.3	8.3	15	> 32	> 0

	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	$0.85 \leq$	$0.0 \leq$	-1.0	-2.1	-1.2	2.9	-3.2	> 0
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	0.87	10	8.7	8.2	17	> 48	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	$0.86 \leq$	$0.0 \leq$	-1.0	-2.4	-1.7	1.3	-4.7	> 0
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	0.87	6.0	6.4	7.7	17	> 48	> 32	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	$0.85 \leq$	$0.0 \leq$	-1.0	-1.3	1.5	16	> 32	> 0
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	0.88	9.2	9.7	11	> 56	> 48	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	$0.86 \leq$	$0.0 \leq$	-1.0	-1.8	0.6	13	> 32	> 0
OLMo 2 32B	$\sigma = 60$, $c = 1$, relevant	0.90	1.9	1.8	2.0	2.3	5.8	> 32	> 0
	$\sigma = 60$, $c = 1$, irrelevant	$0.89 \leq$	0.0	-0.5	-0.2	-0.2	1.7	25	> 0
	$\sigma = 60$, $c = 4$, relevant	0.91	5.8	3.4	3.4	4.5	9.3	> 32	> 0
	$\sigma = 60$, $c = 4$, irrelevant	$0.90 \leq$	0.0	-0.7	-0.3	-0.5	0.3	17	> 0
	$\sigma = 60$, $c = 36$, relevant	0.91	11	4.6	5.3	6.8	15	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant	$0.90 \leq$	$0.0 \leq$	-1.0	-0.6	-0.5	-0.2	15	> 0
	$\sigma = 30$, $c = 1$, relevant	0.92	1.8	1.9	1.7	1.7	4.4	24	> 0
	$\sigma = 30$, $c = 4$, relevant	0.92	4.8	3.3	3.1	4.0	9.5	> 32	> 0
	$\sigma = 30$, $c = 36$, relevant	0.93	7.0	4.4	4.0	5.9	14	> 32	> 0
	$\sigma = 60$, $c = 4$, relevant, 10% wrong	0.96	6.0	4.6	3.8	4.5	5.8	23	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	$0.96 \leq$	0.0	-0.9	0.0	0.0	-0.4	4.1	> 0
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	0.96	12	5.9	6.0	6.3	8.9	31	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	$0.96 \leq$	$0.0 \leq$	-1.0	-0.3	-0.1	-0.8	0.3	> 0
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	0.93	5.3	4.2	3.5	4.0	5.4	27	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	$0.92 \leq$	0.0	-0.5	0.1	0.1	0.0	11	> 0
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	0.93	7.8	5.4	4.9	5.6	7.9	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	$0.92 \leq$	0.0	-0.8	-0.1	0.0	-0.3	7.8	> 0

Table A.2: **ESS by description condition and demonstration count: written readout.** Results cover the three written-readout models in the 2,000-task run without reasoning. Replies without a qualifying number are excluded. Columns and reference values follow Table A.1.

Predictor	Description	number	$n=1$	$n=2$	$n=4$	$n=8$	$n=16$	$n=32$	$n=64$
Qwen3.5 9B (written)	$\sigma=60$, $c=1$, relevant	1.00	0.2	0.3	−0.9	−2.8	−9.3	−24	−51
	$\sigma=60$, $c=1$, irrelevant	1.00	≤ 0.0	≤ -1.0	−2.7	−4.8	−9.7	−23	−49
	$\sigma=60$, $c=4$, relevant	1.00	3.9	2.7	1.3	−0.7	−6.2	−18	−44
	$\sigma=60$, $c=4$, irrelevant	1.00	≤ 0.0	≤ -1.0	−1.7	−3.8	−9.4	−24	−49
	$\sigma=60$, $c=36$, relevant	1.00	6.5	4.0	2.8	6.9	17	>32	>0
	$\sigma=60$, $c=36$, irrelevant	1.00	≤ 0.0	≤ -1.0	−2.8	−5.1	−10	−24	−51
	$\sigma=30$, $c=1$, relevant	1.00	≤ 0.0	0.2	−0.7	−2.5	−8.7	−22	−49
	$\sigma=30$, $c=4$, relevant	1.00	3.4	2.7	1.5	−0.3	−4.5	−14	−40
	$\sigma=30$, $c=36$, relevant	1.00	4.7	4.3	3.0	9.2	14	>32	>0
	$\sigma=60$, $c=4$, relevant, 10% wrong	1.00	3.8	−0.4	−0.7	−1.8	−8.5	−21	−47
	$\sigma=60$, $c=4$, irrelevant, 10% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−4.9	−10	−24	−50
	$\sigma=60$, $c=36$, relevant, 10% wrong	1.00	5.3	0.4	0.6	−0.3	−3.3	−4.2	−6.9
	$\sigma=60$, $c=36$, irrelevant, 10% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−5.2	−10	−25	−53
	$\sigma=60$, $c=4$, relevant, 50% wrong	1.00	4.5	−0.1	−0.4	−1.9	−9.0	−23	−50
	$\sigma=60$, $c=4$, irrelevant, 50% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−6.2	−12	−26	−54
	$\sigma=60$, $c=36$, relevant, 50% wrong	1.00	7.3	0.8	1.0	−0.1	−4.0	−4.2	−12
	$\sigma=60$, $c=36$, irrelevant, 50% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−7.0	−13	−27	−56
Gemma 4 12B (written)	$\sigma=60$, $c=1$, relevant	1.00	0.0	0.0	−0.1	−0.6	−3.2	−5.4	−21
	$\sigma=60$, $c=1$, irrelevant	1.00	≤ 0.0	−0.4	−1.3	−1.3	−4.4	−6.8	−24
	$\sigma=60$, $c=4$, relevant	1.00	0.3	0.5	1.4	1.1	−0.6	−2.7	−13
	$\sigma=60$, $c=4$, irrelevant	1.00	≤ 0.0	−0.4	−1.1	−1.2	−4.1	−6.9	−23
	$\sigma=60$, $c=36$, relevant	1.00	1.8	1.2	2.7	3.6	3.6	4.9	>0
	$\sigma=60$, $c=36$, irrelevant	1.00	≤ 0.0	−0.8	−1.3	−1.3	−4.9	−7.9	−26
	$\sigma=30$, $c=1$, relevant	1.00	≤ 0.0	−0.2	−0.1	−0.5	−3.2	−7.9	−27
	$\sigma=30$, $c=4$, relevant	1.00	0.7	0.8	2.1	1.5	−0.2	−4.6	−18
	$\sigma=30$, $c=36$, relevant	1.00	4.0	1.7	3.4	3.7	2.4	−1.8	−8.9
	$\sigma=60$, $c=4$, relevant, 10% wrong	1.00	0.1	0.7	1.4	0.4	−1.0	−3.1	−7.3
	$\sigma=60$, $c=4$, irrelevant, 10% wrong	1.00	≤ 0.0	−0.2	−0.4	−0.5	−2.2	−5.4	−13
	$\sigma=60$, $c=36$, relevant, 10% wrong	1.00	0.6	1.3	2.3	1.6	0.7	−0.6	−1.3
	$\sigma=60$, $c=36$, irrelevant, 10% wrong	1.00	≤ 0.0	−0.2	−0.5	−0.5	−2.5	−5.8	−15
	$\sigma=60$, $c=4$, relevant, 50% wrong	1.00	1.0	0.6	1.7	1.0	−0.1	−1.7	−4.5
	$\sigma=60$, $c=4$, irrelevant, 50% wrong	1.00	≤ 0.0	−0.3	−0.4	−0.4	−1.3	−4.1	−9.6
	$\sigma=60$, $c=36$, relevant, 50% wrong	1.00	5.1	1.0	2.7	2.3	2.4	1.1	>0
	$\sigma=60$, $c=36$, irrelevant, 50% wrong	1.00	≤ 0.0	−0.3	−0.6	−0.4	−1.9	−4.8	−13
Gemma 4 31B (written)	$\sigma=60$, $c=1$, relevant	1.00	0.1	0.0	−0.1	−0.5	−1.4	−5.1	−13
	$\sigma=60$, $c=1$, irrelevant	1.00	≤ 0.0	−0.1	−0.1	−0.5	−1.0	−4.5	−11
	$\sigma=60$, $c=4$, relevant	1.00	0.8	0.1	0.2	−0.5	−1.0	−4.3	−12
	$\sigma=60$, $c=4$, irrelevant	1.00	≤ 0.0	−0.2	−0.2	−0.4	−1.0	−4.4	−10
	$\sigma=60$, $c=36$, relevant	1.00	2.6	0.6	1.4	0.9	2.9	3.4	−2.9
	$\sigma=60$, $c=36$, irrelevant	1.00	≤ 0.0	≤ -1.0	−0.8	−0.6	−1.1	−2.7	−4.8
	$\sigma=30$, $c=1$, relevant	1.00	≤ 0.0	0.0	−0.1	−0.7	−1.2	−3.0	−8.1
	$\sigma=30$, $c=4$, relevant	1.00	0.5	0.0	0.1	−0.5	−1.3	−3.4	−8.3
	$\sigma=30$, $c=36$, relevant	1.00	4.3	1.5	2.1	2.3	3.3	3.3	−0.4
	$\sigma=60$, $c=4$, relevant, 10% wrong	1.00	0.1	0.0	0.0	−0.7	−1.9	−5.0	−15
	$\sigma=60$, $c=4$, irrelevant, 10% wrong	1.00	≤ 0.0	0.0	−0.1	−0.4	−1.2	−4.1	−9.5
	$\sigma=60$, $c=36$, relevant, 10% wrong	1.00	1.0	0.0	0.6	−0.3	−1.4	−4.8	−12
	$\sigma=60$, $c=36$, irrelevant, 10% wrong	1.00	≤ 0.0	0.0	−0.2	−0.4	−0.9	−3.0	−6.2
	$\sigma=60$, $c=4$, relevant, 50% wrong	1.00	0.2	0.0	0.0	−0.7	−1.7	−4.6	−17
	$\sigma=60$, $c=4$, irrelevant, 50% wrong	1.00	≤ 0.0	0.0	−0.2	−0.4	−1.2	−3.6	−12
	$\sigma=60$, $c=36$, relevant, 50% wrong	1.00	1.0	0.0	0.5	−0.3	−1.5	−4.5	−15
	$\sigma=60$, $c=36$, irrelevant, 50% wrong	1.00	≤ 0.0	0.0	−0.1	−0.4	−1.0	−2.9	−8.5

Table A.3: **Description weights by demonstration count: distribution readout.** Results cover the Bayesian reference and the three distribution-readout models in the 2,000-task run without reasoning. "Number" is defined as in Table A.1; cells are italicized when the numeric response rate is below three quarters at the corresponding n.

Predictor	Description	number	$n=0$	$n=1$	$n=2$	$n=4$	$n=8$	$n=16$	$n=32$	$n=64$
Reference	$\sigma=60$, $c=1$, relevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=60$, $c=1$, irrelevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=60$, $c=4$, relevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=60$, $c=4$, irrelevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=60$, $c=36$, relevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=60$, $c=36$, irrelevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=30$, $c=1$, relevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=30$, $c=4$, relevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=30$, $c=36$, relevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=60$, $c=4$, relevant, 10% wrong		0.90	0.71	0.59	0.45	0.30	0.19	0.10	0.05
	$\sigma=60$, $c=4$, irrelevant, 10% wrong		0.90	0.27	0.14	0.08	0.04	0.02	0.01	0.00
	$\sigma=60$, $c=36$, relevant, 10% wrong		0.90	0.84	0.84	0.82	0.77	0.65	0.51	0.34
	$\sigma=60$, $c=36$, irrelevant, 10% wrong		0.90	0.26	0.14	0.07	0.03	0.01	0.00	0.00
	$\sigma=60$, $c=4$, relevant, 50% wrong		0.50	0.46	0.39	0.32	0.22	0.14	0.08	0.04
	$\sigma=60$, $c=4$, irrelevant, 50% wrong		0.50	0.12	0.07	0.03	0.02	0.01	0.00	0.00
	$\sigma=60$, $c=36$, relevant, 50% wrong		0.50	0.55	0.58	0.60	0.59	0.53	0.42	0.29
	$\sigma=60$, $c=36$, irrelevant, 50% wrong		0.50	0.12	0.06	0.03	0.01	0.00	0.00	0.00
gpt-oss-20b	$\sigma=60$, $c=1$, relevant	0.98	0.94	0.53	0.29	0.23	0.16	0.12	0.12	0.23
	$\sigma=60$, $c=1$, irrelevant	0.98	0.93	0.59	0.25	0.18	0.09	0.05	0.04	0.10
	$\sigma=60$, $c=4$, relevant	0.98	0.92	0.48	0.32	0.26	0.21	0.19	0.18	0.31
	$\sigma=60$, $c=4$, irrelevant	0.97	0.92	0.57	0.25	0.16	0.08	0.04	0.04	0.09
	$\sigma=60$, $c=36$, relevant	0.98	0.94	0.49	0.35	0.30	0.25	0.23	0.22	0.35
	$\sigma=60$, $c=36$, irrelevant	0.98	0.95	0.60	0.25	0.15	0.07	0.04	0.04	0.09
	$\sigma=30$, $c=1$, relevant	0.99	0.98	0.43	0.23	0.17	0.12	0.09	0.11	0.21
	$\sigma=30$, $c=4$, relevant	0.98	0.98	0.38	0.28	0.23	0.19	0.18	0.19	0.31
	$\sigma=30$, $c=36$, relevant	0.99	0.99	0.37	0.32	0.27	0.23	0.22	0.24	0.37
	$\sigma=60$, $c=4$, relevant, 10% wrong	0.96	0.91	0.45	0.33	0.15	0.13	0.12	0.13	0.22
	$\sigma=60$, $c=4$, irrelevant, 10% wrong	0.96	0.91	0.55	0.34	0.11	0.05	0.02	0.02	0.04
	$\sigma=60$, $c=36$, relevant, 10% wrong	0.97	0.92	0.44	0.34	0.17	0.15	0.15	0.16	0.25
	$\sigma=60$, $c=36$, irrelevant, 10% wrong	0.96	0.92	0.53	0.34	0.10	0.04	0.02	0.02	0.04
	$\sigma=60$, $c=4$, relevant, 50% wrong	0.95	0.79	0.46	0.32	0.16	0.12	0.12	0.12	0.20
	$\sigma=60$, $c=4$, irrelevant, 50% wrong	0.94	0.79	0.53	0.32	0.11	0.04	0.02	0.01	0.04
	$\sigma=60$, $c=36$, relevant, 50% wrong	0.95	0.83	0.46	0.32	0.17	0.14	0.15	0.15	0.23
	$\sigma=60$, $c=36$, irrelevant, 50% wrong	0.95	0.83	0.53	0.31	0.11	0.04	0.02	0.02	0.04
Phi-4	$\sigma=60$, $c=1$, relevant	0.87	*0.97*	0.82	0.62	0.43	0.28	0.18	0.13	0.12
	$\sigma=60$, $c=1$, irrelevant	0.86	*0.97*	0.83	0.64	0.42	0.22	0.11	0.06	0.05
	$\sigma=60$, $c=4$, relevant	0.85	*0.98*	0.76	0.55	0.37	0.26	0.19	0.15	0.16
	$\sigma=60$, $c=4$, irrelevant	0.84	*0.98*	0.79	0.56	0.36	0.18	0.09	0.06	0.05
	$\sigma=60$, $c=36$, relevant	0.86	*0.99*	0.74	0.53	0.38	0.29	0.23	0.20	0.20
	$\sigma=60$, $c=36$, irrelevant	0.84	*0.99*	0.78	0.53	0.33	0.16	0.09	0.06	0.04
	$\sigma=30$, $c=1$, relevant	0.87	*0.99*	0.77	0.58	0.38	0.22	0.13	0.09	0.09
	$\sigma=30$, $c=4$, relevant	0.86	*1.00*	0.69	0.51	0.34	0.23	0.16	0.13	0.13
	$\sigma=30$, $c=36$, relevant	0.88	*1.00*	0.65	0.48	0.32	0.23	0.18	0.17	0.16
	$\sigma=60$, $c=4$, relevant, 10% wrong	0.87	*0.94*	0.62	0.46	0.30	0.21	0.15	0.13	0.11
	$\sigma=60$, $c=4$, irrelevant, 10% wrong	0.85	*0.95*	0.62	0.44	0.26	0.13	0.06	0.04	0.03
	$\sigma=60$, $c=36$, relevant, 10% wrong	0.87	*0.95*	0.61	0.46	0.33	0.25	0.20	0.18	0.16
	$\sigma=60$, $c=36$, irrelevant, 10% wrong	0.86	*0.96*	0.61	0.43	0.26	0.13	0.06	0.04	0.03
	$\sigma=60$, $c=4$, relevant, 50% wrong	0.87	*0.92*	0.57	0.43	0.29	0.20	0.15	0.13	0.11

	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	0.85	*0.92*	0.57	0.42	0.24	0.12	0.06	0.04	0.03
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	0.88	*0.93*	0.56	0.44	0.33	0.24	0.21	0.19	0.17
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	0.86	*0.94*	0.57	0.41	0.24	0.12	0.06	0.04	0.03
OLMo 2 32B	$\sigma = 60$, $c = 1$, relevant	0.90	*0.99*	0.72	0.33	0.25	0.16	0.09	0.05	0.05
	$\sigma = 60$, $c = 1$, irrelevant	0.89	*0.99*	0.77	0.29	0.16	0.09	0.05	0.02	0.02
	$\sigma = 60$, $c = 4$, relevant	0.91	*1.00*	0.70	0.37	0.30	0.21	0.14	0.08	0.07
	$\sigma = 60$, $c = 4$, irrelevant	0.90	*1.00*	0.75	0.29	0.15	0.08	0.04	0.02	0.01
	$\sigma = 60$, $c = 36$, relevant	0.91	*1.00*	0.70	0.40	0.34	0.26	0.19	0.12	0.10
	$\sigma = 60$, $c = 36$, irrelevant	0.90	*1.00*	0.76	0.29	0.17	0.08	0.04	0.02	0.02
	$\sigma = 30$, $c = 1$, relevant	0.92	*0.99*	0.69	0.32	0.23	0.14	0.09	0.04	0.04
	$\sigma = 30$, $c = 4$, relevant	0.92	*1.00*	0.65	0.34	0.27	0.19	0.13	0.06	0.06
	$\sigma = 30$, $c = 36$, relevant	0.93	*1.00*	0.64	0.36	0.32	0.23	0.16	0.08	0.09
	$\sigma = 60$, $c = 4$, relevant, 10% wrong	0.96	1.01	0.72	0.42	0.29	0.18	0.11	0.09	0.05
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	0.96	1.01	0.81	0.32	0.14	0.06	0.03	0.02	0.01
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	0.96	1.01	0.71	0.44	0.33	0.23	0.15	0.13	0.07
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	0.96	1.01	0.82	0.33	0.15	0.07	0.03	0.02	0.01
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	0.93	1.02	0.65	0.39	0.27	0.16	0.09	0.07	0.04
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	0.92	1.01	0.71	0.29	0.13	0.06	0.02	0.01	0.01
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	0.93	1.02	0.64	0.42	0.31	0.21	0.13	0.11	0.06
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	0.92	1.02	0.72	0.30	0.14	0.06	0.03	0.02	0.01